

SEMESTER 3

CHEMICAL ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3
PARTICLE TECHNOLOGY

Course Code	PCCHT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Enable the students to identify various methods for particle separation based on size, density or both.
2. Enable the students to learn the principle and working of various equipment used for handling particulate solids and apply in design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Particle diameter and shape factor - particle size analysis - sieve analysis - particle size distribution - cumulative and differential methods of analysis - mean diameters – specific surface area and number of particles - screening - effectiveness and capacity of screen and factors affecting them - types of industrial screens. Principles of free and hindered settling- Sub-sieve analysis - pipette analysis -beaker decantation- elutriation. Classification- sorting classification -sink and float and differential classification-equal settling particles - mechanical and non-mechanical classifiers. Problem solving in size analysis using EXCEL / Matlab / SCILAB / PYTHON etc. -only for selfstudy / microproject / assignment.	11

2	Principles of mineral beneficiation methods - jigging - Wilfley table - froth flotation, principles, additives- batch and continuous thickening - Kynch theory - design of continuous thickener. Laws of comminution - mechanism and efficiency of size reduction – construction, working and principle of important size reduction equipments(jaw crushers, gyratory crushers, crushing rolls , tumbling mills, hammer mill, fluid energy mill, colloid mill and rotary knife cutter)- closed circuit and open circuit grinding - free crushing and choke feeding - wet and dry grinding. Problem solving in thickener design using EXCEL/ Matlab / SCILAB / PYTHON etc. – only for selfstudy / microproject / assignment	11
3	Filtration: theory of constant pressure and constant rate filtration - cake porosity and compressibility - filter aids - optimum filtration cycle - types of batch and continuous filters (plate and frame, rotary drum, leaf filters- construction and working) -washing of filter cakes –principles of centrifugal filtration (Equipment's not required) Problem solving in filtration using EXCEL / Matlab / SCILAB / PYTHON etc. – only for selfstudy / microproject / assignment	11
4	Separation of solids from gases: cyclone separation – electrostatic precipitation –Bag filters- venturi scrubber. Mixing of granular solids and pastes – mixing performance and effectiveness – mixers for noncohesive and cohesive solids –Working of different types of conveyors-Belt, chain, screw and pneumatic.	11

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the particle size distribution, mean particle diameter, specific surface area and number of particles per unit mass using techniques such as sieve analysis, pipette analysis and beaker decantation	K3
CO2	Explain the settling principles to design classifiers and thickeners and understand various mineral beneficiation techniques using settling principles.	K2
CO3	Apply size reduction laws to compute power requirement for size reduction and understand the various size reduction equipment used in industries.	K3
CO4	Apply the concepts of filtration theory to compute filtration variables and select appropriate filtration equipment	K3
CO5	Explain separation techniques for particulates in air, mixing of solids and conveying processes for solids in chemical industries	K2
CO6	Examine and run software programs for solving basic problems in Particle Technology	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										1
CO3	3	3										1
CO4	3	3										1
CO5	3											1
CO6	3	3	3		3				3			3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit Operations of Chemical Engineering	McCabe W.L., Smith J.C. and Parrot P.	McGraw Hill, New York	6th Edition 2001.
2	Mechanical Operations	Anup K Swain, Hemlata Patra, G.K.Roy ,	Mc-Graw-Hill Education	1 st Edition July 2017
3	Mechanical Operations for Chemical Engineers	C.M.Narayan, B.C. Bhattacharya	Khanna Publishers	1990, (ISBN-10 : 8174090363, ISBN-13 : 978-8174090362)
4	Unit Operations	G G Brown	CBS Publishers	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Coulson and Richardson's CHEMICAL ENGINEERING VOLUME 2 Particle Technology and Separation Processes	J. F. Richardson J. H. Harker J. R. Backhurst	Butterworth-Heinemann	5 th Edition, 2002
2	Introduction to chemical engineering.	Walter L. Badger and Julius T. Banchero.	McGraw-Hill book company, Inc., New York	(1955)
3	Unit Operations of Particulate Solids: Theory and Practice	E. Ortega-Rivas	CRC Press, FL	2012
4	Perry's Chemical Engineers' Handbook	Editor-in-Chief, Don W. Green, Associate Editor Marylee Z. Southard	Mc-Graw-Hill Education	9 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/103/103103155/ , https://nptel.ac.in/courses/103107123
2	https://archive.nptel.ac.in/courses/103/103/103103155/ , https://nptel.ac.in/courses/103107123
3	https://archive.nptel.ac.in/courses/103/103/103103155/ , https://nptel.ac.in/courses/103107123
4	https://archive.nptel.ac.in/courses/103/103/103103155/ , https://nptel.ac.in/courses/103107123

SEMESTER S3

CHEMICAL ENGINEERING THERMODYNAMICS

Course Code	PCCHT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Equip students with the fundamental knowledge and analytical skills necessary for designing, analysing, and optimizing chemical processes in various industries.
2. Enable the students to understand and analyse ideal conditions in various aspects of chemical processes and thus develop more efficient and effective engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction, First Law of Thermodynamics - Energy Balance for closed and open Systems- application of first law to analyse flow through horizontal pipe and nozzle. Limitations of First Law. Second Law of Thermodynamics - definition of entropy – calculation of entropy changes in processes involving ideal gases, Third law of Thermodynamics. Fundamental property relations, Maxwell's Equations, Clausius-Clapeyron equation, Entropy-heat capacity relationships - equations for entropy, internal energy and enthalpy in terms of measurable quantities, Joule-Thomson coefficient, Gibbs Helmholtz Equation. Problem solving in application of laws of thermodynamics, and property calculations using EXCEL / Matlab / SCILAB / PYTHON etc. – only for selfstudy / microproject /assignment	11

2	P-V and P-T diagram of pure substances, Cubic Equations of state for real gases (Concept of Redlich Kwong, Peng Robinson and Virial equations) Principle of corresponding states and generalized compressibility chart, Concept of Residual properties, Estimation of Fugacity and activity of pure fluids - effect of temperature and pressure on fugacity and activity. Definition of partial molar properties and chemical potential, Fugacity in solution, Ideal solution, Lewis-Randall rule - Raoult's law, Henry's law, Activity in solutions, Activity coefficient - Gibbs-Duhem equations, property changes of mixing, excess properties. Problem solving in Equations of state, properties of ideal solutions using EXCEL/Matlab/SCILAB/PYTHON etc. – only for selfstudy/microproject/assignment	11
3	Criterion of phase equilibrium – equations for single component and multicomponent equilibria, Gibbs Phase rule, Duhem's theorem. Vapour-liquid equilibrium- phase diagram for binary solutions- VLE in ideal solutions, non-ideal solutions, azeotropes, equations for VLE at low pressures and VLE at high pressures Activity coefficient models (General concepts of Margules equation, Wohl's equation -Wilson equation - NRTL, UNIQUAC and UNIFAC models), calculations based on Van Laar equation Vapour Liquid-Liquid Equilibrium in partially miscible and immiscible systems. Calculations in Vapour liquid Equilibrium using EXCEL/Matlab/SCILAB/PYTHON etc. –only for selfstudy/microproject/assignment	11
4	Chemical reaction equilibria: Extent of reaction, equilibrium constant, standard free energy change, feasibility of reaction, effect of temperature on equilibrium constant, evaluation of equilibrium constant, Equilibrium conversion in gas-phase reactions, Effect of pressure and other parameters on conversion, Simultaneous reactions, liquid phase reactions, pressure of decomposition, Phase-rule for reacting systems. Problem solving in Reaction equilibrium using EXCEL/Matlab/SCILAB/PYTHON etc. – only for selfstudy/microproject/assignment	11

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the law of thermodynamics to derive correlations among thermodynamic properties	K3
CO2	Choose suitable parameters for quantifying deviation from ideal gas/ solution behaviour	K3
CO3	Solve vapour liquid equilibria at varying conditions	K3
CO4	Solve the equilibrium constant and equilibrium conversion of chemical reactions	K3
CO5	Examine and run software programs for solving basic problems in Thermodynamics	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									3
CO2	3	2	2									3
CO3	3	2	2									3
CO4	3	2	2									3
CO5	3	3	3		3				2			3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Chemical Engineering Thermodynamics	Smith J. M. & Van Ness H.V.	McGraw Hill	8 th Edn., 2018
2	A Textbook of Chemical Engineering Thermodynamics	Narayanan K. V	Prentice-Hall of India	2 nd Edn., 2013.
3	Chemical Engineering Thermodynamics	Rao Y.V.C	Universities Press	1997

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering and Chemical Thermodynamics	Milo D. Koretsky	Wiley	2 nd Edn, 2012
2	Chemical and Process Thermodynamics	Kyle B.G.	Pearson	3 rd Edn, 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/103/103103144/
2	https://archive.nptel.ac.in/courses/103/103/103103144/
3	https://archive.nptel.ac.in/courses/103/103/103103144/
4	https://archive.nptel.ac.in/courses/103/103/103103144/

SEMESTER S3

FLUID AND PARTICLE MECHANICS

Course Code	PBCHT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Introduce the concepts of fluid and particle mechanics and its relevance in Chemical Engineering.
2. Impart the fundamental principles of momentum transport for systems with and without particles.
3. Develop the ability to apply principles of fluid and particle mechanics for the design of process flow systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of fluid, Continuum mechanics, Newton's law of viscosity. Physical properties of fluid: Density, viscosity, compressibility & elasticity, surface tension & capillarity. Variation of viscosity and density with temperature and pressure. Rheology of fluids, Rheological Classification of fluids. Pascal's law, Hydrostatic equilibrium in gravity and centrifugal field. Barometric equation. Lapse rate. Principles of continuous gravity decanter and centrifugal decanter. Principles of Manometer, Simple manometer, inclined tube manometer. Introduction to fluid flow: Flow field, Eulerian and Lagrangian approach. Reynolds experiment, Reynolds number, Hydraulic radius and Equivalent diameter, Turbulence, Reynolds stress.	9

2	Basic equations of fluid flow: Continuity Equation, Macroscopic momentum Balance, Navier stokes equation (derivation not required), Bernoulli's Equation. Kinetic energy correction factor. Correction for fluid friction, Pump work, and compressible flow in Bernoulli's equation. Laminar flow of incompressible fluids in conduits, Shear stress and Velocity distribution, Maximum and average velocity-Hagen Poiseuille equation- Friction factor and Reynolds number relationship. Pumps, fan, blower, compressor and its classification. Types of pumps (limited to application, and operation with neat sketch). Centrifugal pumps-Variou losses, Characteristic curves (input power, output power, efficiency, head vs flow rate) (no derivation required), NPSH, Suction lift, Cavitation, Specific speed and Priming, Affinity laws.	9
3	Turbulent flow of incompressible fluids in pipes and conduits: Universal velocity distribution equation (derivation not required), Friction factor and Reynolds number relationship-Nikuradse and Karman equation-Blasius equation (derivation not required), Prandtl one seventh power law-Friction factor chart. Pipe and tubing, Joints and fittings, Valves (diagram and operation of Gate valves and globe valves, diaphragm valves, butterfly valve and ball valves, Check valves), Equivalent length. Friction from changes in velocity or direction-Sudden expansion and contraction-Effect of fittings and valves on friction. Flow measurement devices- types. Flow rate equation for Venturi; Orifice; flow nozzle; Pitot tube; Rectangular, Triangular & Trapezoidal weir; Rotameter	9
4	Flow in boundary-layers, Boundary-layer formation in straight tubes, Boundary-layer separation and wake formation. Flow past immersed bodies -Drag coefficient. Terminal settling velocity in different regimes. Solid-fluid contacting systems. Flow through packed bed - Ergun equation -Kozeny-Carman equation - BurkePlummer equation. Fluidization The phenomenon of fluidization, type of fluidization, Different regimes of fluidization,. Advantages and disadvantages of fluidized beds and packed bed. Industrial applications of fluidized bed and packed bed. Pressure drop calculation, minimum fluidizing velocity, effect of pressure and temperature on fluidized bed behaviour.	9

Suggestion on Project Topics

Project based learning (8hrs)

- The students shall be directed to design a part of flow network consisting of combinations pipe sections, fittings, flow measuring devices from an appropriate process flow diagram.
- The flow system selected may fall into either of the two category such as internal flow and external flow systems.
- The design developed should include the following concepts:
 1. The relevance of transport properties in designing the system as studied in module 1
 2. The principles of pressure measurement using the concepts of hydrostatic equilibrium learned in module 1.
 3. The type of flow such as laminar or turbulent.
 4. How to verify the continuity, momentum, and energy balance equations with reference to the selected flow system as learned in the module 2.
 5. The use of pumping devices in flow systems.
 6. How to calculate the pressure drop in the selected flow system.
 7. The use of friction factor in calculating the pressure drop induced by various pipe fitting and auxiliaries as learned in the module 3.
 8. The working of various flow measuring devices used in flow system and the at least one flow measuring device should be there in the selected system.
 9. How to calculate the pressure drop in flow system falls into the category of external flow as learned in the module 4.
 10. The pressure drop estimation in packed and fluidized beds.

How to do the modelling and simulation of such simple systems selected using any appropriate software/ Experimental validation of such system.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the basic properties and transport laws to fluid in different conditions like statics and dynamics	K3
CO2	Apply the fluid flow principles in the application of the mass, momentum, and energy equations.	K3
CO3	Select pipe fittings, valves, pumps, and flow measuring devices in process industries with the knowledge of the basic principles.	K3
CO4	Compare fluidized bed and a packed bed using the concept of fluid dynamics considering its application.	K4
CO5	Select and analyse a flow system using appropriate simulation software	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3				2			2
CO2	3	3	3	3	3				2			2
CO3	3	3	3	3	3				2			2
CO4	3	3	3	3	3				2			2
CO5	3	3	3	3	3				3	3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit Operations of Chemical Engg.	McCabe W.L., Smith J.C., Harriot P.,	McGraw Hill	7 th Edition 2017
2	Fluid mechanics.	Frank M. White	McGraw Hill.	7 th Edition 2011
3	Fluid mechanics.	Y. Nakayama	Butterworth-Heinemann.	2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Perry's Chemical Engineers' Handbook, 7th Edition,	Perry, RH & Green, DW	McGraw-Hill	
2	Principles of unit operations	Foust A S, Wenzel L A, Clump C W, Maus L, Andersen L B	John Wiley & Sons	2 nd Edition 2015
3	Fluid Mechanics for Chemical Engineers	Noel de Nerves	McGraw Hill	
4	Fluid Mechanics	Streeter V.L.	McGraw Hill	
5	Chemical Engg. Vol. 1,	Coulson J.M. & Richardson J.F.,	Pergamon	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/106/103106158/
2	https://archive.nptel.ac.in/courses/103/106/103106158/
3	https://archive.nptel.ac.in/courses/103/106/103106158/
4	https://archive.nptel.ac.in/courses/103/106/103106158/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11

3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, Power BI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpgclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
	b)	Point out any three merits of NPV method.	CO4	(3)												

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology,	6

	Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental	6

	policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15

4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.

- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

CHEMICAL TECHNOLOGY AND ENVIRONMENTAL ENGINEERING LAB

Course Code	PCCHL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Lab

Course Objectives:

1. To provide experience on preparation, analysis and testing of chemicals used for industrial raw materials and end users.
2. To provide experience in sampling and analysis of environmental quality control.

Expt. No.	Experiments
1	Acid value of oils
2	Iodine value of oils
3	Saponification value of oils
4	Preparation and analysis of soap
5	Determination of available chlorine in bleaching powder
6	Determination of flash and fire point and viscosity of lubricating oil
7	Determination of hardness of water
8	Estimate the total solids and dissolved solids content of the given wastewater Sample.
9	Determination of BOD of wastewater sample
10	Determination of COD of wastewater sample
11	Analysis of oil & grease in wastewater sample
12	Preparation of biodiesel
13	Preparation of magnetite particle
14	Preparation of pigments
15	Study of analytical instruments: Spectrophotometer, pH meter, Gas Chromatograph, High Performance Liquid Chromatography (HPLC), Total Organic Carbon Analyser (TOC) etc

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse and estimate parameters for the selected chemicals.	K4
CO2	Apply skills to use analytical and instrumental methods for measurement of parameters relevant to chemical engineering.	K3
CO3	Apply skills of accuracy in experimentation, interpret the experimental result and suggest its area of application.	K3
CO4	Develop capacity to work in team and exhibit knowledge of safety, health and environment by practicing laboratory ethics	K3
CO5	Make use of students with the handling and analysis of chemicals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									
CO2		3	3									3
CO3		3	3	3	3				3	3		3
CO4			3			3	3	3	3			3
CO5						3	3	3	3			3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Vogels Textbook of quantitative chemical analysis	G H Jeffery, J Bassett, J Mendham, R C Denney	John Wiley & Sons, Inc., New York	Fifth edition, 1989
2	Sawyer and McCarty, "Chemistry for Environmental Engineers",	Clair N. Sawyer, Gene F. Parkin, Perry L. Mccarty	Tata McGraw Hill	1994
3	"Water Supply Engineering",	S.K.Garg,	Khanna Publishers.	2010

Reference Books		
Sl. No	Title of the Book	Name of the Publisher
1	IS 2490 - 1981, IS 3306 - 1974, IS 3307 - 1977, IS 7968 - 1976, IS 2296 - 1974	Bureau of Indian Standards, New Delhi.

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

FLUID & PARTICLE MECHANICS LAB

Course Code	PCCHL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3: 0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Fluid and particle mechanics (PBCHT304)	Course Type	Lab

Course Objectives:

1. To understand the basic principles of flow meters
2. To perform experiments in fluid moving machinery and analyse the principles involved
3. To perform experiments in solid-fluid systems and analyse the principles involved

Expt. No.	Experiments
1	Experiments on Reynolds apparatus for determination of flow regime and construction of fanning friction factor vs. Reynolds No. plot
2	Determination of co efficient of discharge for orifice meter
3	Determination of co efficient of discharge for venturimeter
4	Determination of co-efficient of Pitot tube and construction of velocity profile across the cross section of pipe.
5	Determination of co-efficient of discharge for different types of weirs.
6	Determination of pressure drop for flow through packed bed and verification of Ergun equation.
7	Experiment on fluidization techniques and determination of a) Minimum fluidization velocity; b) Pressure drop profile.
8	Determination of efficiency of a centrifugal pump or Reciprocating Pump.
9	Determination of the friction factor for pipes of different sizes and materials in different ranges of Reynolds number.
10	Calibration of a Rotameter.
11	Determination of viscosity of Newtonian & non-Newtonian fluid by Falling Sphere method.

12	Verification of Bernoulli's Theorem.
13	Determination of coefficient of discharge C_d , coefficient of velocity C_v , and coefficient of contraction C_c for open orifice

***Minimum 10 Experiments**

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Plan and perform experiments in pipes and flow measuring equipments. Also analyse the principles involved.	K4
CO2	Plan and perform experiments in fluid moving machinery and analyse the principles involved.	K4
CO3	Plan and perform experiments in solid-fluid systems and analyse the principles involved.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3								3	3		3
CO2	3								3	3		3
CO3	3								3	3		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit Operations of Chemical Engineering	McCabe W.L. & Smith J.C.	McGraw Hill	7 th Edition 2017
2	Principles of Unit operation	Foust, Wenzel, Clump, Maus & Anderson	Wiley- India	2 nd Edition 2015
3	Fluid Mechanics and Hydraulic Machines	R.K.Bansal	Laxmi Publications	9 th Edition 201

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

CHEMICAL ENGINEERING

SEMESTER S4
MATHEMATICS FOR PHYSICAL SCIENCE AND LIFE SCIENCE – 4
(Group C)

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5, 4.2, 7.5, 8.4, 8.5, 9.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z -tests and the one-sample t -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: *K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create*

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S S Sastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4

HEAT TRANSFER OPERATIONS

Course Code	PCCHT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Provide fundamental understanding of basic principles of heat transfer and various modes of heat transfer.
2. Solve heat transfer problems using the principles and correlations for different modes of heat transfer.
3. Design and estimate heat transfer rate for heat transfer equipment like heat exchangers and evaporators.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic Concepts - Overview of applications of heat transfer in different fields of engineering, modes of heat transfer - conduction, convection and radiation. Material properties of importance in heat transfer - thermal conductivity, thermal diffusivity, specific heat capacity, isotropic and anisotropic materials, thermal conductivity of solids, liquids and gases – comparison between them, effect of temperature on thermal conductivity. Conduction Heat Transfer - General heat conduction equation in cartesian coordinates (derivation required), Fourier heat conduction equation, conduction through systems of constant thermal conductivity - conduction through plane, cylindrical and spherical wall, combined boundary condition systems (conduction-convection systems), conduction through composite slab - multi-layered plane, cylindrical and spherical shells, electrical analogy to heat flow, overall heat transfer coefficient. Thermal Insulation - Analysis of critical radius of insulation for cylinders and spheres, critical thickness of insulation, concept of optimum thickness of insulation, concept of thermal contact resistance. Unsteady State Heat Conduction - Analysis of transient heat flow with	11

	negligible internal resistance-lumped capacity analysis, concept of BiotModulus and Fourier number. Numerical problems of practical importance on the above topics.Problem solving in conduction using EXCEL / Matlab /SCILAB/ PYTHON etc. – only for selfstudy / microproject / assignment	
2	Convection - Mechanism, boundary layer concepts, thermal and velocity boundary layers, boundary layer thickness, relationship between hydrodynamic and thermal boundary layer thickness for flow over flat plates. The convective heat transfer coefficient, dimensionless numbers in convection. Dimensional analysis - Buckingham's pi theorem, its application of dimensional analysis to forced convection. Forced Convection – correlation equations for estimation of convection heat transfer coefficient- laminar and turbulent flow in a circular tube, laminar and turbulent flow over flat plates, numerical problems of practical interest. Natural Convection - Dimensional analysis, natural convection from vertical and horizontal surfaces under laminar and turbulent conditions for flat plates and cylinders under constant heat flux and constant wall temperature conditions, physical significance of Grashof and Rayleigh numbers, numerical problems of practical interest. Analogy between momentum and heat transfer- Reynolds, Prandtl, Colburn analogies. Radiation heat transfer - thermal radiation, spectral emissive power, total emissive power, emissivity, radiative properties- Emission, irradiation, radiosity, absorptivity, reflectivity and transmissivity, concept of black and gray bodies. Black body radiation -laws of black body radiation, view factor, radiation between two large parallel gray planes-derivation of expression for rate of radiant energy exchange. Problem solving in convection and radiation using EXCEL/ Matlab/ SCILAB/ PYTHON etc. – only for selfstudy /microproject/ assignment	11
3	Boiling and Condensation - pool boiling, regimes of boiling, correlation for critical heat flux for nucleate pool boiling-Condensation - Drop and film type condensation – Nusselt equation to determine the heat transfer coefficient in film condensation (no derivation). Evaporation - short tube vertical or calandria type evaporators- Evaporator auxiliaries -steam traps, vacuum devices and entrainment separators. Single	11

	effect and multiple effect evaporators, performance of evaporators, capacity and economy of evaporators, factors affecting the performance of evaporators, effect of liquid head and boiling point elevation, material and energy balances for single effect evaporator, numerical problems of practical interest. Multiple effect evaporators - material and energy balances, Different feeding arrangements in multiple effect evaporators – their merits and demerits. Problem solving in boiling and evaporation, evaporator design using EXCEL/Matlab/SCILAB/PYTHON etc. – only for selfstudy/ microproject/ assignment	
4	Heat exchangers - classification according to construction features - tubular heat exchangers (shell and tube, double pipe), plate-type heat exchangers, extended surface heat exchangers, classification according to flow arrangements - single-pass exchangers, multi-pass exchangers. Concept and types of fouling - fouling factors, determination of overall heat transfer coefficient with and without fouling, derivation of expression for LMTD, concept of correction factor logarithmic mean temperature difference. Heat exchanger analysis using LMTD method in parallel flow, counter flow, cross flow and multi-pass heat exchangers, temperature – distance plots for different flow arrangements in single and multi-pass heat exchangers. NTU method of heat exchanger calculation, determination of area, length, number of tubes required for a given duty in different configurations using LMTD and NTU method of analysis. Problem solving in heat exchanger design using EXCEL /Matlab / SCILAB / PYTHON etc. – only for selfstudy /microproject/assignment	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply appropriate governing equations and analyse conduction heat transfer problems for different geometries under steady state and transient processes.	K3
CO2	Identify and distinguish type of convection, and solve forced and natural convection heat transfer problems using empirical equations	K3
CO3	Apply the concepts behind radiation heat transfer and solve radiation heat transfer problems	K3
CO4	Select the heat transfer processes involved in boiling, condensation and evaporation, and analyse evaporator performance.	K3
CO5	Utilize the basic concepts and apply to the design of heat exchangers.	K3
CO6	Select and run software programs for solving basic problems in heat transfer operations	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2								3
CO2	3	3	3	2								3
CO3	3	3	3	2								3
CO4	3	3	3	2								3
CO5	3	3	3	2								3
CO6	3	3	3	3	3				3	2		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat Transfer: Principles and Applications	Binay K Dutta	Prentice Hall India	2 nd Edition 2022
2	Fundamentals of Engineering Heat and Mass Transfer	R C Sachdeva	New age International Publishers	6 th Edition 2022
3	Unit Operations in Chemical Engineering	McCabe WL, Smith JC&Harriott P	McGraw Hill	7 th Edition 2017
4	Heat Transfer	Hollman JP	McGraw Hill	10 th Edition 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat transfer - A basic Approach	MNecatiOzisik	McGraw-Hill Education	1984
2	Process Heat Transfer	D Q Kern	McGraw Hill	2 nd Edition 2019
3	Chemical Engineering, Vol. I	Coulson JM& Richardson JF	Pergamon Press	6 th Edition 1999
4	Introduction to Heat Transfer	Incropera F P and DeWitt D P	John Wiley New York	2nd Ed (1996).

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://www.digimat.in/nptel/courses/video/103101137/L01.html to http://www.digimat.in/nptel/courses/video/103101137/L13.html
2	http://www.digimat.in/nptel/courses/video/103101137/L16.html to http://www.digimat.in/nptel/courses/video/103101137/L38.html http://www.digimat.in/nptel/courses/video/103101137/L43.html to http://www.digimat.in/nptel/courses/video/103101137/L52.html
3	http://www.digimat.in/nptel/courses/video/103101137/L39.html to http://www.digimat.in/nptel/courses/video/103101137/L42.html
4	http://www.digimat.in/nptel/courses/video/103101137/L56.html to http://www.digimat.in/nptel/courses/video/103101137/L60.html

SEMESTER S4

MASS TRANSFER OPERATIONS 1

Course Code	PCCHT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-1-0-0	ESE Marks	60
Credits	4	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	Process Calculations PCCHT205	Course Type	Theory

Course Objectives:

1. Impart the fundamentals of mass transfer operations in industrial processes.
2. Apply the principles of mass transfer to the design of equipment used in separation processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Molecular diffusion: Mass and molar flux, Fick's law, diffusivity and estimation, steady state diffusion of A through stagnant B and equimolar counter diffusion in binary gases, liquids and multi component gas mixtures. Mass transfer coefficients: Introduction to mass transfer coefficients, dimensionless groups, analogy between mass, heat and momentum transfer. Basic concepts and assumptions involved in theories of mass transfer: penetration and surface renewal theories. Inter-phase mass transfer: Equilibrium, diffusion between phases, two-film theory, local and overall k-type coefficients. Problem solving in molecular diffusion, mass transfer coefficient calculation using EXCEL /Matlab/SCILAB/PYTHON etc. – only for self study/microproject/assignment	11
2	Gas-Liquid contacting equipment: Tray towers: -tray types, wetted wall columns, venturi scrubbers. Packed columns:-packing materials and characteristics, factors affecting column performance-flooding, priming, coning, weeping, loading etc, comparison between plate and packed columns.	13

	Gas absorption: Solubility of gases in liquid, choice of solvent, Material balance in counter current and co-current absorption and stripping, L/G ratio, Minimum gas liquid ratio, Multistage operation, number of plates by graphical construction, Kremser equation, tray efficiency, design of packed columns, transfer unit and general graphical method, dilute solutions and simplified design methods. Problem solving in absorption using EXCEL / Matlab / SCILAB/ PYTHON etc. – only for self study / microproject / assignment	
3	Humidification and dehumidification: Use of humidity chart to find properties of air, Lewis relation, water cooling with air, types of cooling towers, design calculations of cooling tower, Spray chambers for air humidification, principles of gas dehumidification. Adsorption:- types of adsorption, properties of adsorbents adsorption single gases, vapours and dilute liquid solutions, Adsorption isotherms – Freundlich and Langmuir, Fixed bed adsorption, adsorption wave, rate of adsorption and breakthrough curve. Problem solving in humidification and adsorption using EXCEL/Matlab/SCILAB/PYTHON etc. – only for self study/microproject/assignment	10
4	Drying: Equilibrium, bound and unbound moisture content, batch drying, rate of drying, mechanism of moisture movement, continuous drying, parallel and counter current drying, material and enthalpy balances, classification of industrial dryers for batch and continuous drying. Crystallization: Principles of crystallization, purity, yield, super-saturation, nucleation-primary and secondary nucleation, growth of crystals-Delta –L law, Ostwald ripening, crystallisation equipment-Classification of crystallizers, working and schematic diagram of forced circulation, draft tube baffled and Oslo crystallizers, MSMR. Problem solving in drying and crystallisation using EXCEL/ Matlab / SCILAB / PYTHON etc. – only for self study / microproject / assignment	10

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply fundamentals of mass transfer operations and concepts of mass transfer coefficients in interphase mass transfer.	K3
CO2	Select suitable Gas-Liquid contacting equipment for mass transfer operations and identify its operational difficulties.	K3
CO3	Analyze and estimate design parameters of tray and packed columns of gas liquid contacting equipments	K4
CO4	Analyze and estimate design parameters of humidification and adsorption systems	K4
CO5	Analyze and estimate design parameters of drying and crystallization systems.	K4
CO6	Select and run software programs for solving basic problems in mass transfer operations	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3										
CO3	3	3	3									
CO4	3	3	3									
CO5	3	3	3									
CO6	3	3	3		3				3			3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mass Transfer Operations,	Treybal R.E.,	McGraw Hill	3 rd Edition 2017
2	Mass Transfer-Theory and Applications	K.V. Narayanan and B. Lakshmikutty	CBS Publishers	2 nd Edition 2023
3	Principles of Mass Transfer & Separation Processes	Binay K Dutta	PHI Learning Private Limited.	1 st edition 2006
4	Separation Process Principles	Seader, J.D. and Henley, E.J.,	Wiley, New York.	4 th Edition 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mass Transfer-Theory and Practice	N.Anantharaman and K.M.MeeraSheriffa Begum	PHI Learning Private Limited (2011) New Delhi.	
2	Fundamentals of Momentum Heat and Mass Transfer, John	WeltyJ.R., Wilson R.E.&WicksC.E	Wiley	5 th Edition 2010
3	Unit Operations in Chemical Engineering	McCabe W.L., Smith J.C. &Harriot P	McGraw Hill	7 th edition 2022
4	Principles of Unit Operations	Foust A.S. et. al.	John Wiley	2 nd Edition 1980
5	Chemical Engineering, Vol. II	Coulson J.M. & Richardson J.F.	ELBS, Pergamon.	5 th Edition 2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in › courses/
2	https://onlinecourses.nptel.ac.in
3	https://archive.nptel.ac.in › courses
4	https://archive.nptel.ac.in › courses

SEMESTER S4

ENVIRONMENTAL ENGINEERING

Course Code	PBCHT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	4	ESE Marks	40
Credits	3:0:0:1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	4	Course Type	Theory

Course Objectives:

1. Develop the ability to identify and analyse wastewater sources, understand wastewater characteristics and assess the impact.
2. Impart comprehensive knowledge of the principles, technologies, and methodologies in wastewater treatment, solid waste management and air pollution control.
3. Instill the insight of sustainable development and optimal use of water, land and air resources.

SYLLABUS

Module No	Syllabus Description	Contact Hours
1	Introduction – environmental legislation and regulation-water quality standards, Sources and classification of wastewater- Types of water pollutants and their effects, Wastewater sampling-preservation and analysis. Physical and chemical characteristics of wastewater-determination of solids and organic matter, dissolved oxygen (Oxygen Sag Curve), biochemical oxygen demand, chemical oxygen demand. Pretreatment and primary treatment of wastewater (<i>brief overview- diagram, principle and working of equipment only, design not needed</i>) - Screening, grit removal, oil removal and equalization, neutralization, coagulation, flocculation and sedimentation, clarifiers and clariflocculation.	9
2	Secondary wastewater treatment methods (<i>brief overview- diagram, principle and working of equipment only, Modelling and design not needed</i>) – Aerobic biological processes - activated sludge process, trickling filters, oxidation ditch, aeration lagoon, rotating biological contactors, aerobic fluidized bed bioreactors. Anaerobic digestion process - anaerobic filter, anaerobic contact process, anaerobic fluidized bed bioreactors, up flow	9

	anaerobic sludge blanket (UASB). Tertiary treatment- Disinfections: Chlorination and Ozonation – Adsorption sand filters.	
3	Sludge treatment and disposal - sludge thickening, sludge conditioning, sludge dewatering, sludge digestion and composting (<i>brief overview</i>). Solid waste treatment -sources and classification, collection and disposal methods, open dumping, sanitary landfill, incineration, composting. Hazardous waste -types of hazardous waste, health effects, treatment and disposal methods. Noise pollution - effects of noise on people, Noise control methods.	9
4	Air pollution: Units of measurement, sources, classification. Sampling and analysis of air pollutants, Effects of air pollution-global warming and ozone depletion. Meteorological phenomena on air quality-lapse rates, pressure, wind, moisture and dispersion. Air pollution control methods and equipment (<i>brief overview- diagram, principle and working of equipment only, design not needed</i>) - Control devices for particulate contaminants: Gravitational settling, Centrifugal Collectors, Wet Collectors, Fabric filters and Electrostatic Precipitators (ESP). Control devices for gaseous contaminants: Adsorption, Absorption, Condensation, Combustion, Automobile Emission Control.	9

Suggestion on Project Topics

1. Collect sample from any polluted water source or effluent from nearby industry like food processing, metal polishing, pulp and paper etc., analyze the physical, chemical and biological characteristics and suggest a treatment process with required primary, secondary and tertiary equipment to treat the effluent according to the disposable standards specified by legislation. Use modelling and simulation software to design the process.
2. Collect air sample from flue gas, urban polluted area, analyze air quality index and design suitable air quality monitoring/control equipment.
3. Collect any solid waste sample (like e-waste, hazardous waste, MSW), analyze the characteristics and suggest/design treatment process/equipment.

OR

**Divide the project into 3 parts as follows Project I
(15 marks):** Do any one project from the given list

1. Identify socially relevant / environmental issue in your nearby locality, explore it and suggest remedial measures.
2. Identify a waste water source in nearby locality and study the physical, chemical characteristics.

3. Collect air sample from flue gas, urban polluted area, analyze air quality index.
4. Collect any solid waste sample (like e-waste, hazardous waste, MSW), analyze the characteristics.

Project II (10 marks): Do any one project from the given list

1. Design grit chamber and sedimentation tank for a specific wastewater pollution problem.
2. Design an activated sludge reactor for a specified wastewater pollution issue.
3. Design trickling filter and rotating biological contactor separately to address a specific wastewater pollution challenge.
4. Design any one anaerobic reactor for waste water treatment.
5. Design of an anaerobic digester for a given solid waste management issue.
6. Design any two air pollution controlling equipment.

Project III (5 marks):

1. Design of any single pollution controlling equipment for a specific problem using software like DWSIM, BIOWIN, SUMO, etc. or excel.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse characteristics of waste water. Interpret primary treatment methods used for wastewater treatment.	K4
CO2	Choose suitable secondary and tertiary methods used for wastewater treatment.	K3
CO3	Select suitable methods for treatment and disposal of sludge, solid waste and hazardous waste.	K3
CO4	Identify the sources of air pollutants and select suitable control methods.	K3
CO5	Select and Design equipment for pollution control.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1				1	3		3	3	1	3
CO2	3	1				1	3		3	3	1	3
CO3	3	1				1	3		3	3	1	3
CO4	3	1				1	3		3	3	1	3
CO5	3	3	3		3	3	3	2	3	3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wastewater Engineering: Treatment, Disposal and Reuse	Metcalf & Eddy	McGraw-Hill	4 th Edition 2004
2	Environmental Engineering	Peavy, H.S., Rowe, D.R. and Tchobanoglous, G	McGraw-Hill	7 th Edition
3	Environmental Pollution Control Engineering	Rao C.S.	New age International Pub.	2 nd Edition 2006
4	Air Pollution	Rao M N&Rao H V N	Tata McGraw-Hill.	1 st Edition 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pollution Control in Process Industries	S P Mahajan	Tata Mc-Graw-Hill	July 2017
2	Chemistry for Environmental Engineering & Science	Sawyer and McCarty,	McGraw-Hill	5 th Edition 2017
3	Environmental Engineering-A Design Approach	Sincero A.P. &Sincero G.A.,	Prentice Hall	1995
4	Waste Water Engineering	R. Parker <i>etal.</i>	CBS HB	2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/127/105/127105018/
2	https://nptel.ac.in/courses/103107084
3	https://nptel.ac.in/courses/103107084
4	https://nptel.ac.in/courses/103107084

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

1. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

2. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

3. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

4. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

ENERGY ENGINEERING

Course Code	PECHT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-:0-:0-:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To make the students aware of different forms of energy and methods for harnessing different energy resources.
2. To understand the importance of energy conservation and the different methods employed for energy conservation in industry

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Energy:- general classification of energy, world energy resources, energy consumption and future perspectives, Indian energy resources energy alternatives, Need of clean and renewable energy. Conventional energy plants:- Thermal, hydel and nuclear power plants, efficiency, merits and demerits of the above power plants, combined cycle power plants	7
2	Solar energy:- solar insolation and properties, solar thermal systems, flat plate collectors, focusing collectors, solar water heating, solar cooling, solar distillation, solar refrigeration, solar dryers, solar pond, solar thermal power generation, photo voltaic systems , energy plantations, solar energy application in India Ocean energy:- Ocean wave energy conversion, ocean thermal energy conversion, tidal energy conversion Geothermal energy:- Applications and mechanism for electricity production	9

3	Wind energy:- Types of windmills, types of wind rotors, wind electric power generation, wind power in India, nature of wind and site selection, economics of wind farm Biomass energy:- Resources, thermo-chemical and biochemical methods of biomass conversion:- combustion, gasification, pyrolysis, fermentation, anaerobic digestion Magneto hydro dynamics: open cycle and closed cycle systems.	9
4	Fuel cells:- alkaline fuel cell, phosphoric acid fuel cell, molten carbonate fuel cell, solid oxide fuel cell, solid polymer electrolyte fuel cell, Basic concepts of Microbial Fuel Cell Energy from Hydrogen :- Hydrogen as a source of energy, Sources, hydrogen energy storage, Applications, Future perspectives Energy Conservation: - Energy audit and Energy conservation in chemical process plants, Energy conservation in petroleum, fertilizer and steel industries. Modern Technologies in Energy Systems: - Cogeneration (CHP), pinch technology.	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different sources of energy , analyze the energy scenario and understand different conventional energy production systems	K3
CO2	Explain the concepts of solar , ocean and geothermal energy conversion	K2
CO3	Explain the Wind ,biomass and MHD energy conversion technologies	K2
CO4	Analyse the importance of hydrogen as source of energy, understand the working of fuel cells and energy conservation and energy audit	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				3						
CO2	3		3									
CO3	3		3									
CO4	3		3			3	3	3				2
CO5	3	2				3						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Technology (Non-Conventional, Renewable and Conventional)	Rao S. &ParulekarB.B	Khanna Publishers	3 rd Edn, 1994
2	Renewable Energy Sources & Conversion Technology	Bansal N.K., Kleeman M. &Meliss M	Tata McGraw Hill	1990
3	Energy for Sustainable Development	<u>MdHasanuzzaman</u> , NasrudinAbd Rahim (Eds)	Elsevier	2020
4	Non-Conventional Energy Sources	Rai G.D.,	Khanna Publishers	1988

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable Energy Resources,	Twidel, J. and Tony W	Taylor & Francis	2 nd edn, 2006
2	Energy Management and Conservation,	Kreith F., Goswami D.Y	CRC Press	2008
3	Solar Energy	Sukhatme S.P.,	McGraw-Hill Education (India),	3 rd edn. 2009
4	Non-Conventional Energy Systems	Mittal K.M.,	A H Wheeler Publishing Co Ltd	1999
5	Solar Energy Utilisation	Rai G D	Khanna Publishers	1995

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107157/
2	https://archive.nptel.ac.in/courses/103/107/103107157/
3	https://archive.nptel.ac.in/courses/103/107/103107157/
4	https://archive.nptel.ac.in/courses/103/107/103107157/

SEMESTER S4
MATERIAL SCIENCE

Course Code	PECHT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-:0-:0-:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/(Course code)	Course Type	Theory

Course Objectives:

1. To impart the basic concepts of material science.
2. To develop understanding about selection based on properties for various applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Structure of atom - present concept of atom-Rutherford's and Bhore's model-Bonding in solids-Types of solids-crystalline and amorphous solids-crystal systems-Bravais lattices-miller indices-coordination number-crystal defects-determination of crystal structure-X-ray diffraction-electron diffraction methods-properties of engineering materials-mechanical properties -isotropy and anisotropy-elasticity, plasticity, toughness, resilience, tensile strength, ductility, malleability, brittleness, hardness, fatigue, creep, wear resistance-Poisson's ratio-stress-strain relation-true stress and true strain-	9
2	Solid solutions - types of solid solutions-Hume Rothery rules-intermediate phases-mechanical mixtures-phase diagrams-eutectic systems- peritectic system, eutectoid and peritectoid systems iron- carbon diagram-T-T-T diagram-plastic deformation-re-crystallisation-hot and cold working of metals, Heat treatments-elementary study of various metals and alloys like cast iron, carbon steel, alloy steels.	9

3	Non-ferrous metals and alloys- aluminium and its alloys-copper and its alloys-Non ferrous metals and alloys used for high temperature services and nuclear application-organic polymers and its properties -ceramics-classification comparison of ceramic and non-ceramic structures-properties and application of ceramics- composite materials- classification- general characteristics. Introduction to nanomaterials.	9
4	Corrosion -different types, mechanism and factors influencing corrosion-corrosion prevention-inhibitors and their applications-oxidation-aging of rubber-oxidation of metals and radiation damage-factors affecting the selection of materials for engineering purposes, selection of suitable materials for construction in chemical industry.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain qualitatively the structure and bonding schemes of solids.	K2
CO2	Explain physical properties and possible applications of a given material.	K2
CO3	Summarize the stability of materials with all sorts of environments using phase diagrams.	K2
CO4	Identify and compare the properties of alloys, ceramics, polymers, composite materials and their engineering applications.	K3
CO5	Select suitable material for construction in chemical industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Material Science	Van Vlack,	Addison-Wesley	6 th edition, 1989
2	A Text Book of Material Science & Metallurgy	Khanna O.P.	DhanpatRai& Co	2 nd edition , 2014
3	Material Science & Processes	HajraChoudhary,	J K Publisher	1st edition, 1978
4	Chemical Engineers Handbook	Chilton &Perry,	McGraw Hill	9 th edition, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanocomposite science and technology	Linda S. Schadler, Pulickel M. Ajayan, Paul V. Braun	Wiley	Reprint, 2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113106032
2	https://nptel.ac.in/courses/113106032
3	https://nptel.ac.in/courses/113106032
4	https://nptel.ac.in/courses/113106032

SEMESTER S4

TOTAL QUALITY MANAGEMENT

Course Code	PECHT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-:0-:0--0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To achieve excellence in all areas of an organization by systematically eliminating waste, reducing defects and improving customer experiences and satisfaction.
2. To improve the quality of an organization outputs, including goods and services, through continual improvement of internal practices.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of Quality:- Dimensions of Quality, Quality Planning, Quality costs - Analysis Techniques for Quality Costs, Basic concepts of Total Quality Management, Historical Review, Principles of TQM, Leadership – Concepts, Role of Senior Management, Quality Council, Quality Statements, Strategic Planning, Deming Philosophy, Barriers to TQM Implementation.	9
2	TQM Principles - Customer satisfaction – Customer Perception of Quality, Customer Complaints, Service Quality, Customer Retention, Employee Involvement – Motivation, Empowerment, Teams, Recognition and Reward, Performance Appraisal, Benefits, Continuous Process Improvement –PDSA Cycle, 5S, Kaizen, Supplier Partnership – Partnering, sourcing, Supplier Selection, Supplier Rating.	9

3	Statistical quality control - The seven tools of quality, Statistical Fundamentals – Measures of central Tendency and Dispersion, Population and Sample, Normal Curve, Control Charts for variables and attributes, Process capability, Concept of six sigma, New seven Management tools.	9
4	TQM tools - Benchmarking – Reasons to Benchmark, Benchmarking Process, Quality Function Deployment (QFD) – House of Quality, QFD Process, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM) – Concept, Improvement Needs. Need for ISO 9000 and Other Quality Systems, ISO 9000:2000 Quality System – Elements, Implementation of Quality System, Documentation, Quality Auditing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall statistical fundamentals and describe the basic concepts, principles and tools of TQM.	K1
CO2	Explain the principles of TQM, the seven tools of SQC, the tools of TQM and Quality Systems.	K2
CO3	Utilize the hindrances for implementation of TQM principles, application of seven tools of SQC and the tools of TQM.	K3
CO4	Make use of the costs of quality and outline the processes of development of a TQM organisation.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Total Quality Management	Dale H.Besterfiled Inc.	Pearson Education. 2003	Indian reprint 2004
2	Total Quality Management	James R.Evans& William M.DaleH.Besterfiled	Pearson Education	
3	The Management and Control of Quality	Lidsay	South-Western (Thomson Learning).	5th Edition- 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Quality Management – Concepts and tasks	Narayana V. and Sreenivasan, N.S	New Age International.	1996
2	Total Quality Management for Engineers	Zeiri	Wood Head Publishers.	1991

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/@theleaderstalk
2	https://www.youtube.com/channel/UCaKzlWMr9DaP7hfxa_6tUBw/videos

SEMESTER S4
PROCESS INSTRUMENTATION

Course Code	PECHT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3--0--0--0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To study different instruments and techniques used in the chemical industry to measure various process variables and understand the theory behind them.
2. To understand the range of applicability and characteristics of these instruments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction -definition of instrumentation-concept of an instrument Basic principles of measurements - various elements in a measuring instrument - Sensing element, transducing element manipulating element and functioning element, etc- Principles of working with a suitable example, Classification methods of measurements, static and dynamic characteristics of measuring instrument. Simple problems with characteristics	9
2	Temperature measurements - temperature scales. Basic principles and working with a neat sketch of thermometers-mercury in glass thermometers, bimetallic thermometers resistance temperature detectors, thermistors, thermocouples, optical pyrometers, radiation pyrometers, Ranges of different types of temperature measuring instruments, sources of errors and precautions to be taken in temperature measurements.	9

3	Pressure measurement - basic principles and working with neat sketch of Manometers, Bourdon gauge, Bellow element, and diaphragm element, Pressure to electrical transducers- different types. Vacuum measurement - Different types. basic principles and working with neat sketch of McLeod gauge, Knudsen gauge, radioactive vacuum gauge, thermal conductivity gauges, ionization gauge High pressure measuring instruments. Level measurement - direct type and indirect type. Differential pressure method for pressurized vessels. Conductivity meters. Solid level detectors	9
4	Flow measurements - Liquid and gas flow measurements- list of different types of measuring liquids and gas flow, Basic principles, and working with a neat sketch of quantity meters, Coriolis meter, gas meters, magnetic flow meters, variable area meters, rotameters. turbine meters, weirs & notches, head meters, pitot tube, orifice meters & venturi meters. pH measurement using a calomel electrode. Composition analysis- list of methods with their advantages and limitations. (spectroscopic methods, chromatography methods, conductivity methods). Developments of P&I, diagram for flow systems, level, PH control temp control, Heat exchangers, Distillation column, CSTR, PFR, Single effect evaporator.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the functional elements of measurement systems, and categorization of various instruments based on performance characteristics	K2
CO2	Identify the characteristics of an instrument.	K3
CO3	Classify the instruments for different measurement with its range, advantages and limitations	K2
CO4	Illustrate the construction and working principle of various types of instruments to measure pressure, liquid level, temperature, flow, and pH.	K2
CO5	Apply P&ID for commonly used process instruments	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3							1		
CO2	3	3	3							1		
CO3	3									1		3
CO4	3									1		
CO5	3		3			1				1		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Instrumentation	Donald P Eckman	CBS Publishers and Distributors, New Delhi	2004
2	Mechanical and Industrial measurements. 11th edition	Jain R K	Khanna publishers	2009
3	Principles of Industrial Instrumentation. 3rd edition	Patranabis D	Tata- McGraw Hill.	2017
4	Fundamentals of Industrial Instrumentation	AlokBarua	IOP	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Measurement systems, Application and Design. 5th edition	Ernest O Doebelin, Dhanish. N. Manik,	McGraw Hill	2004
2	Instrumentation Devices and Systems	C.S. Rangan,G.R.Sarma and V.S.V. Mani	TataMcGraw Hill.	2017
3	Industrial Instrumentation: Principles and Design	Tattamangalam R. Padmanabhan	Springer	1999
4	Industrial Instrumentation & Control.3rd edition	S. K. Singh	Tata McGraw Hill	2009
5	Art of P&ID	Ford, Bacon and Davis	AICHE	2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103105130 https://nptel.ac.in/courses/108105064
2	https://nptel.ac.in/courses/103105130 https://nptel.ac.in/courses/108105064
3	https://nptel.ac.in/courses/103105130 https://nptel.ac.in/courses/108105064
4	https://nptel.ac.in/courses/103105130 https://nptel.ac.in/courses/108105064

SEMESTER S4

INTRODUCTION TO DATA ANALYSIS

Course Code	PECHT416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-:0-:0--0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To develop fundamental data analysis skills.
2. To interpret and communicate data insights effectively.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Bar charts, histograms, dot plot, pie charts, scatter plots, line charts, bubble plots, box and whisker plots, interval plots, stem and leaf plots. Descriptive statistics: Mean, Standard Error, Median, Mode, Standard Deviation, Sample Variance, Kurtosis, Skewness, Confidence Level, Quartile, Percentile. Development and interpretation of the above terms in any of the softwares (say, MS Excel, Matlab,R etc.)	9
2	Probability and statistical distributions: Probability, Properties and applications of important statistical distributions such as normal, log-normal and t-distributions, Chi-Square and F distributions- Development and interpretation in any of the softwares (say, MS Excel, Matlab, R etc.)	9
3	Hypothesis Testing: Formulation of null and alternate hypotheses, errors in hypothesis Tests, power of hypothesis tests, hypothesis tests on population means and variances Single factor experiments: Introduction to Analysis of Variance (ANOVA), blocking and randomization- Development and interpretation in any of the softwares (say, MS Excel, Matlab, R etc.)	9

4	Linear Regression Analysis: Matrix approach to linear regression, ANOVA in regression analysis, quantifying regression fits of experimental data. Factorial Design of Experiments: Need for planned experimentation Factorial design experiments: involving two factors, effect of interactions, ANOVA in factorial design, general factorial design, partial factorial designs- Development and interpretation in any of the softwares (say, MS Excel, Matlab, R etc.)	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze and interpret the graphical representation of data.	K4
CO2	Analyse the data using descriptive statistics.	K4
CO3	Distinguish the properties and applications of important statistical distributions.	K4
CO4	Analyse the data using inferential statistics.	K4
CO5	Develop and examine the linear regression analysis.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3	2						
CO2	3	3	3	3	3	2						
CO3	3	3	3	3	3							
CO4	3	3	3	3	3						2	
CO5	3	3	3	3	3						2	

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistics for Engineers and Scientists	William Navidi	McGraw Hill	3rd Edition, 2011
2	Design and Analysis of Experiments	Montgomery D. C	Wiley-India	8th edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Response Surface Methodology	Myers R. H., Montgomery D. C. and Anderson C. M.,	New Jersey: Wiley.	3rd edition, 2009.
2	Random Phenomena	Ogunnaike B. A	CRC Press	1st edition, 2010.
3	Introduction to Probability and Statistics for Engineers and Scientists.	Sheldon M. Ross	Academic Press	5 th Edition, 2014.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/110106072
2	https://nptel.ac.in/courses/110106072 , https://archive.nptel.ac.in/courses/103/106/103106120
3	https://archive.nptel.ac.in/courses/103/106/103106120
4	https://archive.nptel.ac.in/courses/103/106/103106120 , https://youtu.be/EMUh8Ksx3Q?si=XM6y_6KYngDwuw7i

SEMESTER S4
ELECTROCHEMICAL ENGINEERING

Course Code	PECHT417	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-:0-:0-:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The course is intended to cover key aspects of electrochemical engineering.
2. The course also imparts knowledge on electrochemical processes and its various applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Review basics of electrochemistry: Faraday's law, Nernst potential galvanic cells, polarography. The electrical double layer, its role in electrochemical processes, electro capillary curve, Helmholtz layer, Stern layer, fields at the interface. Mass transfer in electrochemical systems: Diffusion controlled electrochemical reaction, importance of convention and the concept of limiting current, over potential, primary, secondary current distribution, rotating disc electrode.	9
2	Introduction to corrosion: Electrochemical series, corrosion theories-derivation of potential, current relations of activities controlled and diffusion-controlled corrosion process. Potential, pH diagram, forms of corrosion, definition, factors and control methods of various forms of corrosion.	9

3	Classification and understanding the principles of electrochemical process: Electro deposition, electro refining, electroforming, electro polishing, anodizing, selective solar coatings, primary and secondary batteries, types of batteries, fuel cells.	9
4	Electrodes used in different electrochemical industries: Metals-Graphite – Lead dioxide – Titanium substrate insoluble electrodes – Iron oxide – semiconducting type etc. Metal finishing- cell design. Types of electrochemical reactors, batch cell, fluidized bed electrochemical reactor, filter press cell, Swiss roll cell, plug flow cell, design equation, figures of merits of different types of electrochemical reactors.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basics of electrochemistry and mass transfer phenomena in electrochemical systems.	K2
CO2	Describe the fundamentals and theories of corrosion.	K2
CO3	Classify and understand the principles of electrochemical process.	K2
CO4	Analyze the fundamentals of electrochemical reactors and its applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									
CO2	3	3	3									
CO3	3	3	3									
CO4	3	3	3									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrochemical Engineering	Picket	Prentice Hall	1977
2	Electrochemical systems	Newman, J.S.	Prentice Hall	1973

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrochemical Power Sources – Primary and Secondary Batteries	Barak, M. and Stevenge, U.K.		1980
2	Electrochemical Engineering	Mantell, C.,	McGraw Hill	1972

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/feed/playlists
2	https://www.youtube.com/playlist?list=PLFppcO11RhKjvtKQ-eYKgDG0lNzLAILdq

SEMESTER S4

FLUIDIZATION ENGINEERING

Course Code	PECHT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBCHT304	Course Type	Theory

Course Objectives:

1. Impart the basic concepts of fluidization phenomenon
2. Familiarize the fundamental and practical aspects of fluidization operations for industrial application.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The phenomenon of fluidization – types of fluidization, liquid like behaviour of a fluidized bed, advantages and disadvantages of fluidized bed for industrial operations, fluidization quality, selection of contacting mode for given application. Minimum fluidization velocity, pressure drop versus velocity diagram – uniformly sized particles, concept of minimum bubbling velocity, effect of temperature and pressure on fluidized bed behaviour, sintering and agglomeration of particles at high temperature. Geldart classification of particles, mapping of fluidization regimes. Microproject/assignment – calculation of particle size distribution, minimum fluidization velocity and pressure drop	11
2	Distributors: types of distributors – ideal distributors, perforated plates, Tuyeres and caps, pipe grids and spargers, gas entry region of a bed, gas jets in fluidized beds, pressure drop requirements across distributors. Power consumption. Entrainment and elutriation – concept of FBH, TDH. Solid movement – vertical and horizontal movement, segregation of solid particles, large solids in beds of smaller particles, staging of fluidized beds. Microproject/assignment/self-study- Design of distributor	11

3	Bubbles in dense beds: single rising bubbles, rise rate of bubbles, Davidson model for gas flow at bubbles, wake region, solids within bubbles (overview only), coalescence and splitting of bubbles, bubble formation above distributor. Slug flow Bubbling fluidized bed: estimation of bed properties – gas flow in the emulsion phase, bubble gas flow, bubble size and bubble growth, bubble rise velocity. K-L flow model for bubbling fluidized bed. Gas dispersion and gas interchange in bubbling beds (concept only) Microproject/ assignment/self-study - Simulation/modeling of bubbling fluidized bed.	11
4	Particle- to- gas mass and heat transfer: Mass transfer – single spheres and fixed bed and gas fluidized bed. Interpretation of mass transfer coefficients. Heat transfer – single spheres and fixed bed and gas fluidized bed. Interpretation of heat transfer coefficients. Industrial application of fluidized beds: Physical operations: Heat exchange, coating of metal objects with plastic, drying of solids, adsorption. Synthesis reaction: Fischer–Tropsch synthesis, polymerization of olefins. Cracking: Fluid catalytic cracking (FCC), Fluid coking and Flexi coking, Thermal cracking Fluidized combustion of coal, gasification of coal and coke, calcination. Microproject/assignment/self-study – fitting reported mass transfer and heat transfer data with the bubbling bed model	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (4 points)

- a. Clearly defines the real-world problem involving application of fluidization.
- b. Examine and identifies relevant factors influencing the process.

2. Problem Analysis (5 points)

- a. Present a well-reasoned solution approach.
- b. Compare and justify the proposed solutions with evidence and logical reasoning.

3. Evaluate (5 points)

- a. Thoroughly evaluate the proposed solutions, considering feasibility and practical implications.

4. Implementation of software (2 points)

- a. Demonstrates proficiency in relevant software.

5. Conclusion (K4- 2 points, K5 – 2 points)

- a. Summarizes findings and insights. **(K4)**
- b. Reflects critical thinking and informed decision-making. **(K5)**

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the fluidized bed behaviour and mapping of fluidization regime.	K2
CO2	Examine the pressure drop and other parameters for fluidized beds.	K4
CO3	Summarize flow models for bubbling fluidized beds.	K2
CO4	Solve the heat and mass transfer coefficients and apply in bubbling fluidized beds.	K3
CO5	Recommend fluidized bed system for different applications.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3										3
CO3	3	3	3									3
CO4	3	3	3									3
CO5	3	3	3	3	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Fluidization Engineering”	Kunii, D. and Levenspiel, O.,	John Wiley & Sons, Inc.	Second, 1991
2	“Fluidization”	Leva, M.	McGraw-Hill	First, 1959
3	“Fluidized Particle”	Davidson, J.F. and Harrison, D	Cambridge University Press	
4	Fluidized Bed Technology: Principles and Applications.	Howard, J.R.	CRC press	1st ed., (1989)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Engineering Hand Book,	J.H. Perry	McGraw hill	9th Edition. 2018
2	Unit Operations of chemical Engineering,	Warren McCabe, Julian Smith and Peter Harriott	McGraw Hill publishers	Fifth edition 1993

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103103132
2	https://nptel.ac.in/courses/103103132
3	https://nptel.ac.in/courses/103103132
4	https://nptel.ac.in/courses/103103132

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		<i>Answer all questions. Each question carries 3 marks</i>	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
<i>Answer any one full question from each module. Each question carries 8 marks</i>				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4
ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	6

2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world	6

	case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4
PARTICLE TECHNOLOGY LAB

Course Code	PCCHL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ Course Code	Course Type	Lab

Course Objectives:

1. To analyse and apply knowledge of size analysis and size reduction methods by performing experiments.
2. To impart knowledge of solid-solid and solid-fluid separation equipment's and provide knowledge of their working and constructional features.

A minimum of eight experiments are to be conducted

Expt. No.	Experiments
1	Sieve analysis -Determination of particle size distribution, mean diameters,specific surface area and number of particles per unit mass
2	Determination of the effectiveness of the screen
3	Pipette analysis-Determination of particle size distribution, specific surface area and mean diameters
4	Beaker decantation- Determination of particle size distribution, specific surface area and mean diameters
5	Free settling – Verification of Stokes's law
6	Sedimentation – Determination of area of a thickener
7	Verification of the laws of crushing
8	Ball mill - Determination of the critical speed

9	Leaf filter- Determination of specific cake resistance and compressibility factor
10	Cyclone separator – Determination of collection efficiency
11	Studies on, Mineral jig, Wilfley table and Continuous thickener
12	Studies on Plate & frame filter press, Rotary drum filter
13	Studies on Jaw crusher and Hammer mill
14	Any other experiment on particle size analysis/size reduction/mechanical separation/filtration/mixing of solids as per the syllabus of PCCHT302 Particle Technology

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the basic principles of Particle technology to find solutions of problems by conducting experiments in the laboratory.	K3
CO2	Plan experiments and analyze/interpret data collected from experimental investigation in the laboratory	K3
CO3	Apply modern computing tools necessary for analysis of the experimental data in the laboratory.	K3
CO4	Make use of ethical principles in engineering profession by practicing ethical approaches in experimental investigation, collection and reporting of data and adhering to the safety ethics set by the laboratory	K3
CO5	Organize work in diverse groups and perform laboratory experiments	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3		3								
CO3					3							
CO4				2		3		3				
CO5				2					3			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit Operations of chemical Engineering,	Warren McCabe, Julian Smith and Peter Harriott	McGraw publishers Hill	Fifth edition 1993
2	Coulson and Richardson's Chemical Engineering volume 2, Particle Technology and Separation Process,	Richardson, J.F.; Harker, J.H.; Backhurst, J.R	Elsevier publishers.	Fifth edition 2002
3	Introduction To Chemical Engineering,	T. Banchero and W.L. Badger	McGraw Hill	Indian edition 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit Operations of Particulate Solids: Theory and Practice	E. Ortega-Rivas	CRC Press, FL	2012
2	Perry's Chemical Engineers' Handbook	Editor-in-Chief, Don W. Green, Associate Editor Marylee Z. Southard	Mc-Graw-Hill Education	9 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/103/103103155/ , https://nptel.ac.in/courses/103107123
2	https://archive.nptel.ac.in/courses/103/103/103103155/ , https://nptel.ac.in/courses/103107123

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4
HEAT TRANSFER OPERATIONS LAB

Course Code	PCCHL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ Course Code	Course Type	Lab

Course Objectives:

1. To understand the principles of various modes of heat transfer
2. To familiarize the operations and working of various heat transfer equipment.

Minimum of nine experiments are to be conducted

Expt. No.	Experiments
1	Heat Transfer by Natural Convection
2	Heat Transfer by Forced Convection
3	Thermal conductivity of Metal Rod
4	Heat transfer through Composite Wall
5	Heat transfer in Double pipe heat exchanger
6	Heat Transfer in Shell and Tube Heat Exchanger
7	Emissivity Measurement Apparatus
8	Heat Transfer in Fins
9	Unsteady State Heat Transfer
10	Determination of Critical Radius of Insulation
11	Combined Convection and Radiation Heat Transfer
12	Radiation Heat Transfer
13	Heat Transfer in Agitated Vessels
14	Drop and Filmwise Condensation
15	Any other experiment related to different modes of heat transfer with and without change of phase.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine the rate of heat transfer in various modes of heat transfer.	K4
CO2	Evaluation of the heat transfer coefficients.	K5
CO3	Analyse the working of heat transfer equipments.	K4
CO4	Make use of the experimental data meaningfully.	K3
CO5	Influence teamwork skills.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2							3	1		
CO2	3	2							3	1		
CO3	3	3	2						3	1		2
CO4	3			3					3	1		
CO5									3	1		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat Transfer: Principles and Applications	Binay K Dutta	Prentice Hall India	2 nd Edition 2022
2	Fundamentals of Engineering Heat and Mass Transfer	R C Sachdeva	New age International Publishers	6 th Edition 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat Transfer: Principles and Applications	Datta B.K	Prentice Hall India	
2	Unit Operations in Chemical Engineering	McCabe W.L., Smith J.C. & Harriot P	McGraw Hill	
3	Heat Transfer	Hollman J.P	McGraw Hill	
4	Fundamentals of Engineering Heat and Mass Transfer	R C Sachdeva	New age International Publishers	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://www.digimat.in/nptel/courses/video/103101137/L01.html to http://www.digimat.in/nptel/courses/video/103101137/L13.html

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

CHEMICAL ENGINEERING

SEMESTER S5

MASS TRANSFER OPERATIONS II

Course Code	PCCHT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCHT403	Course Type	Theory

Course Objectives:

1. To impart fundamental concepts of mass transfer operations such as distillation, liquid extraction, leaching and membrane separation processes.
2. To explain how to quantify, formulate, and solve engineering problems involving different mass transfer operations like distillation, liquid-liquid extraction, leaching and membrane separation processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Distillation: Vapour-Liquid Equilibria, boiling- point diagram and equilibrium curves, relative volatility, application of Raoult's law. Distillation methods- flash distillation, Simple distillation, Steam distillation, Fractionation of binary mixtures, Principle of fractionation, Fractionation in plate columns, condensers and reboilers. Material and energy balance. Problem solving in distillation using EXCEL / Matlab / SCILAB/ PYTHON etc. – only for self study / microproject / assignment	11
2	Design of fractionation column: McCabe - Thiele method - basic assumptions, feed quality and feed line, number of plates, feed plate location, total reflux, minimum reflux , optimum reflux, plate efficiency, cold reflux and open steam. Ponchon–Savarit method: Enthalpy composition diagrams, Concept of net flow and difference point, Ponchon–Savarit method– material and enthalpy balance, number of plates, feed plate location, minimum reflux conditions.	13

	Rectification in packed columns, height of packed towers. Azeotropic and Extractive distillation (qualitative treatment only). Problem solving in distillation using EXCEL / Matlab / SCILAB/ PYTHON etc. – only for self study / microproject / assignment	
3	Extraction –applications, ternary equilibria on triangular coordinate system, mixture rule, distribution curve, selectivity, choice of solvent. Single-stage and multistage extraction operations - Calculations for immiscible systems and partially miscible systems. Construction and working of mixer -settler cascades, sieve-tray columns, agitated towers, pulse columns and centrifugal extractors.	10
4	Leaching – Rate of leaching, factors affecting rate of leaching. Working principles of leaching equipment- Shank's system- thickeners, classifiers and moving bed leaching equipment. Leaching equilibrium - constant underflow - variable underflow Single stage and multistage leaching. Membrane separation processes – classification – types of membranes: flat, spiral wound, hollow fibre -concentration polarization – ultrafiltration.	10

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the principles boiling point diagrams, relative volatility and select various types of distillation techniques	K3
CO2	Design a fractionation column using McCabe – Thiele method and apply it for various reflux conditions	K3
CO3	Apply Ponchon - Savarit method to determine the number of stages required for a given separation in a fractionator for different reflux conditions and to understand rectification in packed columns.	K3
CO4	Apply the theory of extraction and leaching and the design of single stage and multi-stage extraction and leaching processes with an understanding of construction and working of equipments.	K3
CO5	Select suitable types of membrane separation processes.	K3
CO6	Select and run software programs for solving basic problems in mass transfer operations	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3									3
CO3	3	3	3									3
CO4	3	3	3									3
CO5	3											3
CO6	3	3	3		3				3			3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mass Transfer-Theory and Applications	K.V. Narayanan and B. Lakshmikutty	CBS Publishers	2 nd Edition 2023
2	Mass Transfer Operations,	Treybal R.E.,	McGraw Hill	3 rd Edition 2017
3	Principles of Mass Transfer & Separation Processes	Binay K Dutta	PHI Learning Private Limited.	1 st edition 2006
4	Separation Process Principles	Seader, J.D. and Henley, E.J.,	Wiley, New York.	4 th Edition 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mass Transfer-Theory and Practice	N.Anantharaman and K.M.MeeraSheriffa Begum	PHI Learning Private Limited (2011) New Delhi.	
2	Fundamentals of Momentum, Heat and Mass Transfer, John	Welty J.R., Wilson R.E. & Wicks C.E	Wiley	5 th Edition 2010
3	Unit Operations in Chemical Engineering	McCabe W.L., Smith J.C. & Harriot P	McGraw Hill	7 th edition 2022
4	Principles of Unit Operations	Foust A.S. et. al.	John Wiley	2 nd Edition 1980
5	Chemical Engineering, Vol. II	Coulson J.M. & Richardson J.F.	ELBS, Pergamon.	5 th Edition 2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in > courses//nnel.iitm.ac.in11htt.iitm.ac.inhttp://nptel.m.ac.in/m.ac.in
2	https://onlinecourses.nptel.ac.in
3	https://archive.nptel.ac.in > courses
4	https://archive.nptel.ac.in > courses

SEMESTER 5

CHEMICAL TECHNOLOGY

Course Code	PCCHT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Familiarize students with chemical industry operations. It helps the student to develop new chemical products and processes, test processing equipment and instrumentation, gather data, and monitor quality.
2. Reveals knowledge about operations of industries such as raw materials and products, unit operations and processes, engineering problems related to operations and marketing strategies currently followed in chemical industries, etc.
3. Enable the students to do research, design, and construct safe, sustainable machinery to produce a wide range of valuable items.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to chemical industries:- Classification of chemical Industries Manufacture of sulphuric acid:- DCDA process, phosphoric acid - wet processes and electric furnace process. Chlor-alkali industries:- soda ash, caustic soda, chlorine.	9
2	Fertilizer industries:- Ammonia, urea, single super phosphate and triple super phosphate. Natural product industries – Soap and detergents, glycerine, pulp and paper, sugar, oiland fats, starch, ethyl alcohol by fermentation.	9

3	Petroleum refining:- Chemical composition of crude, petroleum refinery products, atmospheric distillation unit, octane number. Refinery conversion processes - pyrolysis, catalytic cracking process-fluidized bed and moving bed type, reforming.	9
4	Petrochemicals:- Manufacture of acetylene, butadiene and phenol-cumene process. Polymerization industries:- polyethylene, polypropylene, PVC, polyamides-nylon6 and nylon 6,6, and polyester synthetic fibers.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain manufacture of inorganic chemicals and chlor-alkali products with a suitable PFD.	K2
CO2	Explain the production of fertilizers and natural products .	K2
CO3	Summarise the various process engineering technologies in petroleum refinery operations.	K2
CO4	Explain the reactions and process flow diagrams for the production of polymers, petrochemicals	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Outlines of Chemical Technology, Edited and revised by M. GopalaRao and Marshall Sitting.	C.E. Dryden	Affiliated East-West Press Pvt. Ltd. New Delhi.	3rd Edition, 1997.
2	Chemical Process Industries	G.T. Austin, R.N. Shreve	McGraw Hill	5th Ed., 1984.
3	A Text Book of Chemical Technology- Vol. I and II	Shukla S. D. and G. N. Pandey	Vikas Publishing House	1986.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Engineering Education Development Centre.	Chemtech Vol. I – IV	Indian Institute of Technology, Madras.	1979
2	Encyclopaedia of Chemical Technology.	Kirk-Othmer	John Wiley and Sons	
3	Encyclopaedia of Industrial Chemistry	Ullmann	John Wiley and Sons	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103103218
2	https://nptel.ac.in/courses/103103218 https://nptel.ac.in/courses/103107081
3	https://nptel.ac.in/courses/103107081
4	https://nptel.ac.in/courses/103106109

SEMESTER 5

CHEMICAL REACTION ENGINEERING

Course Code	PCCHT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCHT303	Course Type	Theory

Course Objectives:

1. Enable students to apply the principles of chemical kinetics to describe the rates of chemical reactions for isothermal and non-isothermal reactions.
2. Equip the students with fundamental knowledge for the design of different types of ideal reactors, including batch, continuous-stirred tank reactors (CSTR), plug-flow reactors (PFR for single and multiple reactions).
3. Impart the concepts such as residence time distribution (RTD) and models required to analyze and design real reactors.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of chemical reaction engineering: Classification of chemical reactions, Variables affecting the rate of reaction, Definition of reaction rate. Kinetics of homogeneous reactions: Concentration-dependent and Temperature-dependent terms of rate equation, Temperature dependency from Arrhenius law, Collision theory, and transition state theory (no derivation). Interpretation of batch reactor data: Evaluation of rate equation by integral and differential methods of analysis for constant volume and variable volume systems. Problem solving in chemical kinetics using	11

	EXCEL/Matlab/SCILAB/PYTHON etc. – only for self study/microproject/assignment.	
2	Introduction to reactor design: Classification of reactors. Ideal reactors for a single reaction: Ideal batch reactor, Steady state mixed flow reactor, Steady state plug flow reactor. Design for single reactions: Size comparison of single reactors, Multiple reactor systems, Plug flow reactor in series and parallel, equal-sized mixed reactors in series, mixed flow reactors of different sizes in series, determination of the best system for a given conversion. Recycle reactor, Auto catalytic reactions. Design for multiple reactions: Reactions in parallel, contacting patterns for reactions in parallel, qualitative discussion about product distribution of series reactions. Problem solving in ideal reactors and recycle reactor using EXCEL/Matlab/SCILAB/PYTHON etc. – only for self study/microproject/assignment.	11
3	Non isothermal reactor design: Temperature and pressure effects, single reactions, Heat of reaction from thermodynamics, heat of reaction and temperature, equilibrium constants from thermodynamics. General graphical design procedure, optimum temperature progression. Heat effects: adiabatic operations and non-adiabatic operations, Multiple Steady States in CSTR. Problem solving in non-isothermal reactors using EXCEL/Matlab/SCILAB/PYTHON etc. – only for self study/microproject/assignment.	11
4	Non ideal reactors: Basics of nonideal flow, Residence time distribution studies - C, E & F curves and their relationships. Conversion in nonideal reactors, RTD in ideal reactors: Plug flow reactors, single CSTR. Laminar flow reactor. Models for nonideal flow: dispersion model and tanks in series model. Problem solving in RTD, conversion in real reactors and models for nonideal flow in reactors using EXCEL/Matlab/SCILAB/PYTHON etc. – only for self study/microproject/assignment.	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the principles of chemical kinetics and determine the kinetics/conversion using batch reactor.	K3
CO2	Apply isothermal chemical reactors under ideal conditions for single and multiple reactions.	K3
CO3	Select and design chemical reactors for non-isothermal operations.	K3
CO4	Apply non-ideal nature of reactors and design chemical reactors for real conditions.	K3
CO5	Select and run software programs for solving problems in ideal and nonideal reactor design.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									
CO2	3	2	3									
CO3	3	2	3									
CO4	3	2	3									
CO5	3	2	3	3	3			2	2			3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Reaction Engineering	Levenspiel Octave	John Wiley & Sons	Third Edition. 1999
2	Elements of Chemical Reaction Engineering	H Scott Fogler	Prentice Hall of India	Fifth Edition, 2016
3	Chemical Engineering Kinetics	Smith J.M	McGraw Hill	Second Edition, 1970

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Chemical Reaction Engineering and Kinetics	Ronald W. Missen, Charles A. Mims, Bradley A. Saville	John Wiley & Sons	First Edition, 1999
2	An Introduction to Chemical Engineering Kinetics & Reactor Design	Charles G. Hill	John Wiley & Sons	First Edition, 1977
3	Chemical Reactor Theory- An Introduction	K.G Denbigh & J.C.R Turner	Cambridge University Press.	Third Edition, 1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/103/103103153/
2	https://archive.nptel.ac.in/courses/103/103/103103153/
3	https://archive.nptel.ac.in/courses/103/103/103103153/
4	https://archive.nptel.ac.in/courses/103/103/103103153/

SEMESTER 5

CHEMICAL ENGINEERING DESIGN-I

Course Code	PBCHT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCHT205, PBCHT304, PCCHT402	Course Type	Theory

Course Objectives:

1. To give undergraduates a foundation in designing equipment used in process industries for heat transfer and evaporation operations.
2. To familiarise using codes and standards for designing process equipment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Thermal design of double pipe heat exchanger: Constructional features - Calculation of heat transfer area – estimation of heat transfer coefficient – calculation of the size of double pipe – pressure drop estimation in double pipe flow. Thermal design of shell and tube heat exchanger: Constructional features – baffles and pass partition – TEMA nomenclature – routing of fluids - Calculation of heat transfer area – estimation of tube bundle size - estimation of overall heat transfer coefficient using Bell's method – pressure drop in shell side and tube side. Thermal design of shell and tube condensers for single component isothermal condensation of saturated vapours in the shell. Estimation of heat transfer area – Estimation of bundle size and shell diameter - estimation of in-shell condensation coefficients using Dukler charts/Nusselt equation – estimation of overall heat transfer coefficient.	18

2	Thermal design of evaporators: Estimation of heat transfer area in single effect and multiple effect(double/triple effect forward feed only) evaporators with negligible boiling point rise Effect of boiling point elevation in evaporator calculation. Constructional features and design of short tube calandria evaporator, long tube evaporator, and forced circulation evaporator. Mechanical design of process equipment: Design of thin-walled pressure vessels as per IS 2825 –Shells subject to internal and external pressure- flat ends, ends concave to pressure, and convex to pressure: domed ends and conical ends.	18
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Suggestions on Project Topics

- Identify a tubular exchanger in a plant and design the exchanger. The design shall include the following:
 - a. Selection of exchanger type with justification
 - b. Selection of material of construction for various parts, its specification with approximate cost data.
 - c. Obtain/Prepare physical properties for the fluids in the operating range required for design.
 - d. Manual sizing- First Design.
 - e. Use computational tools to optimize the first design.
 - f. Prepare design data sheet as per standards.

- Identify a process that uses different types of evaporators. Choose a suitable evaporator and design it. The design shall include the following:
 - g. Selection of evaporator type with justification
 - h. Selection of material of construction for various parts, its specification with approximate cost data.
 - i. Obtain/Prepare physical properties for the fluids in the operating range required for design.
 - j. Manual sizing- First Design.
 - k. Use computational tools to optimize the first design.
 - l. Prepare design data sheet as per standards.

- Conduct the mechanical design of tubular exchangers and evaporators as per Indian Standards/ ASME and prepare drawings.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

	Total
Two numerical questions will be given from each module, out of which one should be answered. The questions in each module shall be from different topics. Each question carries 20 marks. (2x20 = 40 marks)	40

The questions should be clear regarding the type of equipment and its alignment if any, operating conditions, and materials handled. Data required for design such as equilibrium data and physical properties in case it cannot be obtained from handbook/data book shall be provided with the question.

Question paper should contain instruction as given below:

- *Assume any missing data suitably.*
- *Apart from scientific calculators the following books and data books are permitted for the exam (Photostat copies of the books are not permitted except the item no.3 and item no.4):*
 1. *Steam tables*
 2. *Perry's Handbook*
 3. *Attested copies or original of IS Codes (2825, 1730, 4049)*
 4. *Attested copies of Dühring's charts, Nomographs, charts and tables used in design taken from TEMA standard/other editions of Handbook. The copies*

should be attested by the course faculty concerned or as directed by university.

- *No other textbooks or materials such as handouts, printed notes etc. are permitted for the exam.*

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Select and design tubular/pipe exchangers for a given heat exchange operation between single-component fluids	K3
CO2	Select and design tubular condensers for condensation of single-component fluids	K3
CO3	Select and design evaporators for a given feed solution and terminal conditions	K3
CO4	Solve mechanical design of pressure vessels as per codes and standards.	K3
CO5	Decide real field design of heat transfer process equipment by using modern tools.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3									3
CO2	3	2	3									3
CO3	3	2	3									3
CO4	3	2	3									3
CO5	3	2	3	3	3				3	3	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Heat Transfer	Kern D. Q.	Tata McGraw Hill	
2	IS Codes: 2825, 1730, 4049		BIS	
3	Chemical Engineers Handbook	Perry. R.H & Green.D.W.	Mc- Graw Hill.	
4	Introduction To Chemical Equipment Design Mechanical Aspects	B C Bhattacharyya	CBS	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Equipment Design	M.V Joshi & Mahajan V.V.	Mac-Milan & Co. India.	3rd Edn,
2	Introduction to Chemical Engineering	Badger &Bancharo	McGraw Hill	1 st Edn.
3	Chemical Engineering, Vol.6	Coulson J.M.& Richardson J.F.	Butterworth Heinemann, (Indian print)	3rd Edn
4	Heat Transfer: Principles and Applications	Dutta B.K.	Prentice Hall India	2 nd Edn.
5	Unit Operations in Chemical Engineering	McCabe W.L., Smith J.C. & Harriott P.	McGraw Hill.	3rd Edn
6	Standards of Tubular Exchanger Manufacturers Association	Shravan H. Gawande, Sunil D. Wankhede	TEMA	8th Edn
7	Applied Process Design for Chemical & Petrochemical Plants	E. Ludwig	Gulf Publication, London	Vol I, II, III
8	Process Heat Transfer: Principles and Applications	R. W. Serth	Academic press	1stEdn.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103107207
2	https://nptel.ac.in/courses/103107207 https://nptel.ac.in/courses/103107143

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution to Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and teamwork	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation, and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S5

BIOCHEMICAL ENGINEERING

Course Code	PECHT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental aspects of Biochemical engineering
2. To develop the kinetics of enzyme catalyzed reactions
3. To acquire comprehensive knowledge of metabolic pathways, bioreactors and biosensors

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Micro Biology, Cell theory, Structure of cells, Cell fractionation, protest kingdom and their distinguishing characteristics. Chemicals of life: repetitive and non-repetitive biopolymers - lipids, sugars and polysaccharides, amino acids and proteins. Protein structure nucleotides RNA and DNA.	6
2	Kinetics of Enzyme catalyzed reactions: simple enzyme kinetics with one or two substrates, Michaelis - Menten Kinetics, Evaluation of parameters in Michaelis – Menten equation, Substrate activation and inhibition, Modulation and regulation of enzyme activity – competitive and uncompetitive inhibition. Enzyme specificity and enzyme specificity hypotheses. Enzymes of industrial importance. Immobilized enzyme technology: enzyme immobilization, medical and analytical applications of immobilized enzymes. Batch cultivation - growth cycle (lag, exponential,	10

	stationery and death phase)	
3	Metabolic pathways and energetics of the cell: Metabolic reaction coupling: ATP, ADP and NAD. Embden-Meyerhof pathway (EMP), Pentose phosphate cycle, Respiration - TCA cycle. Transport across cell membranes - passive transport, active transport and facilitated diffusion. Transport phenomena in Bio process system-Gas-liquid mass transfer in cellular system - basic mass transfer concepts - rates of metabolic oxygen utilization	10
4	Biochemical reactors-specifications Continuous stirred tank bioreactor, bubble column bioreactor, Airlift bioreactors, Fluidized bed bioreactors, Packed bed bioreactors, and photo bioreactors. Types of reactors for sterilization. Fermentation technology, medium formulation, design and operation of a typical aseptic, aerobic fermentation process. Concept of biosensors. Physical and chemical sensors, gas analysis sensors, On-line sensors and offline sensors.	10

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain different types of cells, classification of Kingdom Protista and potential of chemicals of life	K2
CO2	Apply kinetics of enzyme catalysed reactions and explain Immobilized enzyme technology	K3
CO3	Apply basic knowledge of energy and material balances for understanding the metabolic pathways within the cell	K3
CO4	Apply heat and mass transfer principles in design of fermentation and, bioprocess.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3									3		3
CO2	3	3										3
CO3	3					2				2		3
CO4	3					3						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biochemical Engineering Fundamentals	James E Bailey, David F Ollis	McGraw-Hill	2 nd Edition 1986
2	Bioprocess Engineering Basic Concepts	Michael L Schuler, FrikretKhargii	Prentice Hall of India	2 nd Edition 2015
3	Introduction to Biochemical Engineering	D G Rao	Tata Mc Graw Hill	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bioprocess Engineering Principles	Pauline M Doran	Academic Press Inc: CBS Publishers and Distributors Pvt Ltd	2 nd Edition 2013
2	Perry's Chemical Engineers Handbook	Don W. Green, Marylee Z. Southard	McGraw-Hill	9 th Edition 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105054/
2	https://archive.nptel.ac.in/courses/103/105/103105054/
3	https://archive.nptel.ac.in/courses/103/105/103105054/
4	https://archive.nptel.ac.in/courses/102/105/102105064/

SEMESTER S5

OIL AND NATURAL GAS ENGINEERING

Course Code	PECHT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To bring an optimal balance between natural gas production, natural gas processing and gas transportation, an extensive treatise on natural gas engineering, both upstream and gas refining processes with key equipment and facility design.
2. Awareness of the current status of production of natural gas through unconventional sources/technics and the applications of natural gas.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of petroleum geology and exploration:- Introduction, origin, formation, geological occurrence and characteristics of oil and natural gas. Source Rocks, Reservoir Rocks, and Caprocks: Definition, Characteristics, Classification and nomenclature, Concept of Shale oil, Reservoir. Properties, Hydrocarbon migration, Petroleum Exploration.	9
2	Conventional natural gas:- Associated gas, non-associated gas, Gas condensate. Unconventional production of natural gas: Coal bed methane, Natural gas hydrate, shale gas and tight gas sands. Coal bed methane - Introduction, present status, formation and properties of coal bed methane. Natural Gas Hydrate - Concepts and Structures, Evaluation and Prediction, Production Techniques.	9

	Shale Gas - Exploration, Production - Drilling and completion. Separation and Processing: Gas and liquid separation – internal construction of separators, types of separators, factors affecting separation, separator design, stage separation, low temperature separation.	
3	Dehydration of Natural gas: Water content of natural gas streams, hydrate control in gas production, dehydration systems - Glycol dehydration and Solid desiccant dehydration. Acid gas removal: Metal oxide process, Slurry process, Amine process, Carbonate washing process, Methanol based process and Sulphur recovery process. Compression of Natural Gas: Types of Compressors, Selection of reciprocating and centrifugal compressors, Thermodynamics of Compressors, Compression calculations.	9
4	Gas flow measurement: Fundamentals, Methods of measurements, Orifice meters equation, turbine meters, Selection, Recording charts, Uncertainties in flow. Gas Gathering, Transportation and Storage: Gas Gathering System, Steady State and Unsteady State Flow in Pipelines, Transmission of Natural Gas, Specifications. Underground Storage and Conservation of Natural Gas. Liquefied Natural Gas: Gas treating before liquefaction- Liquefaction cycles- Storage of LNG, Transportation- Regasification and cold utilization of LNG. Economics - Plant efficiency - Safety and environmental considerations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different source rocks from which hydrocarbons are generated.	K3
CO2	Explain fluid hydrocarbons migrate from a source rock to reservoir rock, entrapment and accumulation of hydrocarbons.	K2
CO3	Identify sources of natural gas and its applications in variety of fields.	K3
CO4	Explain various methods of natural gas processing for dehydration and sweetening.	K2
CO5	Summarize gas compression, gas gathering and transport installation.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3										
CO3	3	3										
CO4	3	3										
CO5	3	3										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Petroleum Geology	Richard, C. Selley	Elsevier	4 th Edition 2022
2	Fundamental of Natural Gas Processing	Arthur J. Kidnay, William R. Parrish	Taylor and Francis	2 nd Edition 2011
3	Gas Production Operations	Beggs D. H.	OGCI Publications	1985

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Natural Gas: A Basic Handbook	James G. Speight	Gulf Publishing Company,	2 nd Edition 2018
2	Gas Conditioning and Processing	John M. Campbell	Campbell Petroleum Series	Volume 2, 7 th Edition , 1992
3	Natural Gas Production Engineering	Ikoku, Chi	John Wiley and Sons,	1 st Edition 1984
4	Gas Production Engineering	Kumar Sanjay	Gulf Publishing Company, TX, USA,	1987

SEMESTER S5

POLYMER TECHNOLOGY

Course Code	PECHT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To impart the basic concepts of polymer technology,
2. To develop understanding about the different polymers as an engineering material.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to polymers-monomer, functionality, classification of polymer based on source, structure, application, thermal behaviour, mode of polymerization. Kinetics of polymerisation– addition polymerization and condensation polymerisation – free radical polymerization – anionic and cationic polymerization. Copolymerisation. Different types of copolymers – Characteristic features. Methods of polymerization – bulk, solution, suspension and emulsion polymerization.	8
2	Molecular weight of polymers – weight average and number average molecular weight – sedimentation and viscosity average molecular weights. Experimental methods for molecular weight determination – end group analysis, light scattering method – viscometry (Ostwald viscometer) intrinsic viscosity. Molecular weight distribution curve. Factors affecting polymer properties – crystallinity – orientation treatment – solubility of polymers – glass transition temperature – types of polymer degradation – effect of reinforcement on the properties.	10

	Properties of polymers – rheology- viscous flow – apparent viscosity – rubber like elasticity – stress strain behaviour of elastomers – viscoelasticity – stress relaxation and creep – measurement of rheological properties – melt flow index (MFI) – capillary rheometers. Estimation of tensile strength of polymers.	
3	Manufacture, properties and applications of polymers: Thermoplastics– ABS, acrylics, cellulose acetate, fluoropolymers (PTFE), nylons, polycarbonate, PVC, PE, PP, PS, polyurethanes. Thermosetting plastics– epoxy, phenol formaldehyde, urea formaldehyde, polyesters, silicones. Bioplastics – Poly lactic acid (PLA). Elastomers- Butyl rubber –Nitrile rubber.	9
4	Additives for polymer processing - effect of additives used – plasticizers – colourants – heat stabilizers - antioxidants – ultraviolet absorbers – antistatic agents – flame retardants – blowing agents – lubricants and fillers. Plastic processing – injection moulding, compression moulding, calendaring, blow moulding, extrusion, thermoforming, wet, dry and melt spinning methods for fibres, vulcanization of rubber. General study of elastomer processing - brief description of compounding methods.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify types and mechanisms of polymerization	K2
CO2	Summarize the classes, properties and engineering uses/applications of different polymeric materials	K2
CO3	Explain the concepts behind different types of molecular weights and methods for their determination.	K2
CO4	Analyse the rheology and mechanical properties of polymers	K4
CO5	Explain the processing methods and moulding techniques	K2
CO6	Summarize the elastomer processing methods and vulcanization of rubber.	K2

.Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3	3										
CO4	3	3										
CO5	3					3						2
CO6	3					3						2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text book of polymer science.	Billmeyer F.W.	John Wiley	1994
2	Polymer Science	Vasant R. Gowariker, N. V. Viswanathan	New Age.	2005
3	Polymer Science. And Technology	Joel R Fried	Prentice Hall	Third Edition 2014
4	Polymer Science. A Textbook for Engineers and Technologists.	Sebastião V. Canevarolo, Jr.	Elsevier	2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Science And Technology	Premamoy Ghosh	Tata Mc Graw Hill.	2011
2	Principles Of Polymer Systems	Rodrigues F.	Tata Mc Graw Hill	2011
3	Handbook Of Plastic Testing Technology.	Shah V.H.	Wiley	1998
4	Polymers And Their Synthesis	Ilhama A. Zarbaliyeva, Ziyafaddin Hamid Asadov	ELM	2022

5	Fundamental Polymer Science	Mikael S. Hedenqvist and Ulf Gedde	Springer	2nd Edn 2019
6	Fundamentals Of Polymer Science And Technology (Part 1)	Ajay Kumar Manna	Books & Allied (P) Ltd.	2024
7	Fundamentals Of Polymers : Raw Materials To Finish Products	Niranjan Karak	PHI	2009
8	Principles of Polymer Processing	Zehev Tadmor, Costas G. Gogos	Wiley	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/104/105/104105124/ https://archive.nptel.ac.in/courses/103/106/105106205/ https://archive.nptel.ac.in/courses/113/105/113105028/
2	https://archive.nptel.ac.in/courses/104/105/104105124/ https://archive.nptel.ac.in/courses/103/106/105106205/ https://archive.nptel.ac.in/courses/113/105/113105028/
3	https://archive.nptel.ac.in/courses/104/105/104105124/ https://archive.nptel.ac.in/courses/103/106/105106205/ https://archive.nptel.ac.in/courses/113/105/113105028/
4	https://archive.nptel.ac.in/courses/104/105/104105124/ https://archive.nptel.ac.in/courses/103/106/105106205/ https://archive.nptel.ac.in/courses/113/105/113105028/

SEMESTER S5
PROJECT ENGINEERING

Course Code	PECHT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. The objective is to impart, a basic knowledge in project planning, plant process design, economic evaluation and financial control.
2. To describe the importance of process development and process engineering to students.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Classification of Projects, Scope of Project engineering - the role of project engineer. Development of project: R&D, TEFR, Importance of Laboratory development of new process – Bench scale experiments – pilot plant studies – Semi-commercial plant. Process engineering: Conceptual process design- Preliminary process design- Detailed process design (FEED). Flow diagrams – Plot plans - Engineering design and drafting (Detailed Engineering). Plant location and site selection, Preliminary data for construction of projects.	10
2	Planning and scheduling of projects: use of bar chart, PERT/ CPM - Critical path calculations. Site development: Foundation – Erection and site fabrication – Construction – Insulation – Start up and commissioning – Trial runs – Guarantee run and hand over	8

3	. Company formation- Process license – Technology Transfer – Statutory sanctions. Contracts and Contractors: Introduction, detailing of scope of work, factors in selecting scope of work, Type of contracts- Factors in selecting type of contract, contractor selection	8
4	Selection of process equipments: Standard versus special equipment selection criteria, specification sheets. Scope of piping engineering, pipe sizing technique, Codes and standards, Piping design, Safety in plant design. Project financing: Means of finance, Financial institutions, special schemes, Financial institutions in India, key financial indicators and ratios of new projects	10

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Engineering of Process Plants.	Rase & Barrow	John Wiley	
2	Project Engineering of Process Plants	Bhasin, S.D	Chemical Engineering Education Development Centre, I.I.T., Madras	1979
3	Project Engineering	Frederick B. Plummer	BH	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	PERT and CPM	Srinath L. S	Affiliated east press Pvt. Ltd	
2	Chemical engineering handbook	Perry J. H	Mc Graw Hill	
3	Applied project engineering and Management	Ernest E. Ludwig	Gulf Pub. Co	1988
4	Plant design and economics for chemical engineers.	Peter S. Max & Timmer Haus.	Mc Graw Hill	2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/110/104/110104073/
2	https://digimat.in/nptel/courses/video/103103039/L38.html
3	https://nitttrc.edu.in/nptel/courses/video/103103039/L46
4	https://tjsec.digimat.in/nptel/courses/video/103103039/L25.html

SEMESTER S5

NUMERICAL METHODS FOR PROCESS ENGINEERS

Course Code	PECHT526	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. The course is designed to give an overview of numerical methods to students.
2. The course aims at giving adequate exposure to methods for numerically solving algebraic and transcendental equations, system of linear and non-linear equations, polynomial interpolation, differential and partial differential equations.
3. To help students to implement these solutions using Excel/ MATLAB or any relevant programming environments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Motivation and Applications, Accuracy and precision, Errors, Sources of errors, Significant digits and Numerical instability - Numerical solution of polynomial and transcendental equations - Bracketing methods: Bisection method –Regula-Falsi method -Open methods: Newton-Raphson method - Fixed-point iteration - Solution of polynomial and transcendental equations using Excel/ MATLAB (Assignments/ Micro project only)	8
2	Solutions of system of linear algebraic equations: Direct methods – Gauss Elimination and Gauss - Jordan methods - Crout's reduction method - error analysis - iterative methods - Jacobi's iteration - Gauss-seidel iteration - the relaxation method - convergence analysis - solution	8

	of system of nonlinear equations by Newton-Raphson method - Solution of linear/ nonlinear algebraic equations using Excel/ MATLAB(Assignments/ Micro project only)	
3	Polynomial interpolation: finite difference operators: Gregory – Newton forward and backward interpolations -Stirling’s interpolation formula- Lagrange’s interpolation polynomial - divided differences Newton’s divided difference interpolation polynomial - error of interpolation – Interpolation using Excel/ MATLAB(Assignments/ Micro project only) Numerical differentiation: differential formulas in the case of equally spaced points - numerical integration - trapezoidal and Simpson’s rules - Gaussian integration - errors of integration formulas- Differentiation and Integration using Excel/ MATLAB(Assignments/ Micro project only)	9
4	Numerical solution of ordinary differential equations: The Taylor series method - Euler and modified Euler methods - Runge–Kutta methods (4th order only) - Multistep methods - Milne’s predictor - corrector formulas - Adam-Bashforth & Adam-MoultonFormulas- Solution of boundary value problems in ordinary differential equations- solution of ordinary differential equations using Excel/ MATLAB(Assignments/ Micro project only) Solution of partial difference equations: Classification of PDE’s- finite difference methods for solving two-dimensional Laplace’s equation for a rectangular region - finite difference method of solving heat equation and wave equation with given initial and boundary conditions- solution of partial differential equations using Excel/ MATLAB(Assignments/ Micro project only)	11

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine the errors, source of error and its effect on any numerical computations and solve polynomial and transcendental equations using an appropriate numerical method	K4
CO2	Solve linear and non -linear algebraic system of equations using an appropriate numerical method and analyse errors	K4
CO3	Apply different numerical methods for interpolation, differentiation and integration of functions and analyse errors.	K4
CO4	Apply ordinary differential equations by different numerical methods and solve elliptic, parabolic, and hyperbolic partial differential equations using finite difference methods.	K3
CO5	Utilize the numerical methods for computer-aided solutions	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3										3
CO3	3	3										3
CO4	3	3										3
CO5	3	3			3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mathematical Methods in Chemical & Environmental Engineering	Ajay K. Ray	Thomson- Learning	
2	Numerical Methods for Engineers	Steven C. Chapra and Raymond P. Canal	McGraw-Hill	
3	Introduction to Numerical Methods in Chemical Engineering	Pradeep Ahuja	P.H.I	2 nd Edition, 2019
4	Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB	Victor J. Law	CRC Press, Inc.	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Numerical Analysis	Hildebrand F.B.	T.M.H.	
2	Applied Numerical Methods for Digital Computation	James M.L., Smith C.M. & Woford J.C.	Harper & Row	
3	Numerical-Analysis-Using-MATLAB-and-Excel	Steven-T.-Karris	Orchard Publications	3rd Edition, 2007
4	Applied Numerical Analysis	Gerald C.F	Addison Wesley	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/106/103106074/# https://archive.nptel.ac.in/courses/127/106/127106019/#
2	https://archive.nptel.ac.in/courses/103/106/103106074/# https://archive.nptel.ac.in/courses/127/106/127106019/#
3	https://archive.nptel.ac.in/courses/103/106/103106074/# https://archive.nptel.ac.in/courses/127/106/127106019/#
4	https://archive.nptel.ac.in/courses/103/106/103106074/# https://archive.nptel.ac.in/courses/127/106/127106019/#

SEMESTER S5

CORROSION ENGINEERING

Course Code	PECHT527	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enlighten the engineering students about the scope and cost of corrosion.
2. To make them understand the various types of corrosion under normal to severe corrosive environments in manufacturing industrial sectors.
3. To develop the skills among the students about corrosion testing of materials and to familiarize the students in selection of materials, corrosion monitoring and prevention.
4. To develop the specialized skills in corrosion control in different engineering materials and industries.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition and principles of corrosion: Definition and importance of corrosion, Corrosion engineering – scope in industries, Cost of corrosion, corrosion damage, classification of corrosion, Principles of corrosion phenomenon: Corrosion rate expressions, Electrochemical aspects – electrochemical reactions, polarization, passivity, Environmental effects– effect of oxygen and oxidisers, velocity, concentration, temperature, galvanic coupling, Metallurgical and other aspects-metallic properties, economic considerations, importance of corrosion inspection.	9
2	Forms of corrosion: Different forms of corrosion: Galvanic or two metal corrosion, Crevice corrosion, Pitting, Intergranular corrosion, Selective leaching, Erosion corrosion, Stress corrosion, Hydrogen damage.	9

3	Corrosion testing and monitoring and prevention: Corrosion testing and monitoring: Non-electrochemical and electrochemical methods: potentiostat, Tafel extrapolation, linear polarization, galvanostat, impedance spectroscopy, thermogravimetric technique, salt spray test, weight change measurements, Corrosion prevention: Materials selection, alteration of environment, design, cathodic protection and anodic protection, coatings.	9
4	Corrosion and its control in different engineering materials and industries: Corrosion and its control in different engineering materials: concrete structures, duplex, super duplex stainless steels, ceramics, composites and polymers. Corrosion and its control in industries: Power, Process, Petrochemical, ship building, marine and fertilizer industries. Corrosion auditing in industries, Corrosion map of India.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the scope and principles of corrosion	K2
CO2	Summarize the various types of corrosion under normal to severe corrosive environments	K2
CO3	Explain the various corrosion testing and prevention techniques	K2
CO4	Select the appropriate corrosion control techniques for different engineering materials and that are used in different industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				2	1					
CO2	3	3				2	2					
CO3	3	3				2	1					
CO4	3	3				2	1					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Corrosion Engineering	Fontana M. G	Tata McGraw Hill	3rd Edition, 2005
2	Principles and Prevention of Corrosion	Jones D. A	Prentice-Hall	2nd Edition, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of corrosion Engineering	Pierre R. Roberge	McGraw Hill Professional	1999
2	The Fundamentals of Corrosion	Scully J. C	Pergamon Press	2nd Ed

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113108051
2	https://www.youtube.com/watch?v=xmB2gtvvhTw

SEMESTER S5

MATHEMATICAL METHODS IN PROCESS ENGINEERING

Course Code	PECHT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	Basic courses in Heat Transfer, Mass Transfer and Reaction Engineering	Course Type	Theory

Course Objectives:

1. Enable the students to model chemical engineering problems.
2. Equip the students with knowledge on modelling and simulation of processes and use various concepts such as linear algebra, differential calculus etc. to solve chemical engineering problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mathematical formulation of the physical problems:- application of the law of conservation of mass, salt accumulation in stirred tank, starting of equilibrium still, solvent extraction in two stages, diffusion with chemical reaction, application of the law of conservation of energy, radial heat transfer through cylindrical conductors, heating a closed kettle, flow of heat from a fin. Problems in mathematical modelling of reactor/extractor/fins/diffuser using MATLAB/ SIMULINK/ DWSIM- only for self study/project/assignment	9
2	Analytical (explicit) solution of ordinary differential equations encountered in chemical engineering problems:- first order differential equations, method of separation of variables, equations solved by integration factors, certain examples involving mass and energy balances and reaction kinetics, second order differential equations, non-linear	9

	equations, linear equations, simultaneous diffusions and chemical reaction in a tubular reactor, continuous hydrolysis of tallow in a spray column. Formulation of partial differential equations, unsteady state heat conduction in one dimension, mass transfer with axial symmetry, continuity equations, boundary conditions, function specified, derivative specified and mixed conditions, iterative solution of algebraic equations - Jacobi s method, Gauss- Seidel method and successive order relaxation (S.O.R) method. Solution of ODE and PDE and develop performance equation of tubular reactor, unsteady state heat conduction of given system using MATLAB/ SIMULINK/ DWSIM- only for self study/project/assignment	
3	The difference operator:- properties of the difference operator, difference tables and other difference operators, linear finite difference equations, the complimentary solution and particular solution, simultaneous linear differential equations, non-linear finite difference equations, analytical solution. Solution of the following type of problems by finite difference method - calculation of the number of plates required for absorption column, calculation of the number of theoretical plates required for distillation column, number of steps required for a counter-current extraction and leaching operations. Finding number of plates/trays in absorption column/ distillation column using modern software tools- only for self study/ assignments/ projects	9
4	Application of statistical methods:- propagation of errors of experimental data, parameter estimation of algebraic equations encountered in heat and mass transfer, kinetics and thermodynamics by the method of averages, linear least squares and weighted linear least square methods, design of experiments: factorial, fractional factorial methods. <i>Doing simple experiments in laboratory and perform statistical analysis by tools like EXCEL/MINITAB/R/PYTHON – only for self study/ assignments/ projects.</i>	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (4 points)

1. Clearly defines the real-world problem in process engineering.
2. Examine and identifies relevant contextual factors.

2. Problem Analysis (4 points)

1. Break-down and presents a well-reasoned solution approach.
2. Compare and justify the proposed solutions with evidence and logical reasoning.

3. Evaluate (4 points)

1. Thoroughly evaluate the proposed solutions.
2. Considers feasibility and practical implications.

4. Implementation of software (4 points)

1. Select the most feasible solution by implementing the proposed solutions.
2. Demonstrates proficiency in relevant software.

5. Conclusion (K4- 2 points, K5 – 2 points)

1. Summarizes findings and insights. **(K4)**
2. Reflects critical thinking and informed decision-making. **(K5)**

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop mathematical model of simple chemical systems from Mass and Energy balance.	K3
CO2	Solve chemical engineering problems using analytical/numerical techniques.	K3
CO3	Solve problems in Mass transfer operations using finite difference method.	K3
CO4	Apply statistical techniques for analyzing Chemical Engineering problems.	K4
CO5	Recommend and Simulate Chemical Engineering problem / experiments using modern software tools	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									
CO2		3	3									
CO3	3	3	3	3	2							
CO4	3	3	3	2	3							
CO5	3	3	3	3	3				3	2		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mathematical Methods in Chemical Engineering	S. Pushpavanam	PHI	
2	Mathematical Methods in Chemical	Jenson, V.J. and Jeffereys, G.V.	Engineering Academic Press, London and New York	1977.
3	Applied Mathematics in Chemical	Mickley, H.S., Thomas. K. Sherwood and Road, C.E.,	Engineering Tata McGraw-Hill Publications,	2 nd Edition 2007
4	Mathematical methods in Chemical Engineering	Aravind Varma, M, Mordbidelli	Oxford University Press, Indian Edition	1997

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Irvin Kreyszig	New Age International (Pvt) Ltd., New Delhi	10 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105106/
2	https://archive.nptel.ac.in/courses/103/105/103105215/
3	https://onlinecourses.nptel.ac.in/noc22_ch47/preview

SEMESTER S5

CHEMICAL REACTION ENGINEERING LAB

Course Code	PCCHL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCHT503	Course Type	Lab

Course Objectives: Chemical Reaction Engineering Lab familiarize students with main type of chemical reactors. Students are able to analyze the experimental data to obtain the reaction rate expression (reaction order and specific reaction rate constant).

1. Operate lab equipments like Batch, CSTR and PFR reactors
2. Analyze the concentration versus time data and determine the specific rate constant and the order of the reaction.
3. Compare theoretical and experimental conversions in a CSTR and PFR.
4. Estimate RTD and model parameters in a CSTR, PFR and packed bed reactor

Expt. No.	Experiments
1	Isothermal Batch Reactor
2	Isothermal Plug Flow Reactor
3	Isothermal CSTR
4	Packed Bed Reactor
5	RTD Studies in a Packed Bed Reactor
6	RTD Studies in CSTR
7	RTD Studies in PFR
8	Combined Flow Reactor

9	UV Photo Reactor
10	Determination of activation energy
11	Kinetics of reactions
12	Determine kinetics of homogeneous chemical reactions using simulation softwares like MATLAB or SCILAB
13	Simulation of ideal reactors using MATLAB.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop kinetic equation of homogenous chemical reactions and analyse the factors effecting the reactions	K3
CO2	Analyse the effect of temperature in chemical equation and validation of Arrhenius law.	K3
CO3	Examine the kinetics of reaction in ideal reactors- Batch Reactor, PFR and MFR	K4
CO4	Discover non ideality in chemical reactors by calculating residence time distribution	K4
CO5	Inspect and present the experimental data meaningfully and develop teamwork skills.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3						3	3		3
CO2	3	3	3						3	3		3
CO3	3	3	3						3	3		3
CO4	3	3	3						3	3		3
CO5	3	3	3						3	3		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Reaction Engineering	Octave Levenspiel	John Wiley & Sons	3 rd Edition
2	Elements of Chemical Reaction Engineering	H, Scott Fogler	Prentice Hall	5 th Edition
3	Chemical Engineering Kinetics	Smith J.M	McGraw Hill	2 nd Edition
4	Chemical Reactor Theory- An Introduction	K.G Denbigh& J.C.R Turner	Cambridge University Press	3 rd Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Modelling and Computer Simulation	Amiya K. Jana	PHI Learning Pvt. Ltd.	Second edition, 2011
2	Computational techniques for process simulations and analysis using MATLAB	Niket S. Kaisare	CRC Press	2018

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

***Final Marks Averaging:** The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.*

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

MASS TRANSFER OPERATION LAB

Course Code	PCCHL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ Course Code	Course Type	Lab

Course Objectives:

1. Familiarizes the student with various mass transfer operations by doing experiments.
2. Operate lab experiments and understand the working of mass transfer equipments.

Expt. No.	Experiments
1	Diffusion coefficient measurement
2	Simple Distillation
3	Steam Distillation
4	Liquid extraction- Determination of Ternary liquid-liquid equilibria
5	Simple leaching-varying solvent to feed ratio
6	Simple leaching- varying number of stages
7	Cross current leaching
8	Counter current leaching
9	Adsorption Isotherm
10	Atmospheric batch drying
11	Wetted wall column-measurement of mass transfer coefficient
12	Any other experiments related to mass transfer operations applicable in Chemical Engineering.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the fundamental knowledge of mass transfer in related practical problems	K3
CO2	Analyse different mass transfer operations	K4
CO3	Examine separation processes such as simple distillation, steam distillation etc. to estimate the composition in products	K4
CO4	Plan and conduct the experiments and present the experimental data meaningfully	K3
CO5	Develop teamwork skills	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3	3										
CO3	3											
CO4	3	3		3						3		
CO5									3			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mass Transfer Operations	Robert E Treybal	McGraw Hill	3 rd Edition 2017
2	Mass Transfer: Theory and Applications	K V Narayanan and B Lakshmikutty	CBS Publishers	2 nd Edition 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Unit Operations	Foust A.S. et. al.	John Wiley	2 nd Edition 1980
2	Chemical Engineering, Vol. II	Coulson J.M. & Richardson J.F.	ELBS, Pergamon.	5 th Edition 2002

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

CHEMICAL ENGINEERING

SEMESTER S6

CHEMICAL ENGINEERING DESIGN II

Course Code	PCCHT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCHT205, PCCHT302, PCCHT303, PCCHT403, PCCHT501	Course Type	Theory

Course Objectives:

1. To give a foundation for the undergraduates in the design of equipment used in process industries for distillation, gas absorption, liquid extraction and drying
2. To familiarise mechanical design of tall equipment by applying basic principles of mechanics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Process design of steady state isothermal binary component tray distillation columns (Sieve and valve tray): Estimation of theoretical number of stages using McCabe Thiele method –Tray design – tray efficiency - pressure drop, entrainment, downflow flooding and weeping. Process design of steady state isothermal packed bed absorption and stripping column for dilute systems without chemical reaction: Number of transfer units- height of transfer units – column diameter – packing height-liquid distribution-pressure drop. Process design of single solvent sieve tray extraction columns for immiscible system: Number of trays, sieve tray design, height of coalesced layer, Murphree efficiency.	18
2	Design of Direct heat Rotary Dryers: Estimation of air quantity, drying	

	temperature, Number of Transfer Units, Diameter of dryer, Length and hold up time. Mechanical design of tall pressure vessels: design of thin walled multi-course pressure vessels as per IS 2825 – Estimation of thickness of shells subject to external loads (wind and seismic) – Design of skirt support for tall columns.	18
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)	Total
2 numerical questions will be given from each module, out of which 1 question should be answered. The questions in each module shall be from different topics. Each question carries 30 marks. (2x30 = 60 marks)	60
The questions should be clear in respect of type of equipment, operating conditions, materials handled and objective. Data required for design such as equilibrium data and physical properties in case it cannot be obtained from handbook/data book shall be provided with the question. Question paper should contain instruction as given below: • <i>Assume any missing data suitably.</i> • <i>Apart from scientific calculators the following books and data books are permitted for the exam (Photostat copies of the books are not permitted except the item no.2):</i> <i>1. Steam tables</i>	

2. Perry's Handbook (Original or attested bound copy of Chapters 1,2,3,4,5,6,10,11,12,13,14 &15) 3. Attested copies or original of IS Codes (2825, 1730, 4049) 4. Attested copies of Psychrometric charts, Nomographs, charts and tables used in design taken from other editions of Handbook/References. 5. Attested copy of Sieve tray hydraulics and sieve tray mass transfer for liquid extraction from 'Mass Transfer Operations' by Robert E. Treybal, pg 532-538. (The copies should be attested by the course faculty concerned or as directed by university.) No other textbooks or materials such as handouts, printed notes etc. are permitted for the exam.	
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Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Select and design binary tray distillation column	K3
CO2	Select and design packed bed absorption column	K3
CO3	Select and design sieve tray extraction column	K3
CO4	Select and design direct heat rotary dryer	K3
CO5	Select and design thin-walled unfired tall pressure vessels using Indian Standard codes	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3			2		2				3
CO2	3	2	3			2		2				3
CO3	3	2	3			2		2				3
CO4	3	2	3			2		2				3
CO5	3	2	3			2		2				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	IS Codes: 2825, 1730, 4049		BIS	
2	Chemical Engineers Handbook	Perry RH &GreenDW	Mc- Graw Hill.	7 th Edn
3	Mass Transfer Operations	Robert E. Treybal	McGraw Hill Education	3 rd Edn, 2017
4	Chemical Engineering, Vol.6	Coulson JM& Richardson JF	Butterworth Heinemann, (Indian print)	3 rd Edn

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Equipment Design	MV Joshi & Mahajan VV	Mac-Milan & Co. India.	3 rd Edn,
2	Introduction to Chemical Engineering	Badger &Bancharo	McGraw Hill	
3	Introduction to Chemical Equipment Design	B.C Bhattacharya	CBS Publishers & Distributors, New Delhi.	2008
4	Unit Operations in Chemical Engineering	McCabe WL, Smith JC & Harriott P	McGraw Hill.	7 th Edn

SEMESTER S6

TRANSPORT PHENOMENA

Course Code	PCCHT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBCHT304, PCCHT402, PCCHT403	Course Type	Theory

Course Objectives:

1. To impart knowledge in transport of momentum, heat and mass transport and provide insight into the dependence of temperature and pressure on the transport coefficients.
2. To derive simple shell balances to formulate basic conservation equations of transport processes for solving selected engineering problems which can be solved analytically.
3. To give basic axioms of conservation namely conservation of momentum, energy and mass.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Prediction of transport coefficients: viscosity, thermal conductivity and diffusivity. Momentum Transport – Viscosity and the mechanisms of momentum transfer, Generalization of Newton's law of viscosity, pressure and temperature dependence of viscosity of gases and liquids, prediction of viscosity of gases: Rigid sphere model and rigorous models, Numerical problems. Energy Transport: Thermal conductivity and the mechanism of energy transport- prediction of thermal conductivity of gases, effect of temperature and pressure on thermal conductivity of gases, relationship between thermal conductivity and viscosity of gases. Thermal conductivity of solids, the relationship between thermal and electrical conductivity of solids, Numerical problems.	11

	Diffusivity and the Mechanism of Mass Transport: Definition of concentrations, velocities and mass/molar fluxes, interrelationship between fluxes. Fick's law of diffusion, kinetic theory of diffusion in gases at low density, theory of ordinary diffusion in liquids. Prediction of diffusivity of gases, the effect of temperature, pressure and composition on diffusivity, Numerical problems.	
2	Shell momentum balance: boundary conditions - application of shell balance to simple flow systems - falling film -flow of a Newtonian fluid in between two slits formed by two flat plates, flow through tube - flow through annulus - flow of two adjacent immiscible fluids. General transport equations for momentum: derivation of continuity equation and equation of motion in rectangular coordinates - Navier Stoke's equation and Euler equation – transport equations in curvilinear coordinates (no derivation) - application of transport equations to steady flow problems - flow through tube - tangential annular flow - rotating liquid.	11
3	Shell energy balance: boundary conditions - application of shell balance to heat conduction problems - conduction with electric, nuclear and viscous heat sources, use of shell heat balances in variable thermal conductivity systems to derive temperature and heat flux profiles, cooling fins with insulated tip condition. Equations of energy in rectangular and curvilinear coordinates (no derivation) - application to steady state heat transfer problems - tangential flow in annulus with viscous heat generation – flow of nonisothermal film - transpiration cooling.	11
4	Analogies between heat, mass and momentum transfer: Shell mass balance - boundary conditions - diffusion through stagnant gas - diffusion with heterogeneous and homogeneous chemical reaction-diffusion and chemical reaction in porous catalyst – other similar diffusion problems, equation of continuity for binary mixtures in rectangular coordinates.	11

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the mechanisms of momentum, heat and mass transfer.	K3
CO2	Identify the transport coefficients of gases from basic physical variables.	K3
CO3	Analyze industrial problems involving isothermal steady-state momentum transfer in simple geometries using shell momentum balance, equations of change and boundary conditions.	K4
CO4	Examine analytical solutions of selected simple engineering steady-state problems of heat transfer using shell energy balance and equations of change.	K4
CO5	Analyze simple steady-state diffusion problems using shell mass balance.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										2
CO2	3	3										2
CO3	3	3										3
CO4	3	3										3
CO5	3	3										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Transport phenomena	Bird R.B., Stewart W.C and Lightfoot F.N,	John Wiley & Sons..	Second edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Transport Phenomena for Engineers	Theodore L,	International text book Company, U.S.A	
2	Fundamentals of Heat, Momentum and Mass Transfer	Welty, Wicks and Wilson	John Wiley	5 th Edition 2007
3	Momentum, Energy and Mass transfer in continua	John C Slattery	McGraw Hill, Co.	
4	Transport Phenomena a Unified approach	Robert S. Brodkey and Harry C Hersing	McGraw Hill Book Co.	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/106/112106237/
2	https://www.slideshare.net/slideshow/nptel-tp-notes-fullpdf/251701715

SEMESTER S6

PROCESS MODELLING AND SIMULATION

Course Code	PECHT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To impart the basic concepts of modelling of chemical processes
2. To impart the basic concepts of simulation of chemical processes

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Definition of Modelling, Simulation, Advantages and limitations, Applications. Models and classification of models, Alternate classification of models, Mathematical modelling based on Transport phenomena, Basic concepts, Basic modelling principles, Energy equations, continuity equation, equation of motion, transport equations, equations of state, Equilibrium states and chemical kinetics.	8
2	Mathematical models of Lumped Systems: Continuous flow tanks, Mixing vessel without reaction, Isothermal reactor with irreversible and reversible reaction. Non-isothermal steam jacketed reactor, Boiling-open and enclosed vessel, Batch distillation, Ideal binary distillation column - Simulation using MATLAB/SciLab or any other process simulator (Assignments/ Microprojects only)	10
3	Mathematical models of Distributed systems: Jacketed tubular reactor, Counter current liquid-liquid double-pipe heat exchanger, Unsteady Heat conduction through a rod, Tubular permeation process, Gas absorber-Simulation using MATLAB/SciLab or any other process simulator (Assignments/ Microprojects only)	10
4	Simulation of systems: Algorithm for solving Continuous flow tank, Isothermal CSTR in series, Non-isothermal CSTR, Binary distillation column.	8

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the concept of modelling and simulation of chemical engineering systems	K3
CO2	Explain the important physical phenomena and develop model equations for the given system.	K2
CO3	Build model developing ability for various reactors and separation systems.	K3
CO4	Identify the model developing ability for distributed systems.	K3
CO5	Develop the skill to simulate chemical engineering systems and processes.	K3
CO6	Analyze and simulate various unit operations & unit processes using any simulation software	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2								
CO2	3	3	3	2								
CO3	3	3	3	2								
CO4	3	3	3	2	2						3	
CO5	3	3	3	2							3	3
CO6	3	3	3	2								

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Modelling, Simulation and Control for Chemical Engineers,	Luyben W.L	McGraw Hill International Edition	Second, 1989
2	Computational Methods for Process Simulation	W Fred Ramirez	Butterworth-Heinemann Ltd	2 nd Edition 1997
3	Computer Process Modelling and Computer Simulation	Amiya K.Jana	Prentice Hall of India	Third, 2018
4	Process Plant Simulation	B.V. Babu	Oxford University Press	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Dynamics - Modelling Analysis and Simulation	Biquette W.B.	Prentice Hall of India	First, 1998
2	Mathematical Modelling in Chemical Engineering	Franks R.G.E.	John Wiley	First, 1967
3	Process Modelling and Simulation in Chemical, Biochemical and Environmental Engineering	Ashok Kumar Verma	CRC press	2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107096/
2	https://archive.nptel.ac.in/courses/103/107/103107096/
3	https://archive.nptel.ac.in/courses/103/107/103107096/
4	https://archive.nptel.ac.in/courses/103/107/103107096/

SEMESTER S6

PETROLEUM REFINERY ENGINEERING

Course Code	PECHT 632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. This course will present an overview of the modern, integrated petroleum refinery, its feed stocks, products and the processes employed to convert crude oil and intermediate streams into finished products
2. This course provides major insights into both primary and secondary processes like Atmospheric Distillation, Vacuum Distillation, Cracking, Hydrocracking, Catalytic Reforming Processes for LOBS in a typical refinery.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Origin and composition- Origin and formation of crude oil. Oil exploration and drilling. Storage and transportation of crude oil and products. Classification, Composition and Evaluation of oil stock. Status of Petroleum industry in India. Classification of petroleum refinery. Fundamentals of Refinery economics, Refinery complexity index	9
2	Preliminary and Primary petroleum processing- Impurities in crude oil, Dehydration and desalting of crude, Electric Desalter-, Process description, factors affecting the electric desalter. Pipe still furnaces and its operations. Distillation of crude- Prefractionator, Atmospheric topping unit, Vacuum distillation unit	8

3	Secondary conversion process -Thermal Conversion process. Thermal cracking, Visbreaking, Coking (Delayed coking). Types and uses of petroleum coke. Catalytic conversion process-Catalytic cracking-Types of Catalyst. Process description of Fluid Catalytic cracking unit. Process variables. Latest developments in the catalytic cracking process. Process description and applications of Hydro cracking.	9
4	Production of gasoline and treatment of Petroleum products- Catalytic reforming, Alkylation (Sulphuric acid alkylation), Isomerisation with Platinum catalyst. Hydrogen production in Refinery, Acid gas removal and Sulphur recovery units. Treatment of gasoline-Merox sweetening. Production and treatment of L.P.G. Diesel Hydro desulphurisation. Treatment of Kerosene- Edeleanu process. Production and treatment of Lube- Phenol extraction. Dewaxing methods- Chilling and pressing and MEK dewaxing. Properties, test methods and uses of Refinery products such as L.P.G, Gasoline, Jet fuel, Kerosene, Diesel fuel, Lube oil and Bitumen.	10

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall the origin, various exploration and recovery methods of crude oil.	K1
CO2	Outline the modern integrated petroleum refinery, its feedstock, products and the process employed.	K2
CO3	Summarize different petroleum based products and their uses.	K3
CO4	Explain refining process covering operating description and conditions, feedstock and catalyst selection, product yield and the relation between different process parameters.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						2					
CO2	3						2					
CO3	3						2					
CO4	3						2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Petroleum Refinery Process	Baskara Rao B.K.	Oxford& IBM	6 th Edition 2020
2	Petroleum Refining Technology	Dr.Ram Prasad	Khanna Publishers	1998
3	Modern Petroleum Chemistry.	Dr.Kochu Baby Manjooran S	East-West Press,	2008
4	Outlines of Chemical Technology	Dryden C. E.	Shree Hari Publications	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Petrochemical Process technology	I D Mall	Macmillan	1 st Edition 2007
2	Petroleum Refinery Technology and Economics	James H.Garry Glenn E. Handwerk Mark J.Kaiser	CRC Press, Taylor and Francis group	6 th Edition
3	Fundamentals of Petroleum and Petrochemical Engineering	Uttam Ray Chaudhuri	CRC Press	1 st Edition 2020
4	Petroleum Refinery Engineering,	Nelson W.L	McGraw Hill	4 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/content/syllabus_pdf/103102022.pdf
3	https://www.scribd.com/document/410381960/Nptel-Chemical-Engineering-Petroleum-Refinery-Engineering9

SEMESTER S6

HETEROGENEOUS REACTIONS AND CATALYSIS

Course Code	PECHT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To explain basic principles and theories of heterogeneous catalysis, including adsorption, desorption, surface reactions, and diffusion processes.
2. To develop an understanding of reaction kinetics in heterogeneous catalysis.
3. To acquire skills in various techniques for catalyst preparation and characterisation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Catalyst and catalytic reactors: Catalysts, types of catalysts, catalytic properties, steps in a catalytic reaction, adsorption equilibrium constant, desorption, surface reaction, synthesizing rate law, rate limiting step, Langmuir-Hinshelwood approach. Development of design equations for ideal mixed batch reactor, plug flow tubular reactor and perfectly mixed continuous stirred tank reactor for heterogeneous systems. Heterogeneous data analysis for reactor design	9
2	Diffusion and reaction in porous catalysts: effective diffusivity, tortuosity-modelling of diffusion with reaction on a spherical catalysts. Thiele Modulus, internal effectiveness factor, Overall effectiveness factor. Estimation of diffusion and reaction limited regimes - Weisz - Prater criterion for internal diffusion, Mears criterion for external diffusion.	9

3	Fluid Particle Reactions (Non catalytic): Selection of a model: Unreacted core model for spherical particles of unchanging size, model development for diffusion through gas film, ash layer, and chemical reaction controls. Rate of reaction for shrinking spherical particles - chemical reaction controls, diffusion controls, application to design.,	9
4	Selection, design, and Preparation of catalysts: Deposition, impregnation, precipitation, co-precipitation, and sol-gel formation. Characterization of catalysts- pore volume, pore size, and pore size distribution- Mercury porosimetry- Nitrogen desorption method- BET method. Thermogravimetric analysis- Differential thermal analysis- X-ray diffraction method- Electron microscopy- Scanning electron microscopy- Transmission electron microscopy- Atomic emission spectroscopy.	9

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the types, properties of catalysts and various steps involved in catalytic reactions.	K2
CO2	Develop design equations for ideal reactors for heterogeneous systems.	K3
CO3	Develop the kinetics for diffusion and reaction in porous catalysts.	K3
CO4	Develop the kinetic models for fluid-particle reactions.	K3
CO5	Explain various preparation and characterisation methods of catalysts.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3	3									
CO3	3	3	3									
CO4	3	3	3									
CO5	3	3										

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Chemical Reaction Engineering	H Scott Fogler	Prentice Hall of India	Fifth Edition, 2016
2	Chemical Engineering Kinetics	Smith J.M	McGraw Hill	Second Edition, 1970

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Reaction Engineering	Levenspiel Octave	John Wiley & Sons	Third Edition. 1999
2	Introduction to Chemical Reaction Engineering and Kinetics	Ronald W. Missen, Charles A. Mims, Bradley A. Saville	John Wiley & Sons	First Edition, 1999
3	An Introduction to Chemical Engineering Kinetics & Reactor Design	Charles G. Hill	John Wiley & Sons	First Edition, 1977
4	Chemical Reactor Theory- An Introduction	K.G Denbigh& J.C.R Turner	Cambridge University Press.	Third Edition, 1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/101/103101141 https://archive.nptel.ac.in/courses/103/103/103103026 , https://archive.nptel.ac.in/courses/103/102/103102012
2	https://archive.nptel.ac.in/courses/103/101/103101141
3	https://archive.nptel.ac.in/courses/103/101/103101141 , https://archive.nptel.ac.in/courses/103/106/103106117 .
4	https://archive.nptel.ac.in/courses/103/103/103103026

SEMESTER S6

FUEL CELL TECHNOLOGY

Course Code	PECHT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To expose the students to the fundamental knowledge required in the development of fuel cells

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Fuel Cell, Brief History of fuel cells, Types of Fuel Cells, Fuel Cell and conventional processes – comparison, Energy & power relations, units, Application scenarios, Advantages and disadvantages. Overview of Fuel Cell Types: PAFC, PEMFC, AFC, MCFC, SOFC - Major Cell Components, Processes, Operating Conditions, and applications. General Thermodynamics: Enthalpy-Heat potential of fuel, Gibb's free energy-Work potential of fuel, Reversible voltage - NERNST Equation, Voltage and P, T and concentration dependence – examples, Faraday's Laws, Efficiency: thermodynamic, voltage and fuel.	10
2	Reaction Kinetics: Electrochemical reaction fundamentals, electrode kinetics, Charge transfer and activations energy, Exchange current density - slow and fast reactions, Potential and equilibrium - galvanic potential, Reaction rate and potential - Butler Volmer equation & Tafel equation.	9
3	Charge and Mass Transport: Charge transport resistances, voltage losses, Ionic and electronic conductivities, Ionic conduction in different FC	9

	electrolytes: Aqueous, polymeric, and ceramic, Diffusive transport & voltage loss: Limiting current density, Nernstian and kinetic effect. Convective transport: flow channels, gas diffusion / porous layer, gas velocity, pressure, Flow channel configurations	
4	Stack Design: Sizing of a Fuel Cell Stack, Stack Configuration, Uniform distribution of Reactants, Heat removal, Stack Clamping Fuel Cell System Design: Hydrogen-Oxygen Systems, Hydrogen-Air Systems, Fuel Cell Systems with Fuel Processor, System Efficiency	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the fuel cell technology and compare different types of fuel cell systems.	K2
CO2	Explain the fundamentals of electrochemistry, thermodynamics, fluid mechanics, heat and mass transfer, and reaction kinetics, appropriate for the design or review of components of fuel cells and fuel cell systems.	K2
CO3	Explain the fundamentals of charge and mass transport in fuel cells	K2
CO4	Analyze and explain fuel cell stack and fuel cell systems.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				1						3
CO2	3	3	3			1						3
CO3	3	3				1						3
CO4	3	3	3			1						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuel Cell Fundamentals,	Ryan P. O'Hayre, Suk-Won Cha, Whitney Colella & Fritz B. Prinz	John Wiley & Sons, Inc., New Jersey	2006
2	PEM Fuel Cells: Theory and Practice	Frano Barbir	Elsevier	2005
3	Fuel Cells and Their Applications.	Kordesch, K.; Simader, G	VCH	1996
4	Fuel Cell Systems Explained.	Larminie, J.; Dicks A	John Wiley & Sons Ltd: Chichester	1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuel Cell Handbook, 7th Edn.		EG & G Technical Services	Nov 2004
2	Handbook of Fuel Cells- Fundamentals, Technology and Applications. 2003; Vols 1-4	Vielstich, W, Gasteiger, H. A. Lamm, A.	John Wiley & Sons Ltd: NY, Vol-1-4	2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/102/103102015/
2	http://acl.digimat.in/nptel/courses/video/115105127/lec38.pdf
3	https://www.youtube.com/watch?v=62363H_I_Qk
4	http://acl.digimat.in/nptel/courses/video/121106014/lec35.pdf

SEMESTER S6

PROCESS UTILITY AND PIPING ENGINEERING

Course Code	PECHT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To classify the common process utility used in process industries.
2. To develop an understanding of working of different process utility systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Process Utility Systems: Classification of process utility systems, Importance of process utilities in chemical industries and plants. Water as a utility in process industries: Sources of water, hard and soft water, methods of water treatment, recycle and conservation of water. Cooling Tower: Constructional details, Working, Types and performance evaluation.	9
2	Introduction to Fuels:- Properties of Fuel oil, Coal and Gas, Storage, handling and preparation of fuels, Principles of Combustion, Combustion of Oil, Coal and Gas. Compressed air system: Types of air compressors, Compressor efficiency, Compressed air system components, Factors affecting the performance and efficiency of compressors. Vacuum Pumps: Types of vacuum pumps and their performance characteristics.	9
3	Steam System: Steam and its importance, Properties of steam, Problems based on enthalpy calculation for wet steam, dry saturated steam, and superheated steam.	8

	Boilers: Types of steam generators/boilers, Combustion in boilers, Performance evaluation, Analysis of losses, Feed water treatment, Blow down, Boilers Act. Steam handling and distribution: Steam distribution system, Assessment of steam distribution losses, Steam economy, Steam traps, Condensate and flash steam recovery system. Waste Heat Recovery: Classification, Advantages and applications, commercially viable waste heat recovery devices.	
4	Refrigeration and Ventilation: Principles of refrigeration, vapour compression and vapour absorption refrigeration cycles, types of refrigerants and their importance. Production of cryogenic temperatures. Exhaust & Ventilation. Introduction to Piping: Classification of pipes, pipe materials, pipe sizing, pipe wall thickness, schedule number, codes and standards. Piping colour codes as per types of fluid passing through pipes. Piping Components: Pipes, Fittings, Flanges, Gaskets, and Valves. Pipe fittings: Functions and properties, types of pipe fittings and their selection. Gaskets: Functions and properties, types of gaskets and their selection. Valves: Types of valves. Piping insulation: Insulation materials, estimating thickness of insulation, critical thickness of insulation, optimal thickness of insulation.	10

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify utility systems and explain the working of cooling towers.	K2
CO2	Identify the fuel systems and compressed air systems used in industry.	K3
CO3	Examine the steam system used in industry.	K4
CO4	Utilize the refrigeration system and piping system used in process industry.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3									2	
CO2	3	3									2	
CO3	3	3									2	
CO4	3	3									2	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process utility systems	Jack Broughton	Institution of Chem. Engineers, U.K	1st Edition, 1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plant design and Economics for Chemical Engineers	M.S. Peters and Timmerhaus	Mc Graw Hill,	3rd Edition, 2017
2	Process Plant layout and Piping Design	Roger Hunt and Ed Bausbacher,	PTR PrenticeHall Inc	1990
3	Chemical Engineering Plant Design Edition. Design	F.C. Vibrandt and C.E. Dryden,	McGraw Hill,	5 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107211/ https://onlinecourses.nptel.ac.in/noc22_ch24/preview
2	https://archive.nptel.ac.in/courses/103/107/103107211/
3	https://archive.nptel.ac.in/courses/103/107/103107211/
4	https://archive.nptel.ac.in/courses/103/107/103107211/

SEMESTER S6

FERTILIZER TECHNOLOGY

Course Code	PECHT637	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Provide comprehensive and balanced understanding of essential link between synthetic fertilizer industries.
2. Identify engineering problems in fertilizer manufacturing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Fertilizer: Synthetic fertilizers, Classification of fertilizers, Role of essential Elements in plant Growth, Macro elements and Micro elements, Application of fertilizers considering Nutrient, Balance and types of crop. Development of fertilizer industry; Fertilizer production and consumption in India; Nutrient contents of fertilizers; Secondary nutrients; Feedstock and raw materials for nitrogenous, phosphatic and potassic fertilizers.	8
2	Nitrogenous Fertilizers: Introduction to Ammonia: Physical & chemical properties, applications, Synthesis gas by Steam Hydrocarbon reforming, Ammonia converters: Types and classification. Storage and Transportation of Ammonia. Introduction to Nitric acid: Chemical, physical properties and applications, Manufacturing of Nitric acid by ammonia oxidation process, Concentration of Nitric acid . Urea- Physical, chemical properties, Manufacturing of Urea by	10

	Stamicarbon's CO ₂ stripping process, total recycle process, Manufacturing of Ammonium nitrate by Prilling process, Ammonium sulphate from Ammonium carbonate and gypsum, Ammonium chloride by direct neutralization.	
3	Phosphatic and Pottasium Fertilizers: Manufacture and application of MAP, DAP, Single Superphosphate, Triple super phosphate. Physical, chemical properties and uses of Potassium Chloride, Potassium nitrate, Potassium sulphate, Manufacturing of potassium chloride from sylvinit, Preparation of Potassium nitrate, Potassium sulphate.	9
4	Miscellaneous Fertilizer and Bio Fertilizers: Introduction to complex and mixed fertilizers. Manufacturing of NPK, Ammonium Sulphate Phosphate (ASP), Calcium Ammonium Nitrate(CAN), Biofertilizers, Types of Biofertilizers, fixing biofertilizers, Phosphate-solubilizing biofertilizers, Preparation of a biofertilizers.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the classification and applications of fertilizers.	K2
CO2	Summarize the reactions and unit operation steps in manufacturing of various nitrogenous fertilizers.	K2
CO3	Explain the reactions and process flow diagrams for the preparation of phosphatic and potassium fertilizers.	K2
CO4	Explain the preparation of biofertilizers and other miscellaneous fertilizers.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3						2					
CO3	3						2					
CO4	3						2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bio fertilizers in Agriculture	N S Subba Rao	Oxford & IBH Publishing Company	4 th Edition 2024
2	Chemistry & Technology of Fertilizers	Slack A.V.	Interscience, New York	1967

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Technology, Volume I & II	Pandey & Shukla	Vani Books Company	2nd Edition, 2000
2	Chemical Process Industries	G.T. Austin, R.N. Shreve	McGraw Hill	5th Ed., 1984.
	Outlines of Chemical Technology, Edited and revised by M. GopalaRao and Marshall Sitting.	C.E. Dryden	Affiliated East-West Press Pvt. Ltd. New Delhi.	3rd Edition, 1997.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103106108/33
2	https://archive.nptel.ac.in/content/syllabus_pdf/103107086.pdf
3	https://archive.nptel.ac.in/content/syllabus_pdf/103107086.pdf
4	https://archive.nptel.ac.in/content/syllabus_pdf/103107086.pdf

SEMESTER S6
COMPUTATIONAL FLUID DYNAMICS

Course Code	PECHT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBCHT304	Course Type	Theory

Course Objectives:

1. Impart a solid understanding of the fundamental principles of fluid dynamics through governing equations, turbulence models and boundary conditions.
2. Develop the ability to analyze and solve fluid flow problems using numerical techniques.
3. Enable proficient use of computational tools to set up, run, and interpret simulations for a variety of fluid flow problems, ensuring application to real-world engineering scenarios and research projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Need of CFD as an analysis tool, Applications of CFD. CFD software packages and tools. CFD solution procedure: pre-processing, numerical solution – CFD solver, post-processing. Classification of partial differential equations (PDE) and boundary conditions: well-posed problems. Review of governing equations: Continuity equation, Momentum equation, Energy equation, Navier – Stokes equation, generic form of governing equations.	9
2	Basic concepts of CFD: Introduction to discretization, Finite difference approximation of a derivative, discretization of differential equations using finite differences, consistency, convergence and stability (concepts alone), concept of explicit and implicit methods, Crank Nicholson scheme (FTCS only).	9

3	Finite Volume Discretization: Diffusion problem, convection-diffusion problem, properties of discretization schemes – conservativeness, boundedness, transportiveness. Solution to discretized equations: TDMA, Jacobi, Gauss-Seidel, Gauss elimination methods. Detailed study of Navier stokes Equation-Solution of the Navier Stokes Equations. Grid arrangement-collocated and staggered grid. SIMPLE algorithm.	9
4	Turbulence modelling: Need for turbulence modelling. Closure problem. Reynolds averaged Navier Stokes equation. Turbulence modelling. Multiphase flow modelling in CFD: Classification of multiphase flow, Industrial examples, Examples for multiphase reactors, Multiphase flow modelling approaches Eulerian-Lagrangian approach, Eulerian – Eulerian Approach, Volume of fluid approach.	9

Suggestions on Project Topics

Assignment/Project: 20 Marks

Students should evaluate and analyze a real-world problem, assess the proposed solutions, provide a conclusion on which solution is most appropriate for the problem, and implement the chosen solution using any computational tools like ANSYS, Open Foam, Matlab etc.

Level: Analyze

1. Problem Definition and Decomposition:

- **Clarity and Completeness** (2 points): Evaluate the clarity and completeness of the problem definition. Are the objectives well-defined? Is the problem broken down into manageable sub-problems?
- **Identification of Key Parameters** (2 points): Assess the identification and justification of key parameters influencing the fluid dynamics problem. Are all relevant variables and boundary conditions considered?

2. Analytical Approach:

- **Methodological Soundness** (1 points): Evaluate the soundness of the chosen analytical methods. Are the equations and principles correctly applied to the problem?
- **Critical Analysis** (1 points): Assess the depth of critical analysis in the approach. Are assumptions and limitations clearly stated and justified?

3. Data Interpretation:

- ***Accuracy and Consistency*** (2 points): Evaluate the accuracy and consistency of the data interpretation. Are the results logically derived from the data?
- ***Comparison with Theoretical Predictions*** (2 points): Assess how well the results are compared with theoretical predictions or existing literature. Are discrepancies analyzed and explained?

Level: Evaluate

1. Simulation Setup and Execution:

- ***Appropriateness of CFD Tool*** (1 point): Evaluate the appropriateness of the chosen CFD tool for the problem at hand. Is the tool well-suited to model the specific fluid dynamics scenario?
- ***Simulation Accuracy and Robustness*** (2 points): Assess the accuracy and robustness of the simulation setup. Are the mesh, time steps, and convergence criteria appropriately chosen?

2. Results Validation:

- ***Validation against Experimental Data*** (1 point): Evaluate the extent and rigor of validation against experimental or benchmark data. Are the validation methods appropriate and convincingly presented?
- ***Error Analysis*** (1 point): Assess the thoroughness of error analysis. Are sources of error identified and quantified? Are their impacts on the results discussed?

3. Conclusions and Recommendations:

- ***Logical Consistency*** (1 point): Evaluate the logical consistency of the conclusions drawn from the analysis and simulation results. Are the conclusions supported by the data?
- ***Practical Implications and Recommendations*** (1 point): Assess the practical implications and recommendations based on the findings. Are the recommendations feasible and well-grounded in the analysis?

4. Presentation and Communication:

- ***Clarity and Organization*** (1 point): Evaluate the clarity and organization of the project report or presentation. Is the information logically structured and easy to follow?
- ***Technical Writing and Visual Aids*** (2 points): Assess the quality of technical writing and the effectiveness of visual aids (graphs, charts, simulations). Are complex concepts clearly explained and visually supported?

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply fundamental knowledge of science and engineering to solve practical problems.	K3
CO2	Identify the proficiency in solving basic fluid flow problems, including mesh generation, boundary conditions, and post-processing.	K3
CO3	Analyze and compare the performance of different numerical schemes for solving fluid flow equations, considering accuracy, stability, and computational efficiency.	K4
CO4	Critically analyze and evaluate different turbulence models and their applicability to various flow regimes and demonstrate a deep understanding of turbulence theory.	K5
CO5	Estimate advanced multiphase flow modeling techniques in CFD to simulate and optimize processes involving gas-liquid or solid-liquid interactions, demonstrating a deep understanding of interfacial phenomena.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3	3	3							3
CO3	3	3	3	3	3				3	3		3
CO4	3	3	3	3	3				3	3		3
CO5	3	3	3	3	3				3	3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computational Fluid Dynamics – The basics with applications	John D Anderson	McGraw Hill	Indian Edition 2017
2	An Introduction to Computational Fluid Dynamics: The Finite Volume Method	H Versteeg, M Malalasekera	Pearson	2 nd Edition 2008
3	Numerical Heat Transfer and Fluid Flow	Suhas V Patankar	CRC Press	Special Indian Edition, 2017
4	Computational Fluid Mechanics and Heat transfer	Anderson, D. A; Tarmehel	CRC Press	Special Indian Edition, 2016
5	Computational methods for Fluid Mechanics	Ferziger, J. H and Peric, M	Springer	4 th Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computational Fluid flow and Heat transfer	K Muralidhar and Sundararajan	Narosa	1 st Edition 2009
2	Introduction to Computational Fluid Dynamics	Pradeep Niyogi, S K Chakrabarthy	Pearson	1 st Edition 2005
3	Computational Fluid Dynamics for Engineers and Scientists	Sreenivas Jayanti	Springer	1 st Edition 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_ch05/preview
2	https://onlinecourses.nptel.ac.in/noc20_ch05/preview
3	https://onlinecourses.nptel.ac.in/noc20_ch05/preview

SEMESTER S6

PROCESS DYNAMICS AND CONTROL

Course Code	PBCHT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	4	ESE Marks	40
Credits	3:0:0:1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	4	Course Type	Theory

Course Objectives:

Sound knowledge in process control is very important for Chemical Engineers. Process control has become increasingly important in the process industries as a consequence of global competition, rapidly changing economic conditions, and more stringent environmental and safety regulations. Proper application of process control improves the safety and profitability of a process, while maintaining consistently the desired product quality.

1. Model chemical engineering processes and analyze/predict their dynamics both for open- (without control) and closed-loop (with control) cases.
2. Develop control strategies and select the most appropriate input to manipulate, and tune controllers to meet/achieve process objectives.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to process dynamics and control: - definition of terms - Laplace transform - transform of simple functions - derivatives and integral - properties of Laplace transforms - final value theorem - initial value theorem - transition of transforms and functions – examples - inversion by partial fraction - solution of differential equations- linear open loop systems - first order systems - mercury thermometer, liquid level and mixing processes - response of these towards different types of forcing functions. (Study of dynamics by conducting experiments in above studied first order systems and simulation in MATLAB/SCILAB- for self study/project)	9

2	Systems in series: - interacting and non-interacting types and generalization of results. Linear open loop second order systems - manometer - response of these towards different types of forcing functions: step and sinusoidal functions. Control valves- Characteristics- P, PI and PID Controllers - basic principles and transfer functions. (Study of dynamics by conducting experiments in above studied second order systems, simulation in MATLAB/SCILAB- Controllers/Control valve characteristics- for self study/project)	9
3	Closed loop system: – feedback control- servo and regulator problems - block diagram development –Block diagram reduction-Transient response of simple control systems - step response and offset. Introduction to stability of linear systems: - Routh- Hurwitz criterion for stability -Root locus technique-plotting root locus diagram. (Plotting root locus diagrams and stability analysis using MATLAB/SCILAB/SIMULINK-for self study/project)	9
4	Introduction to frequency response: - substitution rule – Bode diagram for first order systems - first order systems in series - bode stability criterion, gain margin and phase margin. Controller tuning:- Ziegler-Nichol's method - comparison of closed loop responses for different controller settings. (Plotting Bode diagrams and stability analysis using MATLAB/SCILAB/SIMULINK-for self study/project)	9

Suggestion on Project Topics

The students shall be directed to carry out any one of the projects listed below.

1. Design and simulation of suitable temperature control system for stirred tank heater.
2. Design and simulation of suitable level or flow control system for a single tank system.
3. Design and simulation of suitable level or flow control system for a multi-tank system.
4. Design and simulation of suitable control system to maintain product concentration in continuous stirred tank reactor.
5. Design and simulation of a temperature control system for a continuous stirred tank reactor used in chemical process.
6. Any other relevant project related to process dynamics and control.

The completed project work should consist of following key components

1. **Fundamental Concept:** The dynamic behaviour of the process (Module 1)
2. **System Modelling:** Develop a model for the selected system from fundamental conservation principles (Module 1 and 2)
3. **Block Diagram development:** Develop a feed back control system using the concept of block diagram development and reduction. (Module 3)
4. **Controller selection:** Select a suitable controller for the process (Module 2 and 3)
5. **Simulation:** Implement the system model and controller in MATLAB or Simulink and simulate the response. (Self-study)
6. **Stability analysis:** Analyse the stability of the system using methods like Routh-Hurwitz Criterion, Root Locus method, Bode stability criterion etc. (Module 4)
7. **Controller tuning:** Optimize the control parameter for desired performance using Ziegler- Nichols tuning. (Module 4)
8. **Validation by experimental data:** Conduct experiment in the laboratory and evaluate the system performance with the experimental data (Conduct experiment in lab and obtain data for validation)
9. **Report preparation and oral presentation:** Prepare a detailed report and presentation before the evaluation committee.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply Laplace transforms to solve linear differential equations and to obtain the transfer function related to first order systems.	K3
CO2	Develop the response of linear open loop second order systems and systems in series for various forcing functions.	K3
CO3	Examine various types of controllers and analyze servo and regulatory problems of closed loop systems by using transient response.	K4
CO4	Analyze the stability of linear systems by using analytical and graphical methods. Design controllers.	K4
CO5	Recommend suitable controller for a processing system to obtain desired parameter.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				3							3
CO2	3				3							3
CO3	3	3			3							3
CO4	3	3	3		3							3
CO5	3	3	3	3	3			3	3	3	3	3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Systems Analysis& Control	Donald R. Coughanowr	McGraw Hill International Edition	2 nd Edition
2	Chemical Process Control - An Introduction to Theory & Practice	George Stephanopoulos	Prentice Hall of India	Fifth Indian Reprint
3	Process Dynamics and Control	Dale E. Seborg, Thomas F. Edgar and Duncan A. Mellichamp	John Wiley & Sons Inc	2 nd Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automatic Process Control for Chemical Engineers	Ceaglske N.H	John Wiley & Sons Inc	2 nd Edition
2	Process Dynamics and Control	Govind Das Nageshwar, Sudheer S. Bhagade	PHI Learning Pvt. Ltd.,	2011
3	Chemical Process Modelling and Computer Simulation	Amiya K. Jana	PHI Learning Pvt. Ltd.	Second edition, 2011
4	Computational techniques for process simulations and analysis using MATLAB	Niket S. Kaisare	CRC Press	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/101/103101142/
2	https://archive.nptel.ac.in/courses/112/107/112107214/
3	https://archive.nptel.ac.in/courses/103/101/103101142/
4	https://archive.nptel.ac.in/courses/112/107/112107214/

PBL Course Elements

L: Lecture	R: Project (1 Hr.), 2 Faculty Members		
(3 Hrs.)	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S6

PROCESS UTILITIES

Course Code	OECHT611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To classify the common process utility used in process industries.
2. To develop an understanding of working of different process utility systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Process Utility Systems: Classification of process utility systems, Importance of process utilities in chemical industries and plants. Water as a utility in process industries: Sources of water, hard and soft water, methods of water treatment. Cooling Tower: Constructional details, Working, Types and performance evaluation.	9
2	Introduction to Fuels: Properties of Fuel oil, Coal and Gas, Storage, handling and preparation of fuels, Principles of Combustion, Combustion of Oil, Coal and Gas. Compressed air system: Types of air compressors, Compressor efficiency, Compressed air system components, Factors affecting the performance and efficiency of compressors. Vacuum Pumps: Types of vacuum pumps.	9
3	Steam System: Steam and its importance, Properties of steam, wet steam, dry saturated steam, and superheated steam. Boilers: Types of steam generators/boilers, Combustion in boilers,	9

	Performance evaluation, Analysis of losses, Feed water treatment, Blow down, Boilers Act. Steam handling and distribution: Steam distribution system, Assessment of steam distribution losses, Steam traps, Condensate and flash steam recovery system. Waste Heat Recovery: Classification.	
4	Refrigeration and Ventilation: Principles of refrigeration, vapour compression and vapour absorption refrigeration cycles, types of refrigerants and their importance. Exhaust & Ventilation. Introduction to Piping: Classification of pipes, pipe materials, pipe sizing, schedule number, codes and standards. Piping Components: Pipes, Fittings, Flanges, Gaskets, and Valves.	9

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify utility systems and explain the working of cooling towers.	K2
CO2	Identify the fuel systems and compressed air systems used in industry.	K3
CO3	Analyze the steam system used in industry.	K4
CO4	Identify the refrigeration system and piping system used in process industry.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3									2	
CO2	3	3									2	
CO3	3	3									2	
CO4	3	3									2	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process utility systems	Jack Broughton	Institution of Chem. Engineers, U.K	1st Edition, 1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plant design and Economics for Chemical Engineers	M.S. Peters and Timmerhaus	Mc Graw Hill,	3rd Edition, 2017
2	Process Plant layout and Piping Design	Roger Hunt and Ed Bausbacher,	PTR PrenticeHall Inc	1990
3	Chemical Engineering Plant Design	F.C. Vibrandt and C.E. Dryden,	McGraw Hill,	5 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107211/ https://onlinecourses.nptel.ac.in/noc22_ch24/preview
2	https://archive.nptel.ac.in/courses/103/107/103107211/
3	https://archive.nptel.ac.in/courses/103/107/103107211/
4	https://archive.nptel.ac.in/courses/103/107/103107211/

SEMESTER S6

POLYMER SCIENCE AND ENGINEERING

Course Code	OECHT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To impart a comprehensive study of polymers and their classification, Molecular Weight and Properties, synthesis and characterization, processing methods and engineering application of polymers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to polymers -Definition and classification of polymers, classification of polymer based on source, structure, application, thermal behaviour, mode of polymerization - natural and synthetic polymers, thermoplastics, thermosets, elastomers, basic concepts of polymerization (addition, condensation), Copolymerisation-Different types of copolymers-Characteristics.	8
2	Molecular weight of polymers – weight average and number average molecular weight – sedimentation and viscosity average molecular weights. Experimental methods for molecular weight determination – end group analysis, light scattering method – viscometry (Ostwald viscometer) intrinsic viscosity. Molecular weight distribution curve. Polymer properties - Polymer architecture (linear, branched, crosslinked, etc.), Crystallinity and glass transition temperature, Mechanical properties, (elasticity, viscoelasticity, yield), Thermal properties and degradation, Electrical and optical properties	10

	Definition and applications of Polymer blends, composites and nanocomposites	
3	Polymer Synthesis and characterization -step growth polymerization, chain growth polymerization, Methods of polymerization, copolymerization and copolymer types – bulk, solution, suspension and emulsion polymerization. Differential Scanning Calorimetry (DSC) and Thermogravimetric Analysis (TGA), applications of spectroscopic methods in polymer characterization	8
4	Polymer processing and engineering applications- Plastic processing technology -injection moulding-compression moulding - calendaring – blow moulding – extrusion – thermoforming – wet, dry and melt spinning methods for fibres –vulcanization of rubber — general study of elastomer processing - brief description of compounding methods. Applications in drug delivery, tissue engineering, and medical devices, conductive polymers and their applications, polymers in microelectronics and photonics, dielectric and insulating materials, polymers in automotive and aerospace applications, polymer coatings and adhesives	10

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the fundamentals, classification and methods of polymerisation.	K2
CO2	Explain methods to find out average molecular weight of polymers and understand the factors affecting polymer properties.	K2
CO3	Illustrate the polymer synthesis methods and characterization techniques	K2
CO4	Apply the processing methods in engineering applications of different polymeric materials.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Science	V R Gowariker, N V Viswanathan, Jayadev Sreedhar	New Age International (P) Ltd	3 rd Edition 2019
2	Text book of polymer science	Billmeyer F.W	John Wiley and sons	3 rd Edition 1984
3	Polymer Science and Technology	Joel R. Fried	Pearson Education	3 rd Edition 2021
4	Introduction to Polymers	Robert J. Young, Peter A. Lovell	CRC Press	3rd Edition 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Science and Technology	Premamoy Ghosh	Tata Mc Graw Hill.Education	2011
2	Principles of polymer systems	Rodriguez, Ferdinand, Claude Cohen, Christopher K. Ober, and Lynden Archer	CRC press	6 th Edition 2015
3	Handbook of plastic testing technology	Shah V.H	Wiley	2 nd Edition 1998

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113105028
2	https://archive.nptel.ac.in/courses/104/105/104105124/
3	https://archive.nptel.ac.in/courses/113/105/113105028/
4	https://archive.nptel.ac.in/courses/113/105/113105028/

SEMESTER 6

POLLUTION CONTROL IN PROCESS INDUSTRIES

Course Code	OECHT613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Provide a comprehensive understanding of environmental regulations and pollution control principles.
2. Impart various technologies for wastewater treatment, sludge treatment and disposal, solid waste management strategies, air pollution management, and the abatement of noise& odour pollution to students.
3. Equip students with knowledge about preventing pollution in various industries, understanding emerging environmental challenges posed by new technologies like biotechnology and nanotechnology, and analysing historical industrial disasters such as the Bhopal tragedy.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of Pollution Control - Environmental Regulations and Standards: National regulations, Key legislations. Pollution Control Strategies - End-of-pipe treatments, Source reduction and waste minimization. Wastewater Pollution and Treatment Methods - Sources and classification of wastewater- Types of water pollutants and their effects. Treatment methods(brief overview- diagram, principle and working of equipment only) - Physical Treatment Methods: Screening, grit removal, Sedimentation, flotation, filtration, Chemical Treatment Methods: Coagulation, flocculation, precipitation, Biological Treatment Methods:	9

	Activated sludge, biofilm reactors, anaerobic digestion.	
2	Sludge Treatment and Disposal - sludge thickening - sludge conditioning - sludge dewatering - sludge digestion and composting (brief overview). Solid Waste Treatment -sources and classification, disposal methods-open dumping, sanitary landfill, incineration-composting (brief overview). Noise pollution –Sources, effects of noise on people, noise control methods.	9
3	Air Pollution: Units of measurement, sources, classification. Sampling and analysis of air pollutants, health effects of air pollution global warming and ozone depletion. Devices for Particulate Contaminants: Gravitational settling, Centrifugal Collectors, Wet Collectors, Fabric filters and Electrostatic Precipitators (ESP) Control Devices for Gaseous Contaminants: Adsorption, Absorption, Condensation, Combustion, Automobile Emission Control. Odour Pollution Management- Sources, Health Impacts, Odour Measurement and Monitoring, control methods.	9
4	Pollution Prevention Techniques: Generic water management techniques. Pollution prevention in the electronic Industry, Power generation industry, Electroplating industry, Fabricated metals products industry. EmergingPollution: Threats and Pollution Caused by Biotechnology and Nanotechnology. Application of biotechnology and nanotechnology in pollution Control. Threats and Pollution Caused by Renewable energy technologies (PV cells and batteries only included). Case study: Industrial pollution and disasters: Bhopal tragedy, Water pollution control and disposal of waste water in HPCL Refinery Vishakhapatnam.	9

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify sources and types of wastewater pollutants and selecting appropriate treatment methods, such as physical, chemical, and biological processes, to mitigate environmental impact effectively.	K3
CO2	Select suitable methods for treatment and disposal of sludge, solid waste.	K3
CO3	Apply the devices for particulate contaminant control in gaseous contaminant	K3
CO4	Apply pollution prevention techniques across various industries and explain the emerging threats and applications of biotechnology, nanotechnology, and renewable energy technologies in pollution control.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3	3				3
CO2	3					3	3	1				3
CO3	3					3	3	1				3
CO4	3					3	3	1				3
CO5	3					3	3	1				3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Engineering	Peavy, H.S., Rowe, D.R. and Tchobanoglous, G	McGraw-Hill	Seventh Edition
2	Environmental Pollution Control Engineering	Rao C.S.	New age International Pub.	Second Edition 2006
3	Air Pollution	M. N. Rao & H.V.N.Rao	Tata McGraw-Hill.	First Edition 2017
4	Industrial Pollution Prevention Handbook	Harry M. Freeman	Mc Graw Hill Education	2014
5	Environmental Pollution and control in Chemical Process Industries	S. C. Bhatia	Khanna Publishers	Second Edition 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Wastewater Treatment	A. D. Patwardhan	PHI Learning Pvt. Ltd	Second Edition
2	Environmental Engineering	N. N. Basak	McGraw Hill Education (India) Pvt. Ltd.	First Edition
3	Solid and Liquid Waste Management: Waste to Wealth	Rajaram Vasudevan, Siddiqui Faisal Zia, Agrawal Sanjeev, Khan Mohammad Emran	PHI	First Edition, 2016
4	Integrated Solid Waste Management	Bhide A.D. and Sundaresan, B.B	CRC Press	First Edition
5	Fundamentals of Air Pollution	Boubel, R.W., Fox, D.L., Turner, D.B., Stern, A.C	Academic Press	Third Edition
6	Industrial Water Pollution Control	Eckenfelder W.W	Mc Graw hill EXCLUSIVE (CBS)	Third Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103107084 https://nptel.ac.in/courses/105106119
2	https://archive.nptel.ac.in/courses/103/107/103107215/ https://archive.nptel.ac.in/courses/105/105/105105160/
3	https://archive.nptel.ac.in/courses/105/107/105107213/
4	https://archive.nptel.ac.in/courses/127/105/127105018/

SEMESTER S6

ENERGY ENGINEERING AND MANAGEMENT

Course Code	OECHT614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. The course makes the students understand the importance of energy conservation and the different methods employed for methods for harnessing different energy resources.
2. Students may gain knowledge on Energy Auditing, Energy conservation, Waste Heat Recovery, Maintenance of Energy Systems,

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Energy Conversion - Energy resources, Global and Indian scenario of energy resources and consumption, Impact of Energy on economy, Energy policies , development and environment. Conventional Energy conversion plants –Thermal, Hydro, and Nuclear power plants , Comparison, advantages and disadvantages, Combined cycle power plants, Energy from Hydrogen :- Hydrogen as a source of energy, Sources, hydrogen energy storage and transportation, Applications, Future perspectives.	9
2	Solar energy - Solar thermal systems- Flat plate collectors- Focusing collectors- Applications of solar energy in India Ocean energy :- wave energy conversion, Ocean thermal energy conversion,	10

	Tidal energy conversion. Wind energy- Types of windmills- Wind electric power generation- Wind power in India, Magneto hydrodynamics, Fuel cells : types , working of PEMFC.	
3	Energy management and Energy audit Energy Management – Definitions and significance, Objectives, Characterizing of energy usage, Energy Management program, Energy strategies and energy planning. Energy Audit – Types, Procedure and Techniques, understanding energy costs, Benchmarking, Energy performance, Optimum performance of existing facilities.	8
4	Energy conservation – Principles , Cogeneration ,Waste heat recovery , Waste to Energy Technologies- Gasification, Pyrolysis, Fermentation, Anaerobic digestion: compressed biogas. Energy Conservation Opportunities – Electrical ECOs, Thermodynamic ECOs in chemical process industry, ECOs in residential and commercial buildings, Energy Conservation Measures.	9

Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize energy resources and energy conversion processes.	K2
CO2	Explain energy conversion from solar, ocean , wind and fuel cells	K2
CO3	Apply energy management and energy audit procedures & techniques	K3
CO4	Identify energy conservation opportunities and apply energy conservation methods in various fields	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2					3	2					
CO2	2					3	2					
CO3	2					3	2	2				
CO4	2					3	2	2				

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Technology	Rao S. & Parulekar B.B	Khanna Publishers	3rd Edn, 1994
2	Renewable Energy Sources & Conversion Tech	. Bansal N.K., Kleeman M. & Meliss M	Tata McGraw Hill	1990
3	Energy for Sustainable Development	Hasanuzzaman, Nasrudin Abd Rahim (Eds)	Elsevier	2020
4	Energy Engineering and Management	Amlan Chakrabarti	Prentice Hall India	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Non-Conventional Energy Systems	Mittal K.M.	Wheeler Publications	
2	Chemical Technology	Venkataswarlu D.I	S. Chand	
3	A Text Book on Energy System and Engineering	Pandey G.N.	Vikas Publishing	
4	Handbook of Energy Audits,	Albert Thumann P. E. and W. J. Younger	Fairmont Press	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107157/
2	https://archive.nptel.ac.in/courses/103/107/103107157/
3	https://archive.nptel.ac.in/courses/103/107/103107157/
4	https://archive.nptel.ac.in/courses/103/107/103107157/

SEMESTER S6

PETROLEUM REFINING TECHNOLOGY

Course Code	OECHT615	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To apply chemical engineering principles to design and analyze refinery processes.
2. To gain knowledge of the various stages of crude oil refining, including distillation, cracking, and reforming.
3. To provide students with a comprehensive understanding of the petrochemical industry, its processes, and the underlying principles

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Origin and formation of crude oil: Oil exploration and drilling. Storage and transportation of crude oil and products. Classification, Composition and Evaluation of oil stock. Status of Petroleum industry in India. Classification of petroleum refinery. Preliminary petroleum processing-Impurities in crude oil, Dehydration and desalting of crude-Electric Desalter-, Process description, factors affecting the electric desalter. Pipe still furnaces and its operations.	9
2	Distillation of crude- Prefractionator: Atmospheric topping unit, Vacuum distillation unit. Thermal Conversion process: Process description of Thermal cracking, Visbreaking, Coking. Process description of fluid Catalytic cracking unit, Hydro cracking. Process description of catalytic reforming. Process description of alkylation process- sulphuric acid	9

	alkylation. Treatment of gasoline- Copper Chloride process and Merox sweetening.	
3	Petrochemical Industries & their feed stocks: Process of Synthesis gas production by steam reforming of Naphtha, Manufacture of Methanol from Synthesis Gas, Production of liquid fuels from Synthesis gas by Fischer – Tropsch process. Petrochemical products and their applications: First, Second and Third generation petrochemical products, Production of Ethylene, Propylene, and Butadiene by Naphtha/Gas cracking,	9
4	Plastomers, Elastomers and Synthetic fibres: Various methods of polymerization and their mechanisms, Production processes of LDPE, LLDPE and HDPE, Basic difference among the three and their applications, Production of PVC and Polystyrene, Production of Polypropylene, Production of SBR, PBR and Butyl rubber, Production of ABS plastics, Production of Polyamide (Nylon 6 and Nylon 6,6), Polyester and Acrylic fibres, Production of Phenol Formaldehyde resins, Detergents: Classification of detergents, Production of Linear Alkyl Benzene (LAB) from Superior Kerosene and Benzene, Sulphonation of LAB for production of Synthetic Detergents. Additives for synthetic detergents. Hard and soft detergents.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the storage, selection and evaluation of crude oil.	K2
CO2	Outline the principles of distillation in the atmospheric and vacuum distillation unit and analyze the preliminary refinery operation.	K2
CO3	Explain the various petrochemical feed stocks and the production of various petrochemicals.	K2
CO4	Summarize the production of aromatics in the petrochemical industries, the major polymerization processes and the various process for the production of synthetic detergents.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						2					
CO2	3						2					
CO3	3						2					
CO4	3						2					

., Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Text on Petrochemicals	B.K.B. Rao	Khanna Pub.	5 th Edition 2004
2	Introduction to Petrochemicals	Sukumar Maity	Oxford and IBH Publishing Co	2002
3	Modern Petroleum Refinery Process	Baskara Rao B.K	Oxford & IBM	6 th Edition 2020
4	Petroleum Refining Technology	Dr. Ram Prasad	Khanna Publishers	1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Petrochemicals	Dr. G. N. Sarkar	Khanna Publishers	1998
2	Modern Petroleum Chemistry	Dr. Kochu Baby Manjoran S		
3	Petroleum Refinery Technology and Economics	James H. Garry Glenn E. Handwerk Mark J. Kaiser	CRC Press, Taylor and Francis group	
4	Fundamentals of Petroleum and Petrochemical Engineering	Uttam Ray Chaudhuri	CRC Press	1 st Edition 2020
5	Petroleum Refinery Engineering,	Nelson W.L	McGraw Hill	4 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/102/103102022
2	https://archive.nptel.ac.in/content/syllabus_pdf/103102022.pdf

SEMESTER 6

HYDROGEN ENERGY: PRODUCTION, STORAGE, TRANSPORTATION AND SAFETY

Course Code	OECHT616	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. This course provides a comprehensive knowledge of hydrogen production, storage and transportation, utilization in various sectors, associated energy conversion devices, sensing and safety.
2. The course will provide a broad knowledge of hydrogen as an energy carrier.
3. The course also covers the classification and working of fuel cells.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Hydrogen energy: Hydrogen Economy- Decarbonisation of the Economy, road to the Hydrogen Economy. Properties of hydrogen as fuel. General introduction to infrastructure requirement for hydrogen production, storage, dispensing and utilization, and hydrogen production plants.	9
2	Hydrogen production pathways: Thermal process-Steam reformation, Advanced Methods of Steam Reforming, Partial Oxidation Method for Hydrogen Production. Electrochemical- Electrolysis, photo electro chemical method. Thermo- chemical water splitting. Biological process: hydrogen production from microbial biomass conversion- gasification, pyrolysis.	9

3	Hydrogen storage, safety and transportation: Physical and chemical properties, general storage methods, compressed storage-composite cylinders, metal hydride storage, carbon-based materials for hydrogen storage. Hydrogen safety aspects, backfire, pre-ignition, hydrogen emission NOx control techniques and strategies, Hydrogen Transportation via Hydrogen Pipelines, Long Distance Hydrogen Transmission, transportation via road, ships and in form of Liquid Organic Hydrogen Carriers (LOHC), hydrogen transported as ammonia, liquefied hydrogen (LH2) transportation, Hydrogen Refuelling Stations.	9
4	Hydrogen usage: Overview of fuel cells, Alkaline fuel cells, Proton exchange membrane fuel cells, Methanol fuel cells, Phosphoric acid fuel cells, Molten carbonate fuel cells, Solid oxide fuel cells, Fuel cell comparison, Hydrogen combustion engines, Hydrogen turbines, Hydrogen gas burner, Gas engine combined heat and power plants.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the importance of hydrogen economy.	K2
CO2	Explain different methods of hydrogen production.	K2
CO3	Classify different techniques of hydrogen storage, transportation and various safety challenges in hydrogen storage.	K2
CO4	Explain the various uses of hydrogen energy.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1				2						2
CO2	2	1				2						2
CO3	2	1				2						2
CO4	2	1				2						2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydrogen Production by Electrolysis	AgataGodula-Jopek	Wiley-VCH, Germany	2015
2	Hydrogen Fuel: Production, Transport and Storage	Gupta, R. B.	CRC Press, Taylor & Francis Group.	2009
3	Fuel Cell Handbook	EG & G Technical Services,	Morgantown, West Virginia, USA	2004
4	Tomorrow's Energy: Hydrogen, Fuel cells and the prospects for a cleaner planet	Peter Hoffman	MIT Press, Cambridge, London, England	2001
5	Hydrogen storage: state-of-the-art and future perspective. Netherlands	Tzimas, E., Filiou, C., Peteves, S.D., & Veyret, J.B.	European Communities.	2003
6	Principles of Fuel Cells	Xianguo Li	Taylor & Francis.	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Hydrogen Storage	Michael Hirscher	Wiley-VCH 2.	2010
2	Electrochemical Methods	A.J. Bard and L.R.Faulkner	John Wiley & Sons, Inc.	
3	Introduction to Fuel Cell Technology	Chris Rayment and Scott Sherwin	Notre Dame, U.S.A.	2003
4	Fuel Cells: From Fundamentals to Applications	S Srinivasan	Springer	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	NPTEL :: Chemical Engineering - NOC:Hydrogen Energy: Production, Storage, Transportation and Safety
2	NPTEL :: Chemical Engineering - NOC:Hydrogen Energy: Production, Storage, Transportation and Safety
3	NPTEL :: Chemical Engineering - NOC:Hydrogen Energy: Production, Storage, Transportation and Safety
4	NPTEL :: Chemical Engineering - Fuel Cell Technology

SEMESTER S6

ENVIRONMENTAL IMPACT ASSESSMENT

Course Code	OECHT617	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To study the various types of environmental pollution
2. To study the impact of various types of pollutants and their assessment techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Legislations, laws and acts relevant to Environmental protection in India -The environment (protection) Act-1986, The Air (prevention and control of pollution) Act 1981, The Water (prevention and control of pollution) Act-1974, National environmental policy 2006. Classification of Pollution and Pollutants- Water pollution: Major Pollutants of Water and their effects, Physical, chemical and biological characteristics of water, Water Quality standards Air Pollution: Primary and Secondary Pollutants, Impact of air pollutants on human, vegetation and environment, Ambient Air Quality Standards, meteorological factors affecting air pollution.	11
2	Solid waste: Classification and sources of Solid Waste, Characteristics of Solid Waste. Land Pollution: Effects of urbanization on land degradation, Impact of Modern Agriculture on Soil, pesticide pollution, Effect on Environment. Noise Pollution: Sources of Noise, Effects of Noise, measurement of noise, Equivalent sound pressure level, Control measures.	11

	Environmental impact assessment, Need for EIA	
3	Screening –scoping - setting – analysis – mitigation, Impact identification and prediction: Matrices – Networks – Checklists –Cost benefit analysis, Analysis of alternatives, Description of the affected environment – environmental indices, Short EIA and comprehensive EIA, Case studies of EIA, Impact analysis	13
4	EIA Report preparation, Environmental Management Plan - preparation, Implementation and review – Mitigation and Rehabilitation Plans, Public Consultation, Post Monitoring and Auditing	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the environmental legislation in India and to understand various types of environmental pollutants affecting air and water	K2
CO2	Summarize the effects of solid waste and noise pollution on the environment.	K2
CO3	Outline the steps involved in EIA and to study how to conduct impact analysis of various pollutants.	K2
CO4	Develop EIA report and to document EIA and to conduct environmental auditing.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3					
CO2	3					3	3					
CO3	3	3				3	3					
CO4	3	3			2	3	3					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Impact Assessment	Canter, L.W	McGraw Hill, New York	1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Engineering	Peavy H S, Rowe, D.R. Tchobanaglou	Mc Graw Hill Education	
2	Air Pollution	Rao M.N. & Rao H.	Tata McGraw-Hill	
3	Environmental Impact Assessment – Practical solutions to recurrent problems	Lawrence, D.P.	Wiley-Interscience	2003
4	Environmental Impact Assessment: A Practical Guide	Marriot, Betty	Mc Graw Hill Education	1997
5	Source book on EIA		World Bank	
6	Environmental Pollution Control Engineering	C S Rao	New Age International Publishers	4 th edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=aBkeecCKV64 , https://www.youtube.com/watch?v=rpfdYoZ-Z5I , https://youtu.be/5hKjurPjzwI
2	https://www.youtube.com/watch?v=k0ktJRoRcOA .
3	https://www.youtube.com/watch?v=_iLdyhgFv1U , https://www.youtube.com/watch?v=DbZq9dlx6dY&t=1381s
4	http://tjsec.digimat.in/nptel/courses/video/124107160/L57.html

SEMESTER S6

PROCESS SIMULATION LAB

Course Code	PCCHL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCCHT205, PCCHT302, PBCHT304, PCCHT402, PCCHT403, PCCHT401, PCCHT501, PCCHT503, PBCHT504, PCCHT601, PBCHT604	Course Type	Lab

Course Objectives: Process Simulation Lab prepares students for real-world challenges, fosters creativity, and enhances their problem-solving abilities. Fundamental concepts of mathematical models in chemical processes; simulation of chemical units such as flash drum, heat exchanger, continuous stirred tank reactor, plug flow reactor, distillation column, absorption column and so on; and sensitivity analysis are covered in detail.

1. To equip students to conceptualize and develop flow sheet for chemical engineering process.
2. To introduce different commercial softwares to simulate the chemical processes.
3. To integrate and apply knowledge from prior courses to solve open-ended problems.
4. To interpret the results of the simulation problems.

Experiments can be done using Excel, MATLAB/ Scilab, UniSim Design/ DWSIM or any other chemical process simulation software.

Expt. No.	Experiments
1	Equations of State: Solve various equations of state and study the effect of pressure and temperature on molar volume.
2	Property Estimation: Calculation of physical and thermodynamic properties of compounds

3	Phase equilibrium: Vapour- Liquid equilibrium for binary mixtures
4	Chemical Reaction Equilibrium: Study of various possibilities of finding the product composition in an equilibrium reaction
5	Mass Balances with Recycle Streams: Study of material balance of chemical processes involving recycle of material streams
6	Compression System: Steady-state simulation of two stage compression system
7	Simulation of Reactor: Steady state simulation of conversion, and equilibrium reactors
8	Simulation of Reactor: Steady state simulation of CSTR
9	Simulation of Reactor: Steady state simulation of plug flow reactor
10	Simulation of Heat Exchanger: Steady state simulation of heat exchanger
11	Simulation of Flash operation: Steady State simulation of flash drum for binary mixture
12	Simulation of Mass Transfer Equipment: Steady-state simulation of shortcut and rigorous distillation column
13	Simulation of Mass Transfer Equipment: Steady-state simulation of absorption column
14	Simulation of Mass Transfer Equipment: Steady-state simulation of extraction column
15	Simulation of Refrigeration system: Steady-state simulation of refrigeration gas plant
16	Simulation of Chemical Process: Steady-state simulation of chemical process involving unit process and/or unit operations.
17	Simulation of Chemical Process: Steady-state simulation of chemical process involving unit process and/or unit operations.
18	Dynamic Simulation: Dynamic simulation of heat exchanger

19	Dynamic Simulation: Dynamic simulation of distillation column
20	Dynamic Simulation: Dynamic simulation of absorption column

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Plan the numerical techniques to solve the engineering problems.	K3
CO2	Select an appropriate property package, operation or a group of operations to simulate a unit operation, a unit process or part of the process plant.	K3
CO3	Predict the physical and thermodynamic properties of a compound/mixtures.	K6
CO4	Analyse and simulate various types of unit operations and unit processes there by simulating an entire plant using process simulators.	K4
CO5	Analyze the dynamic simulation of an operation or a small portion of a process plant to predict the variation of operating parameters on a servo or regulator problem of process control.	K4
CO6	Interpret and make use of the results from commercial process simulation software.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3				2			3
CO2	3	3	3	3	3				2			3
CO3	3	3	3	3	3				2			3
CO4	3	3	3	3	3				2			3
CO5	3	3	3	3	3				2			3
CO6	3	3	3	3	3				2	3		3

1: Slight (Low) 2: Moderate (Medium), 3: Substantial (High), -:No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Chemical Engineering Computing	Bruce.A.Finlayson	Wiley Interscience	
2	Hysys: An introduction to Chemical Engineering Simulation	Mohd. Kamaruddin Abd Hamid.		
3	ProcessPlantSimulation	B.V. Babu	OxfordUniversity Press	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Modelling and Computer Simulation	Amiya K. Jana	PHI Learning Pvt. Ltd.	Second edition, 2011
2	Chemical Engineering Process Simulation	Nishanth G. Chemmanurappil, Chien Hwa Chon, Denny Ng Kok Sum, Rafil Elyas, Cheng-Liang Chen, I Lung Chien, Hao-Yeh Lee, Rene D Elms	Elsevier	First Edition, 2017
3	Chemical Engineering Process Simulation	Editor: Dominic C.Y. Foo	Elsevier	Second Edition, 2022
4	Computational techniques for process simulations and analysis using MATLAB	Niket S. Kaisare	CRC Press	2018

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

***Final Marks Averaging:** The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.*

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

CHEMICAL ENGINEERING

SEMESTER S7
PETROCHEMICAL TECHNOLOGY

Course Code	PECHT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To understand the classification, sources and uses of petrochemical products.
2. It also describes the industrially relevant petrochemical production methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Petrochemical Industry in India, Structure of petrochemical complexes, Feed stock for petrochemicals, profile of petrochemicals and their end products. Introduction to polymers, Elastomers, Polyurethanes, Synthetic fibres. Methane and synthesis gas derivatives- Production of synthesis gas, Methanol, Formaldehyde, Acetic acid, Acetylene.	9
2	Steam cracking for production of Olefins. Ethylene and ethylene derivatives- Production and application of Ethylene oxide, Mono ethylene glycol, Vinyl chloride, Vinyl acetate, Acetaldehyde, Ethanol. (Industrial relevant process only).	9
3	Propylene and Propylene derivatives- Sources of Propylene, Production and application of Propylene oxide, Propylene glycol, Isopropyl alcohol, Acetone, Acrylonitrile, Cumene, Acrylic acid. (Industrial relevant process only). C4 and C5 compounds- Production and application of Butadiene, Isoprene. (Industrial relevant process only).	9

4	Aromatic production- Reforming, Pyrolysis gasoline as aromatic feed stock, Aromatic separation from reformat and Pyrolysis gasoline. BTX Derivatives- Ethyl benzene and styrene, Linear alkyl benzene, Maleic unhydride, Anilline, Bisphenol-A, Phenol.	9
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the structure of petrochemical complexes and their feed stocks.	K2
CO2	Summarize the production and applications of ethylene derived petrochemicals.	K2
CO3	Explain the industrially relevant process for the manufacture of Propylene derived petrochemicals.	K2
CO4	Interpret the production and applications BTX derived petrochemicals.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						2					
CO2	3						2					
CO3	3						2					
CO4	3						2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Petrochemical Process technology	I D Mall	Lakshmi Publications Pvt. Limited	2017
2	Advanced Petrochemicals	Dr.G N Sarkar	Khanna Publisher	1998
3	Petrochemicals	Anjana Sreevastav	S.N Kataria & Sons	1 st Edition 2010
4	Outlines of Chemical Technology	Dryden C. E.	Shree hari publications	3 rd Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Petrochemical Engineering	Saikat Maitra, O.P Gupta	Khanna Publishers	1 st Edition 2018
2	Petroleum Refinery Technology and Economics	James H.Garry Glenn E. Handwerk Mark J.Kaiser	CRC Press, Taylor and Francis group	5 th Edition
3	Fundamentals of Petroleum and Petrochemical Engineering	Uttam Ray Chaudhuri	CRC Press	1 st Edition 2020
4	Modern Petrochemical Technology: Methods, manufacture and applications.	Sati Kulprathipanja James E Rekoske Daniel Wei	Wiley	2 nd Edition 2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_ch64/preview
2	https://onlinecourses.nptel.ac.in/noc23_ch64/preview

SEMESTER 7

COMPOSITE MATERIALS

Course Code	PECHT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Min
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To provide fundamental knowledge of various types of composite materials, their properties and applications.
2. To impart the manufacturing techniques and testing of different composite materials, there by helping select appropriate materials for composite synthesis.
3. To facilitate understanding of hybrid composites, green composites and nanocomposites, which are the current composite materials of research interest

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to composite materials, constituents and classification: Constituents of composite materials: Reinforcements, Matrix: glass, carbon, Kevlar, boron, asbestos, steel, natural fibres and whiskers-reinforcement fibres, Coupling agents, coatings & fillers, Characteristic features and applications. Classification of composite materials based on matrices and reinforcements- Polymer Matrix Composites - Metal Matrix Composites- Ceramic Matrix Composites-Plastic and Rubber Matrix Composites- Smart Composites-Carbon-Carbon Composites- Intermetallic Composites- Laminated Composites-Fibre reinforced composites-Laminar Composites - Particulate Composites	9
2	Manufacturing/Fabrication Techniques: Traditional and novel approaches process fundamentals. Polymer Matrix Composites- Fabrication of Fibres, Plastic Fibre Forms, Pre-pregs, Moulding Compounds- Processes, Lay-Ups, Filament Winding, Pultrusion, and Recycling. Ceramic Matrix Composites:	9

	Hot-Pressing, Infiltration and Impregnation, In Situ Chemical reaction Techniques, CVD, CVI, Sol-gel. Metal Matrix Composites- Liquid Infiltration, Casting, Solid State Processes, Diffusion Bonding and In Situ Technique. Fibre Reinforced Thermoplastics (FRTP) preparation- brief description of coating process- melt compounding process and dry blending process-injection moulding, rotational moulding and cold forming of reinforced thermoplastics.	
3	Composite Mechanics Theory: Basic criterion to be adopted in the selection of matrix and reinforcement-mechanics of composite materials-micromechanics and macro mechanics-mechanism of load transfer-minimum and critical fibre content-critical fibre length-law of mixture rule-unidirectional and fibrous composites-effects of fibre orientation on stiffness and strength-bidirectional and random fibre composites-concepts of unit cell-stress analysis of unit cells-toughness of fibrous composites, microscopic stress-strain curves	11
4	Testing of composite materials: Testing of composite materials and products for quality control- Brief outlines of testing of glass fibre, testing of resins-testing of products. Failure criteria, Laminate Strength, Stress Concentrations. Key damage mode for composites and composite damage tolerance capabilities. Typical in-service damage types for composites. Non-destructive inspection techniques for detecting damage in composites. Basic types of composite repair and their benefits. Advanced composite materials: Synthesis, Properties and Applications of Green Composites, Nanocomposites and Hybrid Composites.	7

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Compare the properties, applications and different types of composite materials, their matrices, reinforcements and additives.	K2
CO2	Explain the manufacturing techniques of Polymer Matrix, Ceramic Matrix, Metal Matrix and Fibre Reinforced Composite	K2
CO3	Summarize the theory, mechanism and testing procedures of composite materials for the selection of matrix and reinforcement materials.	K2
CO4	Illustrate the properties and applications of advanced composites	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	2					2
CO2	3					2	2					2
CO3	3					2	2					2
CO4	3					2	2					2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of composites.	G.Lubin,	Springer Science & Business Media.	2013
2	Handbook of composite reinforcements	Lee, Stuart M	John Wiley & Sons	1996
2	Advances in composite materials	G. Piatti	Applied Science Publishers Ltd., London	1978
3	An Introduction to Composite Materials	Clyne, Trevor William, and Derek Hull.	Cambridge University Press, Cambridge.	2019
4	SPI handbook of Technology and Engineering of reinforced Plastics/Composites	Mohr, J. Gilbert, and Samuel S. Oleesky	Von Nostrand, New York.	1973
5	Mechanics and analysis of composite materials.	Vasiliev, Valery V., and Evgeny V. Morozov.	Elsevier	2001

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nano: The Essentials	T. Pradeep	McGraw-Hill (India) Pvt Limited	2008
2	Nanocomposite science and technology	P. M. Ajayan, L. S. Schadler, P. V. Braun	Wiley-VCH	2005
3	Ceramic matrix composites	K.K. Chawla	Chapman & Hall, London.	1993
4	Composite Materials	K.K. Chawla	Springer-Verlag, New York	1987
5	Polymer Engineering Composites.	M.O.W. Richardson	Applied Science Publishers, London	1977
6	Handbook of Fillers and Reinforcement for Plastics	Katz.H.S. & J.V. Milewski	Von Nostrand, New York.	1978

SEMESTER S7

SAFETY ENGINEERING OF PROCESS PLANTS

Course Code	PECHT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Impart the basic concepts of safety in Process Industries.
2. Develop an understanding about safety practices in industries.
3. Develop an understanding about hazard identification, emergency planning and hazard evaluation.
4. Develop an understanding about safety practices in industries and emergency planning.
5. Incorporate the concepts of Inherent Safety and Process Intensification and models for Fire, Explosion and Toxic gas dispersion.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept and importance of industrial safety. Fundamental safety tenets. Review of Industrial Accidents- Major Chemical Industry Accidents, Bhopal, Flixborough, SEVESO. Cost of accidents. Key safe practices in chemical industry for accident prevention programme. Material safety data sheet. Fire & Safety: Classification of Fire- Pool fire, Jet fire, Flash fire, Explosion-UVCE, BLEVE, Dust explosion, Deflagration, Detonation. Toxic release, Runaway Reaction. Fire pyramid.	9
2	Types of fire extinguishers and its handling. Fixed fire protection systems. Flammability diagram-construction, application. Hazards in a process industry: Hazard triangle, Classification, consequences of chemical hazards, physical hazards, electrical hazards, mechanical hazards and environmental	9

	hazards. Prevention techniques for hazards. Hazardous area classification. Safety in transportation of hazardous chemicals by road-HAZCHEM CODE, TREM CARD, Relief system and Detectors. T.L.V, STEL, TLV-C, IDLH, UFL, LFL.	
3	Identification of Hazards- Inventory analysis, Dow Fire and Explosion Index, Mond Fire, Explosion and Toxicity Index. Emergency planning-onsite and offsite emergency planning, Mock drill. Technique for Hazard Evaluation- Hazard and Operability Study, Preliminary Hazard Analysis, What if Analysis, Fault Tree Analysis, Event Tree Analysis, Failure Modes and Effects Analysis, Layer of protection analysis, Examples.	9
4	Consequence Analysis and Quantitative Risk Assessment- Consequence of Chemical accidents. Models for Fire, Explosion and Toxic gas dispersion. Individual and Societal Risk, FN curves, Probit function. Elements of Emergency Planning Inherent Safety and Process Intensification- The concept of Inherent Safety, Tools for Inherent Process Safety. Inherent Safety Indices. The concept of Process Intensification. Occupational Health and Safety Administration, Safety provisions in Factories Act.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate the basic concepts of safety in Process Industries	K2
CO2	Explain about safety practices in industries.	K2
CO3	Interpret about hazard identification and analysis	K2
CO4	Illustrate the concepts of Inherent Safety and Process Intensification and models for Fire, Explosion and Toxic gas dispersion.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2			3						
CO2	3		2			3						
CO3	3		2			3						
CO4	3		2			3						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety in Chemical Plants/Industry and Its Management	B. K. Bhaskara Rao, Er. R. K. Jain, Vineet Kumar	Khanna Publishers	1 st edition, 2010
2	Industrial Safety, Health and Environment Management Systems	R. K. Jain & Sunil S Rao	Khanna Publishers	4 th Edition, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Encyclopaedia of Occupational Health & Safety		International labour Office, Geneva,	2012
2	Loss Prevention in Process Industries	Frank P. Lees	Butterworth-Heinemann	Vol.1,2&3, 2 nd Edn., 1996
3	Guidelines for Hazard Evaluation Procedure		Centre for Chemical Process Safety. AICHE	1992
4	Methodologies in Hazard Identification and assessment	K.V. Raghavan and A. A. Khan	Manual by CLRI	1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/110/105/110105094/
2	http://www.digimat.in/nptel/courses/video/103107156/L42.html
3	https://archive.nptel.ac.in/courses/103/107/103107156/

SEMESTER S7

PROCESS AUTOMATION

Course Code	PECHT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip students with the skills to map and analyze process flows, define input and output parameters, and implement automation
2. Understand automatic control principles, enabling them to design and optimize control systems effectively.
3. Apply advanced control strategies and integrate them into automated systems to enhance process reliability and performance in industrial applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to automation, Architecture of Industrial Automation Systems, Measurement Systems Characteristics, Level of automation, Automated flow lines and transfer mechanisms, Data Acquisition Management, Introduction to PLC, PLC functions.	8
2	Process Flows- process mapping, flows, input and output parameters, Automation steps in process industries, PDCA (Plan, Do, Check Act) automation cycle. Introduction to Automatic Control: P-I-D Control, PID Control Tuning, Feed forward Ratio Control, Sequence Control.	8
3	Flow Control Valves, Hydraulic Control Systems, Pneumatic Control Systems, Higher Level Automation Systems. Introduction to computer-based industrial automation- Direct Digital Control (DDC), Distributed Control System (DCS) and supervisory control and data acquisition (SCADA) based architectures. SCADA for process	9

	industries, Process Safety Automation: Levels of process safety and control.	
4	Robotic Process Automation (RPA): Definition of RPA, RPA Applications, RPA Tools and Platforms. Business Process Management (BPM): Definition of BPM, BPM Tools and Platforms, BPM vs RPA. Artificial Intelligence (AI) and Machine Learning (ML): Definition of AI and ML, AI and ML Applications in Process Automation, AI and ML Tools and Platforms. Integration and Data Management: Integration Platforms, Data Management Tools, Data Analysis and Reporting Tools.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a comprehensive understanding and practical skills in industrial automation systems	K3
CO2	Apply various automatic control techniques to implement process automation.	K3
CO3	Develop knowledge and skills to design, implement, and manage various automation systems and control mechanisms	K3
CO4	Apply advanced tools such as Robotic Process Automation (RPA), Business Process Management (BPM), Artificial Intelligence (AI), and Machine Learning (ML) for effective process automation.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							
CO2	3	3	3		3							
CO3	3	3	3		3							
CO4	3	3	3		3							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Instrumentation, Control and Automation	S. Mukhopadhyay, S. Sen and A. K. Deb	Jaico Publishing House	2013
2	Chemical Process Control, An Introduction to Theory and Practice.	George Stephanopoulos,	Prentice Hall India	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Control Instrumentation Technology.	Curtis D. Johnson	Pearson New International	8th Edition 2013
2	Distributed Control Systems	Lukas M.P	Van Nostrand Reinhold Co., New York	1986
3	SMEC Industrial Automation tools and Techniques			
4	Handbook of Artificial Intelligence and Robotic Process Automation Policy and Government Applications	Naqvi, J. M. Munoz,	Anthem Press	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105063
2	https://nptel.ac.in/courses/112101098

SEMESTER S7

ADVANCED HEAT AND MASS TRANSFER

Course Code	PECHT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The analysis of the phenomena of heat and mass transfer is important in engineering design applications. Heat and mass transfer processes are modelled by similar mathematical equations in the case of diffusion and convection. It is thus more efficient to consider them jointly.
2. Besides, heat and mass transfer must be jointly considered in some cases like humidification and drying. This course also explains how to solve engineering problems involving conduction, convection, diffusion mass transfer, interphase mass transfer, simultaneous heat and mass transfer.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Heat conduction with heat generation: Plane wall and cylinder with uniform heat generation, applications. Two-dimensional steady state conduction. Transient conduction lumped and distributed systems, derivation for lumped capacity model. Heat Transfer through extended surfaces: One dimensional conduction and convection through rectangular fin very long fin, fin with insulated end, fin of finite length with heat loss from end.	9
2	Convection: Forced and free convection, dimensionless numbers in forced and free convection, expressions for heat transfer coefficient (no derivation) free convection and forced convection through tubes, Boundary layer over flat plate hydrodynamic and thermal boundary layer, Prandtl number and	9

	relative thicknesses of hydrodynamic and thermal boundary layer, boundary layer thickness, approximate and exact analysis of thermal boundary layer.	
3	Diffusional mass transfer: Diffusion in multicomponent mixtures, molecular diffusion in liquids, diffusivity in porous solids, Knudsen diffusion, unsteady state diffusion basic diffusion equation. Convective transport and interphase mass transfer: Interphase mass transfer, two resistance theory, local and overall mass transfer coefficients, gas phase and liquid phase-controlled systems	9
4	Simultaneous diffusion and chemical reaction: homogeneous and heterogeneous reactions, simultaneous heat and mass transfer: condensation of hot vapour on solid surface in presence of non-condensable gas, steady state binary thermal diffusion in a two-bulb apparatus.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO 1	Explain various modes of heat transfer in multi-dimensions.	K2
CO 2	Identify engineering problems involving conduction and convection heat transfer.	K3
CO 3	Develop boundary layer equations for various cases of heat transfer.	K3
CO 4	Solve mass transfer cases involving multi-component diffusion and interphase mass transport.	K3
CO 5	Solve problems in simultaneous diffusion and chemical reaction.	K3
CO 6	Apply the knowledge of heat and mass transfer to design problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3	3								
CO2	3		3	3	2							
CO3	3		3	3	2							
CO4	3		3	3	2							
CO5	3		3	3	2							
CO6	3		3	3	2							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Momentum, Heat and Mass Transfer	J.R. Welty, C.E. Wicks, R.E. Wilson, G.L. Rorrer	Wiley Publication	5 th Edn

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat and Mass Transfer	P.K. Nag	McGraw Hill	3 rd Edn, 2011
2	Heat and Mass Transfer: Fundamentals and Application	YunusA.Cengel and Afshin J.	McGraw Hill	5 th Edn 2017
3	Fundamental of Heat and Mass Transfer	Incropera and Dewitt,	Wiley Publication	6 th Edn

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/108/112108149/
2	https://archive.nptel.ac.in/courses/103/105/103105052/

SEMESTER S7

FOOD PROCESSING AND TECHNOLOGY

Course Code	PECHT747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart the basic concepts of food processing.
2. To gain knowledge of the various preservation and packing methods.
3. To provide students with an understanding of the production of different food products.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction - general aspects of food industry - world food needs and Indian situation, constituents of food - quality and nutritive aspects, food additives - standards – functional characteristics of chemical additives, food deterioration - deteriorative factors and their control, preliminary processing methods – cleaning – dry cleaning methods and wet cleaning methods, sorting, grading, food conversion techniques and equipments used.	9
2	Food Preservation – objectives, importance, preservation by heat and cold - dehydration - concentration - frying - irradiation - micro wave heating - sterilization and pasteurisation - fermentation and pickling, Food packing – aims, types of packaging materials, packaging methods.	9
3	Production and processing of food products - Cereal grains - Rice, Wheat, Bread - Pulses - vegetables - spices – microbial decontamination of spices, harvesting and processing of black pepper, cardamom - fats and oils.	9
4	Beverages - carbonated non alcoholic beverages or soft drinks processing, alcoholic beverages –processing of beer, wine, brandy, arrack, rum, vodka, gin and non-carbonated non alcoholic beverages such as coffee tea and cocoa processing.	9

	Dairy products – milk processing, butter processing, manufacture of ice cream, cheese preparation, meat and meat products – causes of meat deterioration, materials used in meat processing, meat preservation, processing of poultry and fish preservation methods - treatment and disposal of food processing wastes.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the importance of nutritive aspects, food additives and standards and the food processing methods.	K2
CO2	Select suitable food preservation techniques and packing methods.	K3
CO3	Explain the production and utilization of food products from dairy, meat, poultry and fish industries	K2
CO4	Explain treatment and disposal of food processing wastes	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

Text Books												
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				3						
CO2	3	2				3						
CO3	3	2				3						
CO4	3	2				3						

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103107088
2	https://nptel.ac.in/courses/103107088
3	https://nptel.ac.in/courses/103107088

SEMESTER S7

PROCESS INTEGRATION

Course Code	PECHT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	PCCHT402, PCCHT403, PCCHT501, PCCHT503	Course Type	Theory

Course Objectives:

1. Impart the fundamental principles and concepts of process integration and its importance in enhancing efficiency and sustainability in industrial processes.
2. Apply energy targeting methods to determine the minimum energy requirements and design heat exchanger networks by applying pinch analysis.
3. Develop skills in design of mass exchanger networks to achieve efficient mass transfer and resource utilization within industrial systems.
4. Apply comprehensive process integration techniques to Integrate reactor systems and distillation columns.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Process Integration: Definition of Process Integration, Areas of application and techniques available for Process Integration, Onion model of chemical process design, Role of thermodynamic laws. Energy targeting methods of Heat Exchanger Networks: Composite curve method, Problem table algorithm, Grand composite curve.	9
2	Targeting of Heat Exchanger Network: Number of units targeting, Area targeting, Cost targeting, Number of shells targeting. Heat Exchanger Network Design: The pinch design method, Grid diagram, Stream splitting design for single pinch networks.	9
3	Mass Exchanger Network Synthesis: Mass Exchanger Network, Minimum Mass Separating Agents (MSA), Mass exchange networks for minimum external MSA. Water system design: Water use, Targeting maximum water reuse for	9

	single contaminants.	
4	Integration of Reactor systems: Choice of Idealized reactor model and reactor performance. Reactor configurations: Temperature Control, Choice of Reactors. Heat Integration characteristics of reactors, appropriate placement of reactors. Integration of Distillation systems: Distillation sequencing, Heat Integration characteristics of a Distillation column, appropriate placement of distillation column.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Exam	Evaluate	Analyse	Total
5	15	10	10	40

Assignment: 20 Marks

Students should identify a plant/section of a plant/unit and design a heat exchanger network problem or mass exchange network problem and simulate the problem with available software like HINT.

Criteria for evaluation:

Level: Analyze

1. Problem Definition (2 marks)

Choose an industrial plant or a section of a plant. Gather detailed information about the chosen plant or unit, including process flow diagrams, operating conditions, and energy/mass balances.

2. Problem Analysis (5 marks)

Define the goal of the project. Collect necessary data such as temperatures, flow rates, specific heats etc.

3. Identifying Software (3 marks)

Identify the different software tools required to implement the work and learn its usage.

Level: Evaluate

Design network using software

Identify the streams and their properties. (3 marks)

Analyze the system and find the pinch point. (2 marks)

Design the network. (3 marks)

Report the work (K4 - 2 marks)

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and Optimize the heating and cooling utility requirement.	K3
CO2	Analyze and Design heat exchanger network using pinch technology.	K4
CO3	Choose processes for minimization of raw material and waste generation.	K3
CO4	Select suitable reactors for energy efficient operation.	K3
CO5	Evaluate energy optimal integration solutions for distillation columns.	K5
CO6	Select and design an integration problem using available simulation software.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3							3
CO2	3	3		2	3							3
CO3	3	3	3	2	3							3
CO4	3	3		2	3							3
CO5	3	3		2	3							3
CO6	3	3	3	3	3	3	3			3		3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Design and Integration	Robin Smith	John Wiley and Sons. Ltd., New Delhi.	2nd Edition, 2016
2	Heat Exchanger Network Synthesis	Uday V. Shenoy	Gulf Publishing Co, USA,	1st Edition, 1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product & Process Design Principles	Warren D. Seider, J. D. Seader and Daniel R. Lewin	Wiley Publication	3rd Edition, 2015
2	Pinch Analysis and Process Integration - A user guide on process integration for efficient use of energy	Kemp I. C	Butterworth – Heinemann	2nd Edition, 2006
3	Process Integration and Intensification: Saving Energy, Water and Resources	Jirí J. Klemeš and Petar S. Varbanov, Sharifah R. W. W. Alwi and Zainuddin A. Manan	De Gruyter	1st Edition, 2014
4	Process Integration	Mahmoud M. El – Hawalgi	Academic Press	1st Edition, 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107093/
2	https://archive.nptel.ac.in/courses/103/107/103107093/
3	https://archive.nptel.ac.in/courses/103/107/103107093/
4	https://archive.nptel.ac.in/courses/103/107/103107093/

SEMESTER 7

DRUGS AND PHARMACEUTICALS TECHNOLOGY

Course Code	PECHT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To give an insight into drug discovery and development of drugs, organic therapeutic agents uses and economics, understand different unit processes and its application, manufacturing principles and product formulation, conventional drug development process and regulatory procedures and production of selected biopharmaceutical products.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Pharmaceutical Industry, Drug discovery and Development of Drugs, Organic Therapeutic agents uses and Economics. Chemical Conversion Processes - Alkylation, Carboxylation, Condensation, Cyclisation, Dehydration, Esterification, Halogenation Oxidation, Sulfonation, Complex Chemical Conversions and Fermentation. Outlines of preparation, properties, uses and testing of the following pharmaceuticals and fine chemicals, sulfacetamide, paracetamol, methyl orange, riboflavin, nicotinamide, procaine hydrochloride, para-amino salicylic acid, isonicotinic acid hydrazide, aspirin, penicillin, calcium gluconate, ferric ammonium citrate	9
2	Compressed Tablets, Wet Granulation, Dry Granulation or Slugging, Direct Compression, Tablet Presses, Formulation, Introduction to tablet coating: rationale, advantages etc. • Preformulation considerations for tablet coating • Types of coating • Quality control of coated and uncoated tablets Introduction to capsule dosage form: rationale, advantages etc.	7

	Preformulation considerations for capsule dosage form. Hard gelatin capsules: formulation considerations, capsule manufacture equipments, quality control tests, packaging, Soft gelatin capsules: formulation considerations, capsule filling equipments, quality control tests, packaging, Microencapsulation, Fabrication techniques, Evaluation	
3	Monophasics (Oral and Topicals)(solution, syrups, elixirs, linctus, nasal drops, ear drops, etc.),Preformulation, Formulation, Quality Control Biphasic - Suspensions • Preformulation • Principles and Stabilization techniques • FormulationDevelopment • Evaluation • Large scale manufacture and packaging with focus on equipment Biphasic - Emulsions • Preformulation • Theories of emulsions • Formulation • Evaluationincluding stress testing • Large scale manufacture and packaging with focus on equipment.	7
4	Ointments-Preformulation, Formulation ,Evaluation , Large scale manufacture and packagingwith focus one equipment. Creams - Preformulation, Formulation, Evaluation, Large scalemanufacture and packaging with focus on equipment Gels -Preformulation, Formulation, Evaluation, Large scale manufacture and packaging with focus onequipment. Sterilization: Introduction, risk factor, methods of sterilization, heat (dry and moist), heating withbactericide, filtration, gaseous sterilization and radiation sterilization, suitable example to bediscussed and sterilization testing. Analytical Methods and Tests for various Drugs & Pharmaceuticals-principle, instrumentation andapplications of UV/VIS and IR spectroscopy, X-ray diffraction analysis, Fourier transformspectroscopy, chromatography principle and its types, fluorimetry, polarimetry.	13

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain pre-formulation, formulation, and unit operation involved in the manufacturing of tablets	K2
CO2	Explain the coating polymers, technology, and equipment used for coating tablets and describe microencapsulation techniques	K3
CO3	Outline formulations for hard and soft gelatin capsules, machinery used for filling hard gelatin capsules, process for soft gelatin capsules manufacturing, evaluation of capsules	K2
CO4	Summarize pre-formulation, formulation, evaluation, large-scale manufacturing, and packaging of oral controlled release and sustained release products	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3			3			3
CO2	3					3			3			3
CO3	3					3			3			3
CO4	3					3			3			3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pharmaceutical Manufacturing Handbook	Shayne Cox Gad.	John Wiley and Sons, Inc	2008
2	Pharmacokinetics and Pharmacodynamics of biotech drugs,	Bernd Meibohm.	Wiley-VCH,	2006
3	Bentleys Text book of Pharmaceutics	Rawlines, E.A.	Bailliere Tindall, London,	III Edition, 1977.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pharmaceutical Production Facilities: Design & Applications	Graham C.Cole,	Ellis Horwood	1 st Edition 1990
2	Theory & Practice Of Industrial Pharmacy	Leon Lachman ,HerbertA.Lieberman& Joseph Kanig	Lea &Febiger, Philadelphia	3 rd edition, 1987
3	ICH Guidelines 7 Coated Pharmaceutical Dosage Forms	K. H. Bauer	CRC Press, BocaRaton. Med Pharm	
4	Process Systems Engineering for Pharmaceutical Manufacturing Volume 41	Edited by Ravendra Singh , Zhihong Yuan Tsinghua	Elsevier	2018
5	Pharmaceutical Suspensions from Formulation Development to Manufacturing	, Edited by Alok K. Kulshreshtha, Onkar N. Singh, G.Michael Wall	Springer	2010

SEMESTER S7

NON-CONVENTIONAL PETROLEUM RESOURCES

Course Code	PECHT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Identify the formation and geographic distribution of non-conventional hydrocarbon resources.
2. To understand and characterize the source and reservoir rocks.
3. To study the basic principles of conversion of coal and gas to oil.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Non-Conventional Oil : Introduction of Non-Conventional Oil, geology of heavy oil, extra heavy oil, tar, sand and bituminous oil shale, their origin and occurrence worldwide, resources, reservoir characteristics, new production technologies.	8
2	Shale Gas: Introduction and present status of shale gas. Formation and properties of shale gas. Drilling and completion of shale gas. Uses and applications of shale gas. Environmental issues in shale gas exploration. Future prospects of shale gas.	8
3	Coal Bed Methane (CBM): Formation and properties of coal bed methane. Thermodynamics of coal bed methane. Exploration and evaluation of CBM. Hydro-fracturing of coal seam. Production installation and surface facilities. Well operations and production equipment.	8
4	Gas Hydrates and Conversions to Oil: Introduction and present status of gas hydrates. Formation and properties of gas hydrates. Drilling and completion of gas hydrates wells. Gas hydrates accumulation in porous media. Gas extraction from gas hydrates. Uses and applications of gas hydrates. Coal and Gas Conversion to Oil: Introduction, classification and principles, pyrolysis, theoretical aspect of processes involved in conversion. Environmental and Economic considerations of non-conventional oil and gas	12

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain an overview of origin and occurrence of non conventional energy resources.	K2
CO2	Summarize concepts related to exploration, occurrence and origin of Shale Gas Reservoirs.	K2
CO3	Explain the concepts related to formation and properties of Coal Bed Methane.	K2
CO4	Summarize and apply the concepts related to formation and properties of gas hydrates.	K2
C05	Illustrate different processes for the conversion of coal and gas to oil.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				3	3					
CO2	3	2				3	3					
CO3	3	2				3	3					
CO4	3	2				3	3					
CO5	3	2	2			3						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Natural Gas Hydrates: A guide for engineers	Carroll John	Gulf Professional Publications	2014
2	Practical Heavy Oil Recovery	Farouq Ali, S M, Jones J A and Meldau R F	SPE	1997
3	Producing Liquid Fuels from Coal, Prospects and Policy Issues	James T. Bartis, Frank Camm, David S. Ortiz	NETL	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Coal BedMethane: From Prospects toPipeline	Pramod Thakur, Steve Schatzel and KashyAminian	Elsevier	2014
2	Unconventional Gas Reservoirs: Evaluation, Appraisal andDevelopmentIndustries	Rafiqul Islam, M	Gulf Professional Publishing	2014.
3	Petroleum Engineering Handbook, Volume VI	Warner, H.R	Richardson:Society of Petroleum Engineers.	2007
4	Unconventional Petroleum geology	Caineng Zou	Elsevier Science	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105105226
2	https://nptel.ac.in/courses/103103140
3	https://nptel.ac.in/courses/103103140
4	https://nptel.ac.in/courses/103103140

SEMESTER S7

NANOMATERIALS AND NANOTECHNOLOGY

Course Code	PECHT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Equip students with fundamental knowledge on the physics, chemistry, synthesis and characterisation of nanomaterials.
2. Enable the students with the understanding of the applications and potential of nanomaterials and nanotechnology for humankind.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Nanotechnology: History of nanotechnology, Pioneers in the field of nanotechnology. Classification of nano-materials - Zero-, one-, two- and three-dimensional nanostructured materials, Electromagnetic spectrum, particle size and its significance. Physics of nanomaterials: Size effect on thermal, electrical, electronic, mechanical, optical and magnetic properties of nanomaterials, surface area and aspect ratio, band gap energy, quantum confinement effect. Chemistry of nanomaterials-Ionic properties of nanomaterials. Electronic phenomenon in nanostructures. Synthesis methods - top down and bottom up approaches. Top down approach – size reduction techniques like milling and machining. Bottom up approach - Sol-gel methods, Chemical vapour deposition, Physical vapour deposition, Laser ablation methods.	9
2	Synthesis, properties and applications of nanomaterials: Nanometals - gold and silver, different types of nano-oxides - Al_2O_3 , TiO_2 , ZnO and SiO_2 . Special nanomaterials: synthesis, properties and applications –	9

	fullerenes, graphite, graphene, carbon nano-tubes, nano wires, nano rods, nanofluids, nanoclusters, quantum dots. Nanocomposites - Matrix materials- Basics of Metal matrix, Ceramic Matrix and Polymer Matrix nanocomposites - Nano-reinforcements, nanofillers and nanoclays.	
3	Characterisation techniques: Scanning Electron Microscopy (SEM), Energy Dispersive X-ray Spectroscopy (EDS), Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM), Scanning Probe Microscopy (SPM) - Scanning Tunneling Microscopy (STM) UV-visible spectroscopy, Raman spectroscopy, Nuclear Magnetic Resonance Spectroscopy (NMR), Fourier Transform Infrared Spectroscopy (FTIR), X-Ray Diffraction (XRD), Dynamic Light Scattering (DLS), Thermogravimetric analysis (TGA), Differential Thermal Analysis (DTA), Differential Scanning Calorimetry (DSC).	9
4	Nanoelectronics: Introduction to Micro Electro Mechanical Systems (MEMS), Nano Electro Mechanical Systems (NEMS), Nanomanipulation – STM based atomic manipulations, Nanolithography, softlithography, Scanning Probe Lithography, photolithography, E-beam Lithography, Focused ion beam lithography, Dip-pen Lithography. Applications in energy and catalysis: Applications (fundamental concepts only) Nanoscale advances in energy and catalysis - Nanotechnology for sustainable energy, nanotechnology enabled renewable energy technologies. Application of nanomaterials in catalysis. Applications in biotechnology: (fundamental concepts only) - Nanomedicine, Drug delivery, Therapeutic applications, biosensors.	9

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic principles of Physics and Chemistry and synthesis methods of in Nanomaterials.	K2
CO2	Summarize the synthesis, properties and applications of nanomaterials and nanocomposites.	K2
CO3	Outline various characterisation techniques applied to nanomaterials.	K2
CO4	Explain the applications of nanotechnology in different fields of science and technology.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	2					2
CO2	3					2	2					2
CO3	3					2	2					2
CO4	3					2	2					2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Physics and Chemistry of Materials	Joel I. Gersten	Wiley	2001
2	Nano: The Essentials	T. Pradeep	McGraw-Hill (India) Pvt Limited	2008
3	Nanostructures & Nanomaterials: Synthesis, Properties & Applications	G. Cao. Y Wang	World Scientific	2011
4	Materials Characterization Techniques	S Zhang, L. Li and Ashok Kumar	CRC Press	2008
5	Nanocomposite science and technology	P. M. Ajayan, L. S. Schadler, P. V. Braun	Wiley-VCH	2005
6	Nanobiotechnology: Concepts, Applications and Perspectives	C. M. Niemeyer, C. A. Mirkin,	Wiley-VCH	2004
7	Nanotechnology and Nanoelectronics–Materials, Devices, Measurement Techniques	W. Fahrner	Springer-Verlag Berlin, Germany	2004
8	Surface Science: Foundations of Catalysis and Nanoscience	K.W. Kolasinski	John Wiley	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanomaterials: Synthesis, Properties and Applications	S. Edelstein and R. C. Cammarata	Taylor & Francis	1998
2	Physics and Chemistry of Nanostructured Materials	S. Yang and P. Shen	Taylor & Francis	2000
3	Characterization of Nanophase materials	Z L Wang (Ed.)	Wiley-VCH	2000
4	Nanofabrication, Principles, Capabilities and Limits,	Zheng Cui	Springer Science + business media, New York	2008
5	X-rays in Nanoscience - Spectroscopy, Spectromicroscopy, and Scattering Techniques	Guo, Jinghua (Ed.)	John Wiley	2010
6	Handbook of Nanoscience, Engineering and Technology,	William A. Goddard III (Ed), Donald Brenner (Ed), Sergey Edward Lyshevski (Ed), Gerald J Iafrate (Ed)	Kluwer publishers	2007
7	Bionanotechnology: Lessons from nature	David S Goodsell	John Wiley	2004
8	Textbook of Nanoscience and Nanotechnology	B S Murty, P Shankar, Baldev Raj, B B Rath, James Murday	Springer	2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/113/104/113104102/
2	https://archive.nptel.ac.in/courses/118/102/118102003/ https://archive.nptel.ac.in/courses/118/107/118107015/
3	https://archive.nptel.ac.in/courses/102/104/102104069/
4	https://archive.nptel.ac.in/courses/112/107/112107283/ https://archive.nptel.ac.in/courses/118/107/118107015/

SEMESTER S7

QUANTITATIVE RISK ASSESSMENT

Course Code	PECHT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Impart the basic concepts of safety in Process Industries.
2. Equip students with the skills of risk assessment techniques and hazard identification
3. Develop an understanding about Consequence analysis and Quantitative risk assessment

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to process safety, Process safety strategies- Inherent, Passive, Active and Procedural, Layer of protection, Safety integrity level. Hazard, Risk, Chemical hazards-Fire, Explosion, Toxic gas release. Dow fire & Explosion index, Toxicity index, Chemical explosion index, Inventory analysis. Major industrial hazards, Reasons.	9
2	Techniques for hazard Identification;- safety audit, Safety inspection, Preliminary hazard analysis, HAZOP, What if analysis, Fault tree analysis, Event tree analysis, Failure mode and effect analysis, Layer of protection analysis, Bow-tie analysis (methodology and examples).	9
3	Consequence analysis- Steps involved in the consequence analysis, Consequence of fire, explosion and toxic gas release. Source modelling, Models for fire- pool fire, Models for explosion- TNT equivalent, Models for dispersion- Gaussian gas dispersion model. Probit equation for fire, explosion and toxic gas release.	9
4	Quantitative risk assessment- Procedure for QRA, Risk, type of risk, acceptable risk-ALARP, representation of risk. Calculation of Individual risk, Societal risk, FN Curve. Risk reduction strategies. Emergency planning- Onsite and offsite emergency plan, Reliability-way to improve reliability. Tools for inherent safety, Inherent safety indices.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the importance of safety and process safety strategies	K2
CO2	Explain the various hazard analysis techniques and apply them in real life	K2
CO3	Summarize the consequence analysis techniques	K2
CO4	Analyze and apply Quantitative risk assessment techniques	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3			3					
CO2	3	3	3	3			3					
CO3	3	3	3	3			3					
CO4	3	3	3	3			3					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical process safety	Daniel A Crowl & Joseph F Louvar	Prentice Hall	2 nd edn, 1990
2	Managing risks: Systematic loss prevention for executives	V. J., Englewood, Cliffs. N. J.:	Prentice Hall	1987
3	, Process System Risk Management,	Ian T. Cameron and Reghuraman	Elsevier Academic press,	Vol 6, 2005
4	Industrial Accident Prevention”	Heinrich H.W.	McGraw-Hill Company,	1980

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety in process plant design	G.L.Wells	John Wiley & Sons.	
2	Process Safety Management,	Alexis M. Herman, Charles N. Jeffress	OSHA-3132, U.S. Department of Labor	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/107/103107156/
2	https://archive.nptel.ac.in/courses/103/107/103107156/
3	https://archive.nptel.ac.in/courses/103/107/103107156/
4	https://archive.nptel.ac.in/courses/103/107/103107156/

SEMESTER S7

NOVEL SEPARATION PROCESS

Course Code	PECHT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To study various separation techniques like membrane-based techniques and chromatographic separation techniques which are gaining importance in chemical plants.
2. To choose suitable separation process for the required separation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Characteristics and selection of separation process: Classification of separation processes, factors influencing the choice of separation process and equipment. Membrane processes: Introduction, Type and choice of membranes, Plate and frame, tubular, spiral wound and hollow fibre membrane reactors and their relative merits, Membrane filtration, Microfiltration, Nanofiltration, Ultrafiltration, Reverse Osmosis, Dialysis, Models for membrane separations.	9
2	Surfactant Based Separation Techniques: Basic principles, classifications, Surfactants at Inter phases and in bulk, Foam fractionation, Foam flotation, Adsorptive bubble separations, Ion flotation, Micro emulsion /Macro emulsions, Hydrotopes, Solvent ablation.	9
3	Adsorption systems: Interacting solutes, Adiabatic adsorbers, velocity effects. Adsorption- Desorption operations: Thermal desorption of gases, Activated carbon solute recovery, Processing liquid using thermal regeneration, Pressure swing and vacuum swing adsorption, Regeneration with purge and desorbent. Ion exchange: Basics of Ion exchange, Ion exchange resins, Binary ion exchange equilibrium, Ion movement theory,	9

	Applications, Applications without exchange: Ion exclusion, Mass transfer in ion exchange systems.	
4	Chromatographic Separations: Introduction, types of chromatography, Elution chromatography: Principles and Retention theory, Band broadening and separation efficiency, Types of chromatography, Large scale elution (cyclic/batch) chromatography, Selective adsorption of biological macromolecules, Simulated counter-current techniques, Comparison with other separation methods.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain various separation processes and the factors influencing the choice of separation techniques.	K2
CO2	Illustrate models and solutions for membrane separation processes	K2
CO3	Solve problems involving surfactant and adsorption based separation	K3
CO4	Summarize the treatment of process liquids by ion exchange process and it's applications.	K2
CO5	Explain the treatment of process liquids by Chromatographic Separations and it's applications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Separation Process Principles. New York,	Seader, J D, and Ernest J Henley	Wiley	1998
2	Separation Processes.	King C. J.	Tata McGraw–Hill Publishing Co. Ltd.,	1982
3	Basic Principles of Membrane Technology.	Marcel Mulder	Springer Publications	2 Ed.2007
4	Handbook of Separation Process Technology.	R W Rousseau	John Wiley & Sons	2009
5	Rate- Controlled Separations.	Wankat, P. C	Springer.	1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Membrane Technology in the chemical industry.	Nunes S P, Peinemann K V,	Wiley-VCH	2 nd Edition, 2006
2	Membrane Process.	Rautanbach and Albrecht R.,	John Wiley and Sons.	1989
3	“Membrane Processes in separation and Purification”,	C.respo. J G, Bodekes K W	Kluwer Academic Publications, Netherland,	1994.
4	Transport processes and Unit Operations.	Geankoplis C J	PHI, New Delhi,	4 th Edition, 2006.
5	Handbook of Separation Techniques for Chemical Engineers.	Philip Schweitzer.	Tata McGraw Hill New York.	3 rd edition, 1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105061/
2	https://archive.nptel.ac.in/courses/103/105/103105061/
3	https://nptel.ac.in/courses/103105461 , https://archive.nptel.ac.in/courses/103/105/103105061/
4	https://archive.nptel.ac.in/courses/103/105/103105061/

SEMESTER 7

ECONOMICS AND MANAGEMENT OF CHEMICAL INDUSTRIES

Course Code	PECHT757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course focuses on economic and cost analysis of engineering projects, giving insights on tools used for cost estimation, profitability analysis and taking decisions among alternatives.
2. The basic purpose of this course is to provide a sound understanding of concepts and principles of engineering economy and to develop proficiency with methods for making rational decisions regarding problems likely to be encountered in professional practice.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Equivalenc and cost comparisons: - Time value of money and equivalence, equations used in economic analysis, compound interest and continuous interest, unacost, capitalized cost, cost comparison with equal and unequal duration of service life, inflation Depreciation and taxes: Nature of depreciation , methods of determining depreciation, straight line, declining balances, double declining balance, sum of years digits, sinking fund and units of production methods	9
2	Analysis of Cost Estimation: Factors affecting investment and production costs, Estimation of capital investment, Types of capital cost estimates:order of magnitude estimate, study estimate, preliminary estimate, definitive estimate, detailed estimate,	9

	Cost Indexes: material cost indices, labour cost indices, location index Cost components in Capital investment, Techniques of cost estimates: - conference techniques, comparison techniques graphic relationship, tabular relationship, detailed item estimate, unit rate techniques, percentage of delivered equipment cost, William's six tenth factor, lang factor method, hand factor method, Chilton method, miller method, Peter's and Timmerhaus ratio factor method, Estimation of revenue, Estimation of product cost, direct and indirect production cost, variable production costs, administration expenses, items for total product cost estimates, elements of complete costs, start up costs , plant overhead costs	
3	Profitability analysis: Mathematical methods for profitability evaluation, payout time without interest, return on average investment, Return on original investment, net present value, net present value index, DCF rate of return, incremental analysis. Breakeven and minimum cost analysis: Variable cost and fixed cost, Break even analysis, economic production chart for 100% capacity, above 100% capacity and dumping, non-linear economic production chart	9
4	Accounting : Principles of accounting, accounting definition, steps of accounting, trial balance, profit and loss accounts, balance sheet , financial ratios related to balance sheet and profit and loss account, canons of ethics of engineers.	9

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of time value of money, unacost, capitalized cost etc. for comparing projects of equal and unequal duration	K3
CO2	Utilize the concept of depreciation and apply different methods for calculating depreciation in real life projects	K3
CO3	Apply different techniques and tools for cost estimation of the projects	K3
CO4	Compare different methods for calculating profitability for project selection and ranking	K4
CO5	Examine the concept of break even analysis for selecting / designing a project	K4
CO6	Explain the principle of accounting, prepare financial statements and analyze the financial status of the company	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table: (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3								3	
CO2	3	3									3	
CO3	3	3	3								3	
CO4	3	3	3								3	
CO5	3	3	3								3	

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cost and Optimisation Engineering	Jelen F.C	McGraw Hill	3rd edn, 1990
2	Plant Design & Economics for Chemical Engineers	Peters & Timmerhaus	McGraw Hill	5th edn, 2017
3	Jelen's cost and optimisation Engineering	Kenneth King, Humphreys	Atlantic	3rd end, 1991

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Engineering Economics	Schweyer,	McGraw Hill	1955
2	Chemical Engineering Cost Estimation	Tyler		
3	Chemical Engineering & Cost Estimation	Aries & Newton	New York, McGraw-Hill	1955
4	Chemical Process Economics	Happel, Marcel Decker	New York : M. Dekker	1975

SEMESTER S7

PROCESS OPTIMIZATION

Course Code	PECHT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Enhance the modelling and formulation skills of practically relevant optimization problems in Chemical Engineering
2. Introduce the optimization techniques and enable to select appropriate optimization technique which is best suited to the chemical engineering problem
3. Give an insight to non-traditional optimization techniques and chemical product design/formulation and optimization

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Process optimization & overview of mathematical functions: Scope and hierarchy of optimization, Typical chemical engineering applications of optimization. Statement of an Optimization Problem and its essential features, Classification of Optimization Problems and its essential features. Mathematical modelling of typical chemical engineering optimization problems. Nature and classification of mathematical functions, Graphical representation of univariate and bivariate functions (using MATLAB / Python etc. – Assignment). Unimodal functions, determination of convexity and concavity of single and multivariate functions.	9
2	Analytical & Numerical methods for unconstrained optimization problems: <i>Analytical methods:</i> Necessary and sufficient conditions for optimum of univariate unconstrained functions.	9

	Numerical methods: One dimensional gradient-free search methods (Fixed & accelerated step size, Dichotomous search, Fibonacci search, golden-section method and quadratic interpolation), One dimensional gradient search methods (Newton's method and Quasi-Newton method).	
3	Analytical methods for constrained multivariate optimization problems: <i>Nonlinear programming with equality constraints:</i> method of direct substitution, Lagrange multiplier method, <i>Nonlinear programming with inequality constraints:</i> Kuhn-Tucker conditions for local optimality, Complex method, Rosen's gradient projection method. Numerical methods for unconstrained & constrained multivariate optimization problems: <i>Unconstrained multivariate optimization problems:</i> Univariate search, Powell's method, method of steepest descent, Fletcher-Reeve's conjugate-gradient method, Newton's method. <i>Constrained multivariate optimization problems:</i> Basic concepts and graphical representation of Linear programming, graphical solution, Simplex method and two-phase simplex method.	9
4	Optimization of Chemical Processes: Optimum pipe diameter for an incompressible fluid, Optimization of Heat Exchanger Network, Optimal reflux ratio for a staged Distillation column, Optimal design of an Ammonia Reactor. (Solving chemical engineering. Optimization problems in Excel/Python/MATLAB/Scilab or any optimization software- Assignments/ Microproject only) Chemical Product Design and Optimization: Introduction to the current product design process, typical applications of chemical product design, product formulation and optimization. Case studies- optimization of disinfectant formulation, optimization of mechanical properties of natural fiber polymer composites. Non-traditional Optimization Techniques: Introduction to Simulated annealing, Genetic algorithms and Particle Swarm Optimization (Algorithm only).	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the basic concepts in optimization and develop a mathematical representation of the optimization problem	K3
CO2	Construct the objective functions for constrained and unconstrained optimization problems.	K3
CO3	Apply appropriate optimization techniques to solve optimization problems.	K3
CO4	Discover suitable optimization techniques for solving chemical engineering problems of practical interest.	K4
CO5	Solve optimization problems using appropriate software tools.	K3
CO6	Explain the algorithm for non-traditional optimization techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3		3								3
CO3	3	3	3	3								3
CO4	3	3	3	3	3							3
CO5	3	3	3	3								3
CO6	3											3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Optimisation of Chemical Processes	Edgar T. F., Himmelblau D. M.	McGraw Hill	2 nd Edition
2	Optimization in Chemical Engineering	Suman Dutta	Cambridge University Press	2016
3	Applied optimization with MATLAB Programming	P.Venkataraman	John Wiley & Sons, Inc.	2 nd Edition, 2009
4	Operations Research: An Introduction	Hamdy A. Taha	Pearson Education Limited	Global Edition, 2017
5	Engineering Optimization: Methods and Applications	A. Ravindran, K. M. Ragsdell and G. V. Reklaitis	John Wiley & Sons, Inc.	2 nd Edition, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Optimization: Theory and Practice	Singiresu S. Rao	John Wiley & Sons, Inc	4 th Edition, 2009
2	Chemical Product Formulation Design and Optimization Methods, Techniques, and Case Studies	Ali Elkamel, Hesham Alhumade, Navid Omidbakhsh, Keyvan Nowruzi, Thomas Duever	WILEY-VCH	2023
3	Practical Chemical Process Optimization With MATLAB® and GAMS®	Ioannis K.Kookos	Springer	2022

4	Optimization of Chemical Processes: A Sustainable Perspective	José María Ponce Ortega, Rogelio Ochoa-Barragán, César Ramírez-Márquez	Springer	2024
5	Optimization with LINGO-18 Problems and Applications	Neha Gupta, Irfan Ali	CRC Press	1st Edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105139/
2	https://archive.nptel.ac.in/courses/103/105/103105139/
3	https://archive.nptel.ac.in/courses/103/105/103105139/ https://archive.nptel.ac.in/courses/103/103/103103164/#
4	https://archive.nptel.ac.in/courses/103/105/103105139/ https://archive.nptel.ac.in/courses/103/103/103103164/#

SEMESTER S7

FUEL CELL TECHNOLOGY

Course Code	PECHT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To expose the students to the fundamental knowledge required in the development of fuel cells

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Fuel Cell, Brief History of fuel cells, Types of Fuel Cells, Fuel Cell and conventional processes – comparison, Energy & power relations, units, Application scenarios, Advantages and disadvantages. Overview of Fuel Cell Types: PAFC, PEMFC, AFC, MCFC, SOFC - Major Cell Components, Processes, Operating Conditions, and applications. General Thermodynamics: Enthalpy-Heat potential of fuel, Gibb's free energy-Work potential of fuel, Reversible voltage - NERNST Equation, Voltage and P, T and concentration dependence – examples, Faraday's Laws, Efficiency: thermodynamic, voltage and fuel.	10
2	Reaction Kinetics: Electrochemical reaction fundamentals, electrode kinetics, Charge transfer and activations energy, Exchange current density - slow and fast reactions, Potential and equilibrium - galvanic potential, Reaction rate and potential - Butler Volmer equation & Tafel equation.	9
3	Charge and Mass Transport: Charge transport resistances, voltage losses, Ionic and electronic conductivities, Ionic conduction in different FC electrolytes: Aqueous, polymeric, and ceramic, Diffusive transport & voltage	9

	loss: Limiting current density, Nernstian and kinetic effect. Convective transport: flow channels, gas diffusion / porous layer, gas velocity, pressure, Flow channel configurations	
4	Stack Design: Sizing of a Fuel Cell Stack, Stack Configuration, Uniform distribution of Reactants, Heat removal, Stack Clamping Fuel Cell System Design: Hydrogen-Oxygen Systems, Hydrogen-Air Systems, Fuel Cell Systems with Fuel Processor, System Efficiency	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fuel cell technology and compare different types of fuel cell systems.	K2
CO2	Outline the fundamentals of electrochemistry, thermodynamics, fluid mechanics, heat and mass transfer, and reaction kinetics, appropriate for the design or review of components of fuel cells and fuel cell systems.	K2
CO3	Explain the fundamentals of charge and mass transport in fuel cells	K2
CO4	Develop and explain fuel cell stack and fuel cell systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				1						3
CO2	3	3	3			1						3
CO3	3	3				1						3
CO4	3	3	3			1						3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuel Cell Fundamentals,	Ryan P. O'Hayre, Suk-Won Cha, Whitney Colella & Fritz B. Prinz	John Wiley & Sons, Inc., New Jersey	2006
2	PEM Fuel Cells: Theory and Practice	Frano Barbir	Elsevier	2005
3	Fuel Cells and Their Applications.	Kordesch, K.; Simader, G	VCH	1996
4	Fuel Cell Systems Explained.	Larminie, J.; Dicks A	John Wiley & Sons Ltd: Chichester	1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuel Cell Handbook, 7th Edn.		EG & G Technical Services	Nov 2004
2	Handbook of Fuel Cells- Fundamentals, Technology and Applications. 2003; Vols 1-4	Vielstich, W, Gasteiger, H. A. Lamm, A.	John Wiley & Sons Ltd: NY, Vol-1-4	2003

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/102/103102015/
2	http://acl.digimat.in/nptel/courses/video/115105127/lec38.pdf
3	https://www.youtube.com/watch?v=62363H_I_Qk
4	http://acl.digimat.in/nptel/courses/video/121106014/lec35.pdf

SEMESTER 7
ADVANCED MATERIALS

Course Code	OECHT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs 30Min
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart knowledge of properties and applications of various types of advanced materials

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Classification and structure property relations of materials: Importance of Engineering materials, classification of materials, classification of materials- basic categories (metals, ceramics, and polymers), Composite materials and advanced materials: semiconductors, biomaterials, smart materials, and nanoengineered materials. Amorphous and semi-crystalline materials, mechanical, thermal and electrical properties of materials and their dependence on the microstructure, elastic and plastic behaviour of metals, ceramics and polymers, changes in the microstructure of materials due to mechanical and thermal load, length scales relevant for materials and their characterization.	8
2	Metal alloys: Types of metal alloys, processing and applications of metal alloys. Ceramics: types processing and applications of ceramics advanced ceramics, mechanical properties, processing and applications. Polymers: Classification, properties, processing and applications. Composites: particle-reinforced composites, Fiber-reinforced composites- Characteristics, synthesis and applications of composites.	10

	Biomaterials: Types of Biomaterials: Natural and Synthetic, Properties and Biocompatibility, Applications in Medical Devices, Implants, and Tissue Engineering.	
3	Nanomaterials and nanocomposites: advanced nano-materials - synthesis, properties and applications – fullerenes, graphene, graphite, carbon nano-tubes, nano wires, nano rods, nanofluids, nanoclusters. Nanocomposites - Matrix materials- Basics of Metal matrix, Ceramic Matrix and Polymer Matrix nanocomposites - Nano-reinforcements, nanofillers. Introduction to bionanotechnology (fundamental concepts only) - Nanomedicine, Drug delivery, Therapeutic applications. Applications of biosensors,	9
4	Smart materials: Types of smart materials, thermochromic and photochromic materials, Shape Memory Effect, Shape Memory Alloys, Shape Memory Polymers, , piezoelectric ceramics: Principles of Piezoelectricity, Perovskite Piezoceramic Materials, Single Crystals vs Polycrystalline Systems, Piezoelectric Polymers, properties and applications, magnetostrictive materials: Principles of Magnetostriction, Rare earth Magnetostrictive materials, Giant Magnetostriction and Magneto-resistance Effect, properties and applications, Electro-active Materials, Electronic Materials, Electro-active Polymers, Ionic Polymer Matrix Composite (IPMC)	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the important factors that affect the behaviour of materials under different conditions.	K2
CO2	Compare the physical properties of metals, alloys, polymers, ceramics, and composites significant in various engineering applications.	K2
CO3	Explain different types of advanced nanomaterials, nanocomposites , their methods of synthesis and applications.	K2
CO4	Explain the key theory and operation principles of smart materials, their synthesis, properties and applications.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Materials Science and Engineering An Introduction	William D. Callister, Jr.	Wiley	Tenth, 2018
2	The science of Engineering of Materials	Donald. R Askeland	Cengage learning	Sixth, 2010
3	Nanostructures & Nanomaterials: Synthesis, Properties & Applications	Joel I. Gersten	Wiley	2001
4	Nanocomposite science and technology	Pulikel M. Ajayan	Wiley	2005
5	Engineering analysis of smart material systems.	Donald J. Leo	John Wiley & Sons	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bionanotechnology	David S Goodsell	John Wiley & Sons,	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/113/104/113104009/
2	https://onlinecourses.nptel.ac.in/noc19_mm21/preview
3	https://nptel.ac.in/courses/112104173

SEMESTER S7

PROCESS SAFETY ENGINEERING

Course Code	OECHT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

- Impart the basic concepts of safety in Process Industries.
- Apply engineering fundamentals on fire safety by fire prevention and flammability diagram.
- Identify various hazards associated with chemical process industries.
- Develop an understanding about safety practices in industries and emergency planning.
- Incorporate inherent safety, Hazard analysis techniques, and awareness about government agencies, regulatory bodies, codes, and standards that govern the global, societal, and environmental impact.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to safety: Concept and importance of industrial safety. Fundamental safety tenets. Safety in the site selection and lay out. Review of Industrial Accidents- Major Chemical Industry Accidents, Bhopal, Flixborough, SEVESO. Cost of accidents. Key safe practices in chemical industry for accident prevention programme. Material safety data sheet (MSDS). Work permit system, Personal Protective Equipment's (PPE).	9
2	Fire & Safety: Classification of Fire- Pool fire, Jet fire, Flash fire, Explosion-UVCE, BLEVE, Dust explosion, Deflagration, Detonation. Toxic release, Runaway Reaction. Fire pyramid. Types of fire extinguishers and its handling. Fixed fire protection systems. Flammability diagram- construction, application. Hazards in a process industry: Classification, consequences of	9

	chemical hazards, physical hazards, electrical hazards, mechanical hazards and environmental hazards.	
3	Prevention techniques for hazards: Hazardous area classification. Safety in transportation of hazardous chemicals by road-HAZCHEM CODE, TREM CARD Relief system and Detectors. T.L.V, STEL, TLV-C, IDLH, UFL, LFL. Hazard rating of chemical plants- Dow index and Toxicity index. Emergency planning-onsite and offsite emergency planning, Mock drill.	9
4	Safety analysis: Safety Inspections, Safety Audits, safety Analysis, Hazard Survey and analysis, HAZOP, Bow tie diagram, Fault tree analysis, failure mode and effect analysis, Event tree analysis, examples. Safety integrity level (SIL). The concept of inherent safety. Occupational Health and Safety Administration, Safety provisions in Factories Act.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the basic concepts of safety in process industries.	K2
CO2	Apply engineering fundamentals on fire safety by fire prevention and flammability diagram.	K3
CO3	Develop an understanding about safety practices in industries and emergency planning.	K3
CO4	Select inherent safety, Hazard analysis techniques, and awareness about government agencies, regulatory bodies, codes, and standards that govern the global, societal, and environmental impact.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2			3						
CO2	3		2			3						
CO3	3		2			3						
CO4	3		2			3						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety in Chemical Plants/Industry and Its Management	B. K. Bhaskara Rao, Er. R. K. Jain, Vineet Kumar	Khanna Publishers	1 st edition, 2010
2	Industrial Safety, Health and Environment Management Systems	R. K. Jain & Sunil S Rao	Khanna Publishers	4 th Edition, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Encyclopaedia of Occupational Health & Safety		International labour Office, Geneva,	2012
2	Loss Prevention in Process Industries	Frank P. Lees	Butterworth-Heinemann	Vol.1,2&3,2 nd Edn.,1996
3	Guidelines for Hazard Evaluation Procedure		Centre for Chemical Process Safety. AIChE	1992
4	Methodologies in Hazard Identification and assessment	K.V. Raghavan and A. A. Khan	Manual by CLRI	December 1990
5	Chemical Process Safety Fundamentals with applications	Daniel A. Crowl/ Joseph F. Louvar	Prentice Hall international series.	2 nd Edition
6	Chemical Process Industrial safety	K.S.N Raju	McGraw Hill	2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/110/105/110105094/
2	http://www.digimat.in/nptel/courses/video/103107156/L42.html
3	https://archive.nptel.ac.in/courses/103/107/103107156/

SEMESTER S7

INDUSTRIAL INSTRUMENTATION

Course Code	OECHT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To study about different instruments and techniques used in chemical industry for measurement of various process variables and understand the theory behind them.
2. To understand the range of applicability and characteristics of these instruments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction-definition of instrumentation: concept of an instrument-functional elements and functions of an instrument. classification of instruments. Performance characteristics of an instrument like static and dynamic type. Temperature measurement- electrical,non-electrical, contact and non-contact methods, thermometers of three types like liquid-filled, vapour pressure and gas-filled type, bimetallic thermometers, resistance thermometers, thermocouple type-thermoelectric principles like Seebeck effect, Peltier effect & Thomson effect and the laws of thermoelectricity-thermocouple output measurement. Radiation methods-radiation and optical pyrometry. Thermistors-resistance characteristics and their application in temperature measurement	9
2	Pressure measurement: manometers of U-tube type, well type and inclined type. Prandtl and air type micromanometers. Barometer method for atmospheric pressure measurement. Low pressure measurement by kenetometer, McLeod gage, thermal conductivity gauge, Pressure measurement using bourdon tube, flat and corrugated diaphragms, and capsules. Measurement of pressure in corrosive fluids using liquid seal and diaphragm seal. Transducers of electrical and mechanical type. Density	9

	measurement using constant volume hydrometer and air pressure balance method, gas density detector and gas specific gravity measuring system.	
3	Flow measurement: using head type flowmeters based on differential pressure measurement orifice meter, venturi meter, flow nozzle and pitot tube. Open channel meters like weirs, flumes. Electromagnetic flowmeters. Variable area meters like rotameter and cone and float type. Mechanical flowmeters of positive displacement type like rotating disk and turbine type & anemometers. Level measurement-direct type and indirect type. Differential pressure method for pressurized vessels. Solid level detectors.	9
4	Moisture content and humidity: definition, moisture content determination by thermal drying. Instruments for measuring humidity like hygrometer, psychrometer, dew point apparatus. pH measurement using calomel electrode. Composition analysis using spectroscopic methods like absorption, emission and mass spectrometers. Analysis of solids by X-ray diffraction. Gas analysis by thermal conductivity, polarography & chromatography.	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the various techniques used for the measurement of industrial parameters.	K2
CO2	Illustrate the working of various instruments used for measurement of temperature, pressure and density in process plants.	K2
CO3	Summarize various types of flow measurement and level measurement transducers and their installation.	K2
CO4	Summarize the humidity and moisture measurements adopted in industrial environment.	K2
CO5	Illustrate analytical instruments.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3				2							
CO3	3				2							
CO4	3				2							
CO5	3				2							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Industrial Instrumentation	D. Patranabis	Tata McGraw Hill,	2nd Edition, New Delhi, Reprint 2009.
2	Industrial Instrumentation & Control	S. K. Singh	Tata McGraw Hill,	3 rd Edition, Reprint 2009.
3	Industrial Instrumentation	K.Krishnaswamy&S.Vijayachitra	New age International	Reprint 2008.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanical & Industrial Measurements	R.K.Jain	Khanna Publishers	11th Edition, 2004.
2	Measurement Systems Application & Design	Ernest O. Doebelin, Dhanish. N. Manik,	TMH	5th Edition, 2004.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105130/
2	https://archive.nptel.ac.in/courses/103/105/103105130/
3	https://archive.nptel.ac.in/courses/103/105/103105130/
4	https://archive.nptel.ac.in/courses/103/105/103105130/

SEMESTER S7**ADVANCED WASTEWATER TREATMENT**

Course Code	OECHT725	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives: This course will enable students to

1. Gain knowledge of advanced wastewater technologies
2. Understand the design and operation of the wastewater treatment process
3. Study on Electrochemical wastewater treatment processes and advanced oxidation processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Advanced Waste Water Treatment: Need of Advanced wastewater treatment, Technologies used for Advanced waste water treatment-Classification of Technologies. Nutrient Removal –Nitrogen Removal: Nitrification , Denitrification, Simultaneous nitrification and denitrification. Phosphorus Removal : Introduction, Phosphorus removal by Chemical Precipitation: Principles of process, Chemicals applied, Chemistry of phosphorus precipitation, Process configuration, Phosphorus removal by Biological Precipitation: Principles of the process, Microorganisms involved in the process, Process configurations.	9
2	Wastewater Treatment: Primary treatment: Screening, Grit removal, Neutralization, equalization, Sedimentation, Flotation (oil & grease removal), Air stripping. Secondary treatment: principles of waste treatment, basic kinetic equation, continuous flow treatment models, oxygen requirement in aerobic process, production of sludge. Conventional biological process: Activated Sludge Process (ASP)- development of material balance equations, Trickling Filters , UASB and RBC anaerobic filters. Low-cost wastewater treatment: Aerated lagoons, stabilization ponds,	9

	oxidation ditches.	
3	Advanced Treatment Options: Tertiary treatment – ion exchange, Membrane separation Techniques: Brief description of MF, UF, NF membranes. Reverse osmosis principle, Membrane materials, Types of membranes – Plate & frame, tubular, hollow fibre, spiral wound membranes. Adsorption: Introduction, Fundamentals of adsorption, Type of adsorbents ,Activated carbon adsorption, Granular carbon adsorption.	9
4	Electrochemical Wastewater Treatment Processes: Introduction, Electro-coagulation : Factors affecting Electrocoagulation, Electrode materials , Reactor configurations. Electro-floatation : Factors affecting electro floatation, Comparison with other technology, Reactor configurations, Electro-oxidation : Electro oxidation process, Reactor configurations.Advanced Oxidation Processes: Theory of advanced oxidation, Types of oxidizing agents, ozone based and non- ozone based processes, Fenton and photo-Fenton Oxidation. Solar Photo Catalytic Treatment Systems.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of advanced technologies in Wastewater treatment	K3
CO2	Analyse primary and secondary treatment methods	K4
CO3	Select the most appropriate processes for the tertiary treatment of wastewater	K3
CO4	Explain different types of electrochemical treatment options for wastewater treatment	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3					2					2
CO2	3	3					2					3
CO3	3						2					3
CO4	3						2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Wastewater Engineering – Treatment and Reuse	Metcalf & Eddy	4th edition. Tata McGraw-Hill	2003
2	Water and Wastewater Technology	Mark.J.Hammer&Mark.J. Hammer J	Prentice Hall of India. Ltd.	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Engineering	Peavy, Rowe & Tchobanoglous	McGraw-Hill	2015
2	Unit Processes in Water and Wastewater Engineering	Casey, T.J.,	Wiley Interscience	1997
3	Membrane Separation Processes	Kaushik Nath	PHI	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/agZwWMN0C2c
2	https://youtu.be/6nM1bShwRNw
3	https://youtu.be/_LSKHSQmwwE , https://youtu.be/RW6jiE-K6rA ,https://youtu.be/WR2Pinj81Nk
4	https://youtu.be/0Dpv7-s-uqM , https://youtu.be/7h5cuXiAQak , https://youtu.be/ni6NtEjrFLA

SEMESTER S7

AIR POLLUTION CONTROL

Course Code	OECHT726	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To understand the atmospheric pollutant's, standards, regulations, emission sources and their fate.
2. To imbibe the fundamentals of atmospheric stability and air pollutant transport, characteristics of emission source and their control methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Structure and composition of Atmosphere – Definition, Scope and Scales of Air Pollution- Sources and classification of air pollutants-Effect on human health, vegetation, animals, property, aesthetic value and visibility- Ambient Air Quality and Emission standards–Ambient and stack sampling and Analysis of Particulate and Gaseous Pollutants.	9
2	Effects of meteorology on Air Pollution - Fundamentals, Atmospheric stability- Dry adiabatic lapse rate derivation, Inversion, Wind profiles and stack plume patterns- Atmospheric Diffusion Theories–Dispersion models, Plume Rise-Numerical Problems.	9
3	Gas Particle Interaction – Working principle, Design and performance equations of Gravity Separators, Centrifugal separators, Fabric filters, Particulate Scrubbers, Electrostatic Precipitators – Operational Considerations- Factors affecting Selection of Control Equipment	9
4	Working principle, Design and performance- equations of absorption, Adsorption, condensation, Incineration, Bio scrubbers, Venturi scrubber- Bio filters.	9

	Sources types and control of indoor air pollutants- sick building syndrome types – HVAC system-IAQ issues and impacts on occupants— Developing an IAQ profile-Diagnose IAQ problem- Control-Quantification and Measurement.	
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define the sources, classifications, effects of air pollutants.	K1
CO2	Explain the ambient air quality standards as well as sampling and analysis of air pollutants from emission sources.	K2
CO3	Outline the concepts of atmospheric dispersion characteristics and nature based on lapse rate and inversion.	K2
CO4	Explain the selection strategies and types of equipment and its design to control particulates and gaseous pollutants	K2
CO5	Explain the indoor air pollution sources, measurements, standards and control methods.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	3					3
CO2						3	3	3				3
CO3	2	3					3					2
CO4			3				3					2
CO5							3					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Air Pollution Control Engineering,	Lawrence K. Wang, Norman C. Pareira, Yung Tse Hung,	Humana Press Inc.	Second, 2004
2	Air Pollution Control Engineering,	Noel de Nevers	Mc Graw Hill	Second, 1999
3	Air Pollution and Control Technologies,	Anjaneyulu. Y,	Allied Publishers (P) Ltd.	First, 2002
4	Environmental Pollution Control Engineering	C.S.Rao	Wiley Eastern Ltd,	Third, 2018
5	Handbook of Air Pollution Prevention and Control,	Nicholas P. Cheremisinoff	Butterworth and Heinemann, Elsevier Science (USA), 2002	2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Air Pollution	David H.F. Liu, Bela G. Liptak	Lweis Publishers	First, 2000
2	Air Pollution (Vol.I – Vol.VIII)	Arthur C.Stern	Academic Press	First, 2006
3	Air Pollution Engineering Manual	Wayne T.Davis,	John Wiley & Sons, Inc.	First, 2000

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105107213
2	https://nptel.ac.in/courses/105107213
3	https://nptel.ac.in/courses/105107213
4	https://nptel.ac.in/courses/105107213

SEMESTER 7

DESIGN OF EXPERIMENTS

Course Code	OECHT727	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Mathematics: Probability and statistics	Course Type	Theory

Course Objectives:

1. To help the students to explore innovative strategies for constructing and executing experiments including factorial and fractional factorial designs.
2. To help the students in extraction of meaningful data from well-designed minimal number of experiments at the lowest possible material costs.
3. To help the students to apply statistical models in analysing experimental data and optimise the response of interest from an experiment.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to DoE: Strategy of experimentation, Typical applications of Experimental design ,basic principles of experimental design : replication, randomization and blocking, Guidelines for designing the experiments Basic statistical concepts:- Concept of random variable, probability, density function, cumulative distribution function, measure of central tendency,measures of variability, sampling and sampling distribution, Statistical distributions: Normal, Log normal & Weibull distributions, Hypothesis testing.	9
2	Experiments with single factor:The analysis of Variance Analysis of fixed effect model, Model adequacy checking, contrasts, Orthogonal contrasts, Regression models and ANOVA, Violation of Normality Assumption: Kruskal-Wallis test. Randomized block designs, Latin square designs, Balanced Incomplete Block Designs	9

3	Factorial designs: Definition, Estimating model parameters, Fitting response curves and surfaces. The 2^k Factorial Design, Blocking and Confounding in the 2^k Factorial Design; Focus of 2^2 and 2^3 designs, Blocking and Confounding in the 2^k Factorial Design	9
4	Regression analysis : Linear and multiple linear regression, Estimation of parameters in linear regression models, Hypothesis testing in multiple regression, Confidence intervals in multiple regression, Regression model diagnostics, Testing for lack of fit Response surface methodology: Introduction, Method of steepest ascent, Response surface designs for first-order models, analysis of a second order response surface, Central composite design	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain basic principles of design of experiments.	K2
CO2	Examine statistical analysis of single experiments and do post hoc analysis.	K4
CO3	Analyze the experimental data using statistical methods.	K4
CO4	Choose an appropriate design for the given the research problem	K5
CO5	Examine regression analysis and response surface methodology and apply them in real life applications	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3	3	3					3		
CO3	3	3	3	3	3					3		
CO4	3	3	3	3	3							3
CO5	3	3	3	3	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design and Analysis of Experiments, , Inc. 2013	Douglas C. Montgomery	John Wiley & Sons	8th Edition, 2013
2	Statistics for Experimenters: Design, Innovation, and Discovery,	Box, G. E., Hunter, W.G., Hunter, J.S., Hunter, W.G	Wiley	2nd Edition, 2005
3	Response Surface Methodology: Process and Product Optimization using Designed Experiments	R. H. Myers, D. C. Montgomery	Wiley	4 th edn, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Statistical Quality Control	D. C. Montgomery.	John Wiley & Sons	8 th edn, 2019
2	Design of Experiments in Chemical Engineering:	Živorad R. Lazić	Wiley	1 st edn, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/110/105/110105087/
2	https://archive.nptel.ac.in/courses/110/105/110105087/
3	https://archive.nptel.ac.in/courses/110/105/110105087/
4	https://archive.nptel.ac.in/courses/110/105/110105087/

SEMESTER 8
CHEMICAL ENGINEERING

SEMESTER S8

SOLID WASTE MANAGEMENT

Course Code	PECHT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Understand the fundamentals of solid waste management and its importance in environmental protection
2. Learn about the various methods of solid waste disposal and composting.
3. Understand the principles of waste minimization, reuse, and recycling

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Solid wastes:- Sources, nature and characteristics - types of solid waste, Residential, Commercial, Hazardous wastes, and Industrial wastes, Properties of Solid wastes, Waste generation, Sampling and analysis, Characteristics of solid wastes - Energy content, Chemical content, Estimation of chemical composition of a solid waste sample generation rates - Estimation of solid waste quantities - Factors affecting generation rates.	9
2	Collection of solid waste: On-site storage systems. Disposal methods such as sanitary landfill –methods, leachate in landfills – control of leachate movement, Gas movement – control, Design and operation of landfills, Landfarming, deep well injection etc.	9
3	Composting Factors affecting composting, Aerobic composting and anaerobic digestion, Bio gas plant operations, Incineration, Municipal incinerators, Grates, Furnaces, Pyrolysis of solid waste.	7
4	Recovery, Recycle and Reuse- Material and Energy recovery operations. Overview of solid waste management practices in India. Industrial and Hazardous solid waste management- Biomedical waste, Treatment storage	11

	disposal facility (TSDF) Integrated Waste Management (IWM), E-waste, Spent Oil waste, Basics of Data base Management System (DBMS), Geographic Information System (GIS) and Remote Sensing data in planning and management of MSW. Plastic waste management and extended producers responsibility (EPR):- Plastic waste definition, plastic waste management rules 2022, EPR Act, Benefits and challenges of EPR implementation, Emerging trends and innovations in EPR and plastic waste management.	
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain municipal solid waste management systems with respect to its physical, chemical and biological properties.	K2
CO2	Develop the design and operation of landfills.	K3
CO3	Compare disposal methods of MSW by applying specific criteria.	K2
CO4	Summarize the recovery and recycling methods of waste management.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3				2					
CO2	3		3				3					3
CO3	3	3	3									3
CO4	3	3					3					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Engineering	Howard S.Peavy, Donald R.Rowe, George Tchobanoglous	McGraw Hill	1985
2	Handbook of Solid Waste Management	Frank Kreith, George Tchobanoglous	McGraw Hill	2002
3	Plastics Waste Management: Processing and Disposal	Muralisrinivasan Natamai Subramanian	Wiley-Scrivener	2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Solid waste Engineering	P.Aarne Vesilind and William Worrell	Cengage Learning, Asia Pte Limited	2012
2	Environmental Engineering	Gerard Kiely	McGraw Hill	2008
3	Handbook of Solid Waste Management and Waste Minimization Technologies	Nicholas P. Cheremisinoff	Butterworth-Heinemann	2002
4	Solid Waste Management	Luis F. Diaz, George M. Savage, Linda L. Eggerth, Larry , Rosenberg	United Nations Environment Programme, .Paris	2005
5	Solid and liquid waste management: Waste to wealth	Rajaram Vasudevan , Siddiqui Faisal Zia , Agrawal Sanjeev , Khan Mohammad Emran	PHI Learning Pvt Ltd	2016
6	Plastic recycling & Waste Management	Dr. Andras Szeberenyi, Bhgah Y. Adam, Er. Harish Sharma, Dr Dhivya A	JSR PUBLICATIONS	2023

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105103205
2	https://nptel.ac.in/courses/105105160

SEMESTER S8

HYDROGEN ENERGY: PRODUCTION, STORAGE, TRANSPORTATION AND SAFETY

Course Code	PECHT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. This course provides a comprehensive knowledge of hydrogen production, storage and transportation, utilization in various sectors, associated energy conversion devices, sensing and safety.
2. The course will provide a broad knowledge of hydrogen as an energy carrier.
3. The course also covers the classification and working of fuel cells.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Hydrogen energy: Hydrogen Economy- Decarbonisation of the Economy, road to the Hydrogen Economy. Properties of hydrogen as fuel. General introduction to infrastructure requirement for hydrogen production, storage, dispensing and utilization, and hydrogen production plants.	9
2	Hydrogen production pathways: Thermal process-Steam reformation, Advanced Methods of Steam Reforming, Partial Oxidation Method for Hydrogen Production. Electrochemical- Electrolysis, photo electro chemical method. Thermo- chemical water splitting. Biological process: hydrogen production from microbial biomass conversion- gasification, pyrolysis.	9
3	Hydrogen storage, safety and transportation: Physical and chemical properties, general storage methods, compressed storage-composite cylinders, metal hydride storage, carbon-based materials for hydrogen storage. Hydrogen safety aspects, backfire, pre-ignition, hydrogen emission	9

	NOx control techniques and strategies, Hydrogen Transportation via Hydrogen Pipelines, Long Distance Hydrogen Transmission, transportation via road, ships and in form of Liquid Organic Hydrogen Carriers (LOHC), hydrogen transported as ammonia, liquefied hydrogen (LH2) transportation, Hydrogen Refuelling Stations.	
4	Hydrogen usage: Overview of fuel cells, Alkaline fuel cells, Proton exchange membrane fuel cells, Methanol fuel cells, Phosphoric acid fuel cells, Molten carbonate fuel cells, Solid oxide fuel cells, Fuel cell comparison, Hydrogen combustion engines, Hydrogen turbines, Hydrogen gas burner, Gas engine combined heat and power plants.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the importance of hydrogen economy.	K2
CO2	Explain different methods of hydrogen production.	K2
CO3	Compare different techniques of hydrogen storage, transportation and various safety challenges in hydrogen storage.	K2
CO4	Explain the various uses of hydrogen energy.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1				2						2
CO2	2	1				2						2
CO3	2	1				2						2
CO4	2	1				2						2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydrogen Production by Electrolysis	Agata Godula-Jopek	Wiley-VCH, Germany	2015
2	Hydrogen Fuel: Production, Transport and Storage	Gupta, R. B.	CRC Press, Taylor & Francis Group.	2009
3	Fuel Cell Handbook	EG & G Technical Services,	Morgantown, West Virginia, USA	2004
4	Tomorrow's Energy: Hydrogen, Fuel cells and the prospects for a cleaner planet	Peter Hoffman	MIT Press, Cambridge, London, England	2001
5	Hydrogen storage: state-of-the-art and future perspective. Netherlands	Tzimas, E., Filiou, C., Peteves, S.D., & Veyret, J.B.	European Communities.	2003
6	Principles of Fuel Cells	Xianguo Li	Taylor & Francis.	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Hydrogen Storage	Michael Hirscher	Wiley-VCH	2010
2	Electrochemical Methods	A.J. Bard and L.R.Faulkner	John Wiley & Sons, Inc.	
3	Introduction to Fuel Cell Technology	Chris Rayment and Scott Sherwin	Notre Dame, U.S.A.	2003
4	Fuel Cells: From Fundamentals to Applications	S Srinivasan	Springer	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	NPTEL :: Chemical Engineering - NOC:Hydrogen Energy: Production, Storage, Transportation and Safety
2	NPTEL :: Chemical Engineering - NOC:Hydrogen Energy: Production, Storage, Transportation and Safety
3	NPTEL :: Chemical Engineering - NOC:Hydrogen Energy: Production, Storage, Transportation and Safety
4	NPTEL :: Chemical Engineering - Fuel Cell Technology

SEMESTER S8

COLLOID AND INTERFACIAL SCIENCE

Course Code	PECHT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To study the basic concepts and tools for the analysis of colloidal and interfacial properties.
2. To study theory behind surface and interfacial tension.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic concepts of colloids and interfaces.- examples of interfacial phenomena-capillary action, tears of wine.-solid-fluid interfaces- colloids and interfaces-classification of colloids- electrical charge on colloid particles- stability of colloids-Shulze-hardy rule - kinetic and thermodynamic stabilities- preparation of colloids- parameters of colloid dispersions.	9
2	Properties of colloid dispersions: sedimentation under gravity-sedimentation in centrifugal field- Brownian motion-osmotic pressure-optical-electrical and rheological properties-surfactants and their properties-emulsions and micro emulsions-foams	9
3	Surface and interfacial tension: surface and interfacial tension- contact angle and wetting-shape of the surfaces and interfaces- radius of curvature-young-laplace equation-kelvin equation-measurement of surface and interfacial tension-maximum bubble pressure method-spinning drop method-measurement of contact angle.	9
4	Intermolecular and surface forces- Vander waals forces-experimental determination of vanderwaals forces- Hamaker constant- Estimation of surface tension from Hamaker constant- DLVO theory- non DLVO forces- Adsorption at solid-fluid interfaces-Henry, Freundlich and Langmuir	9

	adsorption isotherm-BET theory- adsorption hysteresis-Direct experimental techniques for measurement of adsorption.	
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Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the fundamental principles of colloids and interfacial science and the stability of colloids.	K2
CO2	Explain the properties of colloids-emulsion, micro emulsions and foams.	K2
CO3	Solve surface tension and/or interfacial tension for various systems based on the data given related to various methods	K3
CO4	Identify the Vander Waal's forces and adsorption isotherms.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Colloid and Interface Science	Ghosh, P	PHI Learning, New Delhi.	2009
2	Principles of Colloid and Surface Chemistry	Hiemenz and R. Rajagopalan	Academic Press, New York,	3 rd Ed., 1997
3	Physical Chemistry of Surfaces.	A.W. Adamson and A.P. Gast	Wiley-Interscience	6 th Ed., 1997
4	Interfacial Phenomena : Equilibrium and Dynamic Effects.	Miller, C. A. and P.Neogi.	CRC Press, NY	2 nd Ed., 2019.
5	Fundamentals of Interfacial Engineering	Stokes, R. J. and Evans, D.F	Wiley-VCH, N.Y.	1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Intermolecular and Surface Forces.	J.N. Israelachvili	Academic Press, New York	2 nd Ed., 1992.
2	Colloidal Dispersions: Suspensions, Emulsions, and Foams.	I.D. Morrison and S. Ross	Wiley	2002
3	The Colloidal Domain: Where Physics, Chemistry, Biology, and Technology Meet.	D.F. Evans and H. Wennerström,	Wiley-VCH	2 nd Ed., 1999.
4	Surfactants and Interfacial Phenomena	M.J. Rosen	Wiley-Interscience	3 rd Ed., 2004.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/106/105106204/ , https://nptel.ac.in/courses/103104045 .
2	https://archive.nptel.ac.in/courses/103/106/105106204/ , https://nptel.ac.in/courses/103104045 .
3	https://archive.nptel.ac.in/courses/103/106/105106204/ , https://nptel.ac.in/courses/103104045 .
4	https://archive.nptel.ac.in/courses/103/106/105106204/ , https://nptel.ac.in/courses/103104045 .

SEMESTER S8

PROCESS SAFETY MANAGEMENT

Course Code	PECHT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Explain importance of PSM and the basic requirements in PSM standards
2. Explain the element required by OSHA standard: Process Safety Information and Process Hazard Analysis
3. Explain the elements required by OSHA standard: Operating Procedures, Employee Participation, Training, Contractors, Pre-start-up Review.
4. Explain the elements required by OSHA standard: Mechanical Integrity, Hot work Permit, Management of Change, Incident Investigation, Emergency Planning and Response, Compliance Audits and Trade Secrets

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Process Safety Management (PSM), Benefits of an effective PSM program, PSM standards, Basic requirements in PSM standards, 14 elements of OSHA, Tools used to implement PSM standards, Industries applicable	9
2	Hazards of chemicals in the processes: Toxicity, Permissible exposure limits, Physical data, Reactivity data, Corrosivity data, Thermal and chemical stability data, hazardous effects of inadvertent mixing of different materials, Materials of construction, Electrical classification, Material and energy	9

	balances, Safety systems, Diagrams: Block Diagram, Process Flow Diagram (PFD) and Piping and Instrumentation diagram (P&ID). Introduction to the PHA techniques: Checklist, What-if analysis, Hazard and Operability study (HAZOP), Failure Modes and Effects Analysis (FMEA), Fault tree analysis and event tree analysis, Bow-tie analysis, Quantitative Risk Assessment (QRA)	
3	Operating Procedures: Initial start-up, Normal operations, Temporary operations, Emergency shutdown, Emergency operations, Normal shutdown, Start-up, Employee participation and training: Initial training, Refresher training, Employer responsibilities, Contractors: contractor employer responsibilities, Pre-start-up Review	9
4	Mechanical Integrity, Pressure vessels and storage tanks, Piping systems, Relief and vent systems and devices, Emergency shutdown systems, Controls, Pumps, Hot work Permit, Management of Change, technical basis for the proposed change, impact of the change on employee safety and health, modifications to operating procedures, necessary time period for the change, authorization requirements for the proposed change, Incident Investigation, Emergency Planning and Response, Compliance Audits, Trade Secrets	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain importance of PSM and the basic requirements in PSM standards	K2
CO2	Explain the element required by OSHA standard: Process Safety Information and Process Hazard Analysis	K2
CO3	Explain the elements required by OSHA standard: Operating Procedures, Employee Participation, Training, Contractors, Pre-start-up Review.	K2
CO4	Explain the elements required by OSHA standard: Mechanical Integrity, Hot work Permit, Management of Change, Incident Investigation, Emergency Planning and Response, Compliance Audits and Trade Secrets	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3	3	3	3	3					
CO2	3		3	3	3	3	3					
CO3	3		3	3	3	3	3					
CO4	3		3	3	3	3	3					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Process Safety Management		OSHA Academy, United States Department of Labor	
2	Process Safety Management	Alexis M. Herman, Charles N. Jeffress	OSHA-3132, U.S. Department of Labor	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Safety Management Guide		Canadian Society for Chemical Engineers	4 th Edition
2	Process Safety Management General Awareness Training	Gary Whitmore		
3	Accident Prevention Manual for Industrial Operations		N.S.C. Chicago	1982
4	Industrial Accident Prevention	Heinrich H.W.	McGraw-Hill Company, New York	1980
5	Safety Management in Industry	Krishnan N.V.	Jaico Publishing House, Bombay	1997
6	Safety at Work	John Ridley	Butterworth & Co., London	1983
7	Industrial Safety	Blake R.B.	Prentice Hall, Inc., New Jersey	1973

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-ch19/
2	https://archive.nptel.ac.in/content/storage2/courses/103103027/pdf/mod9.pdf
3	https://archive.nptel.ac.in/noc/courses/noc19/SEM2/noc19-ch19/

SEMESTER S8

AI-ML APPLICATION IN CHEMICAL ENGINEERING

Course Code	PECHT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Provide background on AI and ML in chemical engineering and allied systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Artificial Intelligence: Types of problems AI addresses like Computer Vision, Natural Language Processing, Robotics, Expert Systems, Object detection and Image segmentation. Artificial Intelligence in Chemical Engineering Background: Early Attempts, Expert System Era, Neural Network Era, Effects and Lacunae of Expert System Era and Neural Network Era on Chemical Engineering , Deep Learning and Data Science Era. Applications of AI in chemical engineering: areas like fault diagnosis, Process control, Process design, development and selection like Separation Design, Heat-Exchanger Network Synthesis, Thermodynamic Model Selection and Physical Property Estimation, Oil reservoir image segmentation. Early applications of AI techniques in chemical engineering. The FALCON, CATDEX, BIOEXPERT and the ExSep algorithms.	9
2	Introduction to Machine Learning: Exploring sub-discipline of AI: Machine Learning, Supervised learning, Unsupervised learning, Reinforcement learning, Classification problems, Regression problems, Clustering problems, Introduction to neural networks and deep learning. Deep Artificial neural networks (DNN): The working of the Rosenblatt perceptron, multilayer perceptrons. Activation functions and their	9

	importance in incorporating nonlinearities into the predictive models. The feedforward process in ANN layers with dense fully connected layers. The error (loss) functions as a measure of the ANN performance. Backpropagation algorithm for neuron learning.	
3	Convolution neural networks (CNN): The drawbacks of Deep ANN. How CNN take into account the spatial patterns. The working of the CNN in pattern recognition. The role of kernels, pooling, padding and stride in CNN learning. How the kernels help in reducing the learning parameters (weight sharing). Some outstanding deep networks proposed like AlexNet, VGGNet, Inception, Google Net and ResNets. The problems they faced and how they resolved the problems.	9
4	Case Studies: Oxidative coupling of methane (OCM) over the Mn/Na ₂ WO ₄ /SiO ₂ catalyst in a fluidized bed reactor, Control of Reactive Distillation Column, Use of Qualitative Models in Discrete Event Simulation to Analyze Malfunctions in Processing Systems, Fault Detection and Diagnosis Using Artificial Neural Networks, A Modular Approach to Multiple Faults Diagnosis, AI in Chemical Engineering Recent Trends and Future Outlook: Emergence in large scale systems of self-organizing intelligent agents.	9

Course Assessment Method

(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop fundamental understanding of the application of AI in chemical and allied engineering.	K3
CO2	Apply Machine Learning (ML) principles to Chemical Engineering	K3
CO3	Apply Deep Learning Networks to solve Chemical Engineering problems	K3
CO4	Choose the use of AI and ML applications in Chemical Engineering problems by case studies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3							
CO2	3	3	3	3	3							
CO3	3	3	3	3	3							
CO4	3	3	3	3	3							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Inside Deep Learning: Math, Algorithms and Models.	Raff, Edward.	Manning Publications,	2022
2	Artificial Intelligence in Chemical Engineering,	Quantrile, Thomas, Liu, Y. A,	Academic Press,	1991
3	Artificial Intelligence in Process Engineering,	Michael L. Mavrovouniotis	Academic Press	1990

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning.	Ian Goodfellow, Yoshua Bengio, Aaron Courville.	Adaptive Computation and Machine Learning series	2015
2	The Promise of Artificial Intelligence in Chemical Engineering: Is It Here, Finally?	Venkat Venkata subramanian,	AIChE, Vol. 65, No. 2	2019
3	Application of AI in Chemical Engineering.	Zeinab Hajjar, Shokoufe Tayyebi and Mohammad Hosein Eghbal Ahmadi		2018,
4	Optimization of fluidized bed reactor of oxidative coupling of methane.	Eghbal-Ahmadi M-H, Zaerpour M, Daneshpayeh M, Mostoui N.	International Journal of Chemical Reactor Engineering.	2012; 10:1-21
5	Fuzzy identification of reactive distillation for acetic acid recovery from waste water.	Araromi DO, Sonibare JA, Emuoyibofarhe JO.	Journal of Environmental Chemical Engineering.	2014; 2:1394-1403
6	Control of Different Reactive Distillation Configurations	Shih-Bo Hung and Ming-Jer Lee	AIChE, Vol. 52, No. 4	2006
7	Control of Reactive Distillation Column: A Review,	Neha Sharma and Kailash Singh	International Journal of Chemical Reactor Engineering, Vol. 8	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=H3xX4fz35Zc
2	https://www.youtube.com/watch?v=UgusRy8Oz3c
3	https://www.youtube.com/watch?v=ipPL6_j8MvE
4	https://www.youtube.com/watch?v=iCTU-IZ6rPQ&list=PLoSULBSCtofflIdbr898SDp5gIqo8XL-t

SEMESTER S8

PROCESS SYSTEMS ENGINEERING

Course Code	PECHT867	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Impart skills to design and optimize chemical processes by applying principles of mass and energy balances, mathematical modeling, and advanced simulation tools.
2. Provide a comprehensive understanding of process control dynamics, including the design and implementation of control systems to maintain process stability and safety.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Systems Engineering: Systems and their origin, examples of problems in Systems Engineering. Foundations of Systems Engineering: Scope and Formulation of Engineering Problems, Goals, Objectives, Specifications and Constraints, Types of Models; Hierarchical decomposition of systems, Types of Problems; Forward solution and inversion of models	9
2	Structural Analysis of Systems: Graphs and digraphs: Representation of systems, Partitioning and Precedence. Ordering of systems, Structural analysis of modeling equations, Structural controllability and observability of systems, Applications to engineering problems	7
3	Steady State Analysis of Systems: Formulating steady-state models and simulations, Degrees of freedom and design specifications, The Sequential-Modular Strategy, The Equation-Oriented Strategy, Applications to engineering problems. Optimization of Systems: Theory and Algorithms: Basic concepts and definitions, Linear programming, Unconstrained nonlinear optimization,	10

	Nonlinear Programming, Combinatorial optimization, Applications to engineering problems	
4	Simulation of Dynamic Systems: Basic concepts: Systems described by ODEs and DAEs, formulating dynamics simulations; consistent initialization, Numerical integration of ODEs and DAEs, Modeling-simulation of hybrid Discrete/Continuous systems, Applications to engineering systems Model-Based Process Control: The nature of feedback control, The concept of model-based control systems, Design and analysis of model-based control systems applications.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate a thorough understanding of the foundational concepts in systems engineering	K2
CO2	Apply principles of mass and energy balances and mathematical modeling for the design, analysis, and optimization of chemical processes	K3
CO3	Demonstrate mastery in structural analysis of systems to solve practical engineering problems effectively	K2
CO4	Analyze and interpret the dynamic behavior of chemical processes and propose suitable control measures.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							
CO2	3	3	3		3							
CO3	3	3	3		3							
CO4	3	3	3		3							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Systems engineering. Theory and practice,	Alberto Sols,	Comillas publication	2013
2	Systems Engineering	Kossiakoff, A., Sweet, W.N., Seymour, S.J., and Biemer S.M., John Wiley Sons Inc., New Jersey	Wiley	Second ed. 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Process Modelling, Simulation and Control for Chemical Engineers,	Luyben W.L	McGraw Hill International Edition	Second, 1989
2	Computational Methods for Process Simulation	W Fred Ramirez	Butterworth-Heinemann Ltd	2 nd Edition 1997
3	Process Dynamics - Modelling Analysis and Simulation	Biquette W.B.	Prentice Hall of India	First, 1998

SEMESTER S8

DESIGN AND ANALYSIS OF EXPERIMENTS

Course Code	PECHT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic course in Probability and statistics	Course Type	Theory

Course Objectives:

1. To give the students reasonably good background in statistical decision
2. To help the students in extraction of meaningful data from well-designed minimal number of experiments at the lowest possible material costs.
3. To help the students in all domains of their life by imparting them a vision for critical appraisal and analysis of data

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to DoE: Strategy of experimentation, Typical applications of Experimental design , basic principles of experimental design : replication, randomization and blocking, Guidelines for designing the experiments Basic statistical concepts:- Concept of random variable, probability, density function, cumulative distribution function, measure of central tendency, measures of variability, sampling and sampling distribution, Statistical distributions: Normal, Log normal & Weibull distributions, Hypothesis testing. Use of R/ Ms-Excel/MINITAB/Design Expert for statistical analysis of data and evaluate them (self study)	8
2	Experiments with single factor :The analysis of Variance Analysis of fixed effect model, Model adequacy checking, contrasts, Orthogonal contrasts, Regression models and ANOVA, Violation of	9

	Normality Assumption: Kruskal-Wallis test. Randomized block designs, Latin square designs, Balanced Incomplete Block Designs Use of MS- Excel /R/ MINITAB/Design Expert etc for the analysis of ANOVA and evaluate them (self study)	
3	Factorial designs: Definition, Estimating model parameters, Fitting response curves and surfaces. The 2^k Factorial Design, Blocking and Confounding in the 2^k Factorial Design; Focus of 2^2 and 2^3 designs, Blocking and Confounding in the 2^k Factorial Design Use of R/MINITAB/Design Expert/ Quantum Boost/ Modde for designing experiments using Factorial design and do analysis and evaluation (self study)	9
4	Regression analysis : Linear and multiple linear regression, Estimation of parameters in linear regression models, Hypothesis testing in multiple regression, Confidence intervals in multiple regression, Regression model diagnostics, Testing for lack of fit Response surface methodology: Introduction, Method of steepest ascent, Response surface designs for first-order models, analysis of a second order response surface , Central composite design • Use of MS- Excel/MINITAB,/Design Expert /Quantum Boost/ Modde/R for regression analysis and evaluation (self study) • Use of MINITAB,/Design Expert /Quantum Boost/ Modde/R for analysis and evaluation using response surface methodology (self study)	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (2 points)

- Check for clarity and completeness of the problem definition.
- Check for clear objectives and divide into sub objectives

2. Problem Analysis (4 points)

- Examine the problem and identify the variables and responses
- Divide the problem to manageable parts

3. Implementation of software (2 points)

- Use suitable software for getting the solution
- Check for the appropriateness of tools used

4. Evaluate (4 points)

- Evaluate the proposed solution.
- Assess the correctness of solution.

5. Validation of the results (4 points)

- Validate the results using available data.
- Do the error analysis of the results obtained.

6. Conclusion and presentation (4 points)

- Summarize the results obtained
- Do critical thinking of the results obtained and evaluate.
- Submit a technical report

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain basic principles of design of experiments.	K2
CO2	Examine statistical analysis of single experiments and do post hoc analysis.	K4
CO3	Analyze the experimental data using statistical methods.	K4
CO4	Choose an appropriate design for the given the research problem	K5
CO5	Choose regression analysis and response surface methodology & apply them in real life applications and evaluate the results	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	3	3	3	3					3		
CO3	3	3	3	3	3					3		
CO4	3	3	3	3	3							3
CO5	3	3	3	3	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design and Analysis of Experiments, , Inc. 2013	Douglas C. Montgomery	John Wiley & Sons	8th Edition,2013
2	Statistics for Experimenters: Design, Innovation, and Discovery,	Box, G. E., Hunter, W.G., Hunter, J.S., Hunter, W.G	Wiley	2nd Edition, 2005
3	Response Surface Methodology: Process and Product Optimization using Designed Experiments	R. H. Myers, D. C. Montgomery	Wiley	4 th edn, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Statistical Quality Control	D. C. Montgomery.	John Wiley & Sons	8 th edn, 2019
2	Design of Experiments in Chemical Engineering:	Živorad R. Lazić	Wiley	1 st edn, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/110/105/110105087/
2	https://archive.nptel.ac.in/courses/110/105/110105087/
3	https://archive.nptel.ac.in/courses/110/105/110105087/
4	https://archive.nptel.ac.in/courses/110/105/110105087/

SEMESTER S8

ENVIRONMENT MANAGEMENT SYSTEMS

Course Code	OECHT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To study environmental policies for sustainable development.
2. To create awareness to practice waste management through continuous improvement.
3. To develop practices that enables an organization to reduce its environmental impacts and increase its operating efficiency.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to environment Management (EM)- Goals of EM- EM tools, Ecosystem and its characteristics, Processes of Ecosystem- Biomass, Energy and energy flow, Policy and Legal Aspects of EM- Introduction to Environmental Policies, Environmental Policies and Programmes in India, Environmental Laws and Legislations, Environmental Legislations in India	9
2	Environmental Management System Standards, EMS Standards: ISO 14000, General overview, Implementation of EMS conforming to ISO 14001, Environmental management product design for the environment (ISO 14062), Product stewardship, principles of clean production, packaging, sustainable procurement, the social responsibility function of corporations, ecolabelling, ecological and carbon footprints (ISO 14064-6), Environmental management techniques, Ecosystem approach to risk assessment	9
3	Life Cycle Assessment (LCA), Stages in LCA of a Product, Procedures for LCA, Different Applications of Life Cycle Assessment (LCA), Environmental Auditing (EA), Elements of Audit Process, Waste Audits and	9

	Pollution Prevention Assessments, EA in Industrial Projects, Waste Audits and Pollution Prevention opportunities in Textile , Sugar, Pulp & Paper, Electroplating, Tanning industry, Dairy, Cement and Chemical industries, Liability Audits and Site Assessment, Auditing of EM.	
4	Environmental Design (ED) for Manufactured Products, ED for Developmental Planning, Environmental Economics, Economics and the Environment, Environmental valuation, Economics of Natural Resources, Ecological Economics, Applications of EMS.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain Environmental Management Systems (EMS) and their main steps, relevant policies, other environmental management tools	K2
CO2	Summarize different environment management system standards	K2
CO3	Explain steps involved in environmental auditing and to make use of environmental auditing in various industries.	K3
CO4	Explain Environmental Design and environmental economics.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3					
CO2	3					3	3					
CO3	3				2	3	3		2			
CO4	3					3	3					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Management	Vijay Kulkarni and Ramachandra T.V	Commonwealth of Learning, Canada and Indian Institute of Science, Bangalore, Printed by TERI Press, New Delhi	2006
2	Environmental Management Systems	Christopher S. and Mark Y	Earthscan Publications, First South Asian Edition	2007, 3 rd edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Environmental Engineering and Science	Gilbert M.M.	Pearson Education	2004, 2 nd edition
2	Environmental Planning and Management	Madu C.N.	Imperial College Press	2007
3	Environmental Management Strategies: The 21st Century Perspective	Gabriele Crognale	Prentice Hall Ptr Environmental Management Series,	Vol 5
4	Basic Concepts in Environmental Management	Kenneth M.M.	Boca Raton, FL, Lewis	1999
5	ISO 14000 Environmental Management	David L.G. and Stanley B.D.	Prentice Hal	2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/120/108/120108004/ , https://onlinecourses.nptel.ac.in/noc22_ag10/preview
2	https://archive.nptel.ac.in/courses/120/108/120108004/ , https://onlinecourses.nptel.ac.in/noc22_ag10/preview
3	https://archive.nptel.ac.in/courses/120/108/120108004/ , https://onlinecourses.nptel.ac.in/noc22_ag10/preview
4	https://archive.nptel.ac.in/courses/120/108/120108004/ , https://onlinecourses.nptel.ac.in/noc22_ag10/preview

SEMESTER S8

FUEL ENGINEERING

Course Code	OECHT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To understand the atmospheric pollutant's, standards, regulations, emission sources and their fate.
2. To understand the fundamentals of atmospheric stability and air pollutant transport, characteristics of emission source and their control methods.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Global and Indian Energy production & consumption patterns and energy resources. Solid Fuels: Biomass, Wood and Charcoal. Classification & Rank of Coal, Peat ,Lignite, Sub-Bituminous coal, Bituminous coal, Anthracite coal, Cannel & Bog head coal. Physical Properties of coal, Proximate & Ultimate Analysis of Coal, Cleaning, washing & Storage of coal. Theory of coal Pyrolysis and Carbonization: Low Temperature Carbonization(LTC), High Temperature Carbonization(HTC), Properties of solid fuels.	9
2	Liquid fuels :Origin and classification of petroleum, crude exploration, petroleum refining processes, transportation, storage and handling of liquid fuels, properties & testing of petroleum products. Liquid fuel from coal: Bergius and Fischer Tropsch process. Other Synthetic Liquid fuels.(Benzol,shaleoil,Gashol, power alcohol Colloidal fuel).	9

3	Gaseous fuels: Classification of gaseous fuel; Physico-chemical principles, Calorific Value, Wobbes index, and flame speed. Types of gaseous fuels: natural gases, methane from coal mines, manufactured gases, producer gas, water gas, biogas, refinery gas, LPG, cleaning and purification of gaseous fuels. Manufactured fuels: Agro fuels, Bio-Fuels: types of bio-fuels, production processes and technologies, bio fuel applications.	9
4	Combustion stoichiometry and thermodynamics, calculation of heat of combustion, theoretical & actual combustion processes - air fuel ratio, estimation of dry and wet flue gases for known fuel composition, calculation of the composition of fuel & excess air supplied, flue gas analysis.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain energy resources, global and Indian energy production and consumption.	K2
CO2	Summarize the origin, classification, analysis, properties of solid fuels and their applications	K2
CO3	Explain the origin, classification, analysis, properties of liquid, fuels and their applications	K2
CO4	Explain the origin, classification, analysis, properties of gaseous and manufactured fuels and their applications. Outline basics of combustion stoichiometry and thermodynamics.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	3					3
CO2						3	3	3				3
CO3	2	3					3					2
CO4			3				3					2
CO5							3					3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuels & Combustion	Dr. Samir Sarkar	Orient Longmans	Third edition, 2009
2	"Modern Petroleum Technology", Vol. 1, Upstream	Dave, R.A.(Ed)	John Wiley and sons.	Sixth,2001
3	"Modern Petroleum Technology", Vol. 2, Downstream	Lucas, A.G. (Ed.)	John Wiley and sons.	Sixth,2002
4	"Combustion",	Glassman, I	Academic Press	Second,2014.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Modern Petroleum Refining Processes"	Rao, B.K.B	Oxford & IBH Publishing Co. Pvt. Ltd	Fourth ,2018
2	Fundamentals of Renewable Energy Systems,	D. Mukherjee and S. Chakrabarti,	New Age International Publishers	First, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/105107213
2	https://nptel.ac.in/courses/105107213
3	https://nptel.ac.in/courses/105107213
4	https://nptel.ac.in/courses/105107213

SEMESTER S8

NANOMATERIALS AND NANOTECHNOLOGY

Course Code	PECHT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3: 0: 0: 0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Equip students with fundamental knowledge on the physics, chemistry, synthesis and characterisation of nanomaterials.
2. Enable the students with the understanding of the applications and potential of nanomaterials and nanotechnology for humankind.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Nanotechnology: History of nanotechnology, Pioneers in the field of nanotechnology. Classification of nano-materials - Zero-, one-, two- and three-dimensional nanostructured materials, Electromagnetic spectrum, particle size and its significance. Physics of nanomaterials: Size effect on thermal, electrical, electronic, mechanical, optical and magnetic properties of nanomaterials, surface area and aspect ratio, band gap energy, quantum confinement effect. Chemistry of nanomaterials-Ionic properties of nanomaterials. Electronic phenomenon in nanostructures. Synthesis methods - top down and bottom up approaches. Top down approach – size reduction techniques like milling and machining. Bottom up approach - Sol-gel methods, Chemical vapour deposition, Physical vapour deposition, Laser ablation methods.	9
2	Synthesis, properties and applications of nanomaterials: Nanometals - gold and silver, different types of nano-oxides - Al_2O_3 , TiO_2 , ZnO and SiO_2 . Special nanomaterials: synthesis, properties and applications –	9

	fullerenes, graphite, graphene, carbon nano-tubes, nano wires, nano rods, nanofluids, nanoclusters, quantum dots. Nanocomposites - Matrix materials- Basics of Metal matrix, Ceramic Matrix and Polymer Matrix nanocomposites - Nano-reinforcements, nanofillers and nanoclays.	
3	Characterisation techniques: Scanning Electron Microscopy (SEM), Energy Dispersive X-ray Spectroscopy (EDS), Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM), Scanning Probe Microscopy (SPM) - Scanning Tunneling Microscopy (STM) UV-visible spectroscopy, Raman spectroscopy, Nuclear Magnetic Resonance Spectroscopy (NMR), Fourier Transform Infrared Spectroscopy (FTIR), X-Ray Diffraction (XRD), Dynamic Light Scattering (DLS), Thermogravimetric analysis (TGA), Differential Thermal Analysis (DTA), Differential Scanning Calorimetry (DSC).	9
4	Nanoelectronics: Introduction to Micro Electro Mechanical Systems (MEMS), Nano Electro Mechanical Systems (NEMS), Nanomanipulation – STM based atomic manipulations, Nanolithography, softlithography, Scanning Probe Lithography, photolithography, E-beam Lithography, Focused ion beam lithography, Dip-pen Lithography. Applications in energy and catalysis: Applications (fundamental concepts only) Nanoscale advances in energy and catalysis - Nanotechnology for sustainable energy, nanotechnology enabled renewable energy technologies. Application of nanomaterials in catalysis. Applications in biotechnology: (fundamental concepts only) - Nanomedicine, Drug delivery, Therapeutic applications, biosensors.	9

Course Assessment Method (CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic principles of Physics and Chemistry and synthesis methods of in Nanomaterials.	K2
CO2	Summarize the synthesis, properties and applications of nanomaterials and nanocomposites.	K2
CO3	Outline various characterisation techniques applied to nanomaterials.	K2
CO4	Explain the applications of nanotechnology in different fields of science and technology.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					2	2					2
CO2	3					2	2					2
CO3	3					2	2					2
CO4	3					2	2					2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Physics and Chemistry of Materials	Joel I. Gersten	Wiley	2001
2	Nano: The Essentials	T. Pradeep	McGraw-Hill (India) Pvt Limited	2008
3	Nanostructures & Nanomaterials: Synthesis, Properties & Applications	G. Cao. Y Wang	World Scientific	2011
4	Materials Characterization Techniques	S Zhang, L. Li and Ashok Kumar	CRC Press	2008
5	Nanocomposite science and technology	P. M. Ajayan, L. S. Schadler, P. V. Braun	Wiley-VCH	2005
6	Nanobiotechnology: Concepts, Applications and Perspectives	C. M. Niemeyer, C. A. Mirkin,	Wiley-VCH	2004
7	Nanotechnology and Nanoelectronics–Materials, Devices, Measurement Techniques	W. Fahrner	Springer-Verlag Berlin, Germany	2004
8	Surface Science: Foundations of Catalysis and Nanoscience	K.W. Kolasinski	John Wiley	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanomaterials: Synthesis, Properties and Applications	S. Edelstein and R. C. Cammarata	Taylor & Francis	1998
2	Physics and Chemistry of Nanostructured Materials	S.Yang and P.Shen	Taylor & Francis	2000
3	Characterization of Nanophase materials	Z L Wang (Ed.)	Wiley-VCH	2000
4	Nanofabrication, Principles, Capabilities and Limits,	Zheng Cui	Springer Science + business media, New York	2008
5	X-rays in Nanoscience - Spectroscopy, Spectromicroscopy, and Scattering Techniques	Guo, Jinghua (Ed.)	John Wiley	2010
6	Handbook of Nanoscience, Engineering and Technology,	William A. Goddard III (Ed), Donald Brenner (Ed), Sergey Edward Lyshevski (Ed), Gerald J Iafrate (Ed)	Kluwer publishers	2007
7	Bionanotechnology: Lessons from nature	David S Goodsell	John Wiley	2004
8	Textbook of Nanoscience and Nanotechnology	B S Murty, P Shankar, Baldev Raj, B B Rath, James Murday	Springer	2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/113/104/113104102/
2	https://archive.nptel.ac.in/courses/118/102/118102003/ https://archive.nptel.ac.in/courses/118/107/118107015/
3	https://archive.nptel.ac.in/courses/102/104/102104069/
4	https://archive.nptel.ac.in/courses/112/107/112107283/ https://archive.nptel.ac.in/courses/118/107/118107015/

SEMESTER S8

BIOPROCESS ENGINEERING

Course Code	OECHT834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide the students with the basics of bioprocess engineering.
2. To develop bioengineering skills for the production of biochemical product using integrated biochemical processes

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Isolation of Microorganisms: Isolation, preservation, and improvement of industrially important microorganisms, screening methods, culture preservation. Strain improvement: mutagenesis, protoplast fusion and r-DNA technology, culture collection and biological resource centres. Fermentation: General requirements of a fermentation process, classification of fermentation processes. Media Optimization	9
2	Sterilization: Medium & air sterilisation methods, del factor, batch & continuous sterilization. Design of depth filter and estimation of efficiency, in-situ sterilization in fermenter, thermal death kinetics of cells and spores, extinction probability, batch and continuous steriliser design aspects, sterilisation of liquid wastes.	9
3	Role of diffusion in bioprocessing: oxygen uptake in cell cultures, oxygen transfer in bioreactors, measurement of volumetric oxygen transfer coefficient Ideal reactor operation, batch, fed batch and continuous operation of mixed bioreactors, chemostat with immobilized cells, chemostat with cell recycle.	9

4	Study of structured models for analysis of various bioprocess: compartmental models, models of cellular energetic and metabolism, single cell models, plasmid replication and plasmid stability model. Dynamic simulation of batch, fed batch, steady and transient culture metabolism.	9
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Course Assessment Method
(CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the development of inoculums and general needs of fermentation	K2
CO2	Summarize medium and air sterilization methods	K2
CO3	Summarize the mass transfer effects in bioreactors	K2
CO4	Explain the modelling and simulation of bioprocesses	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3									3		3
CO2	3	3										3
CO3	3					2				2		3
CO4	3					3						3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bioprocess Engineering Basic Concepts	Michael L Schuler, FrikretKhargii	Prentice Hall of India	
2	Bioprocess Engineering Principles	Pauline M Doran	Academic Press Inc: CBS Publishers and Distributors Pvt Ltd	2 nd edition 2013
3	Biochemical Engineering Fundamentals	James E Bailey, David F Ollis	McGraw-Hill	2 nd edition 1986

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Modern Bioprocessing	Sarfaraz K. Niazi, Justin L. Brown	Taylor & Francis	2017
2	Perry's Chemical Engineers Handbook	Don W. Green, Marylee Z. Southard	McGraw-Hill	9 th edition
3	Bioprocess Technology: Kinetics and Reactors	Anton Moser	Springer Verlag, 2012	
4	Principles of Fermentation Technology	P F Stanbury Dr. A Whitaker	Elsevier,	2 nd edition 1995

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/103/105/103105054/
2	https://archive.nptel.ac.in/courses/103/105/103105054/
3	https://archive.nptel.ac.in/courses/103/105/103105054/
4	https://archive.nptel.ac.in/courses/102/105/102105064/

SEMESTER S8
FIRE ENGINEERING

Course Code	OECHT835	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. This course is intended to impart knowledge on physics and chemistry of fire, fire prevention techniques, fire detection and control in process industries.
2. It also outlines the elements of emergency planning and preparedness for a fire incident.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Fire Processes:- Elements of fire, Classes of fire, NFPA diamond, Principles of fire spread, Measures of flammability: Flash point, Fire point, Autoignition temperature, LFL and UFL ,Flammability limit dependence on temperature and pressure, Estimating flammability limits, Flammability diagram: Construction and application. Products of combustion: Flame, heat, smoke, fire gases, Development of fire- incipient, smoldering, flame and heat stages, Special kinds of combustion-Flash fire, Pool fire, Jet fire, Boil over, Slope over, Dust explosion, BLEVE, UVCE.	9
2	Fire Prevention Techniques:-Control of Ignition Sources: Electrical Area classification, Personal ignition sources, Hot work, Static Electricity: Fundamentals, Electrostatic discharges, streaming current, controlling of static electricity: Methods to prevent electrostatic ignitions. Controlling oxygen levels: Limiting oxygen concentration, Inerting- Vacuum and pressure purging-Advantages and Disadvantages of the Various Pressure and Vacuum purging Procedures.	9

3	Fire detection and fire alarm systems:- Fire detection- Need and importance of automatic fire detection system, principle of detection, classification of detectors; Heat detectors , Smoke detectors, Flame detectors, Flammable gas detection. Fire alarm system- classification of alarm system, Automatic alarm system, various components and modules; Addressable and non-addressable system.	9
4	Principles of Fire Extinguishments:- Basic concept of fire fighting with water, carbon dioxide, powders, foams, halons. Classification of different types of fire fighting systems- Fixed and movable systems. Description, working principle, method of operation of different types of portable fire extinguishers-water type, foam type, dry powder type, CO2 type. Automatic water sprinkler system-Total flooding system.	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fundamentals of fire processes.	K2
CO2	Choose suitable techniques to prevent fire in chemical process industries.	K3
CO3	Explain the working principle of different types of fire detectors and alarm system .	K2
CO4	Interpret the concept of fire extinguishment with different extinguishers.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3									2		
CO2	3									2		
CO3	3									2		
CO4	3									2		

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Safety	Daniel A Crowl & Joseph F Louvar	Prentice-Hall	2 nd Edition
2	Fire protection and prevention the essential handbook	Barendra Mohan Sen	UBS publishers and Dist	1 st edition 2019
3	A handbook of Fire Tecchnology	R S Gupta	Universities Press	2 nd edition 2010
4	Fire Safety Management Handbook	Daniel E. Della-Giustina	CRC press.	3 rd edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Process Safety Engineering	S.K Biswas & Umesh Matur	CRC Press	1 st edition 2022
2	Evaluation of Fire Safety	Rasbash D., Ramachandran G., Kandola B., Watts, J., Law M	John Wiley & Sons Ltd., England	2004
3	NFPA Fire protection handbook			21 st edition
4	SFPE Handbook of Fire Protection Engineering	Morgan J. Hurley, Daniel T. Gottuk , John R. Hall Jr.	Springer	5 th edition 2016

SEMESTER S8

WASTE TO ENERGY TECHNOLOGY

Course Code	PECHT867	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. Develop a comprehensive understanding of various waste-to-energy technologies, including their principles, processes, and applications
2. Evaluate the effectiveness and feasibility of different waste-to-energy technologies in converting waste into usable energy.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Waste to Energy Technology: Introduction to Waste to Energy Technology, Importance and benefits of Waste to Energy, Global and local perspectives on waste management and energy recovery. Types and Sources of Waste: Classification of waste (municipal, industrial, agricultural, etc.), Sources and characteristics of waste suitable for energy conversion, Waste collection, segregation, and preprocessing methods.	9
2	Thermal Conversion Technologies: Incineration: process, advantages, and disadvantages, Incineration technologies and their applications. Pyrolysis and Gasification: Pyrolysis: types, process, and applications, Gasification: principles, types, and benefits, Comparative analysis of pyrolysis and gasification	9
3	Biological Conversion Technologies: Overview of biological conversion technologies in WTE. Importance and role of biological conversion in waste management Anaerobic digestion: process, types, and products, Technologies, and	9

	systems for anaerobic digestion. Fermentation and Composting: Fermentation: ethanol production from waste, Composting and other biological treatments, Comparative analysis of fermentation and composting	
4	Chemical Conversion Technologies: Transesterification: Fundamentals of transesterification, Advantages and challenges of transesterification, Chemical reactions involved, Catalysts used (acid, base, and enzyme catalysts). Biodiesel production from waste oils and fats: Raw materials and feedstock, Process flow and equipment. Hydrothermal processing: Principles of hydrothermal processing, Supercritical water and its properties, Hydrothermal liquefaction (HTL) and hydrothermal carbonization (HTC) Conversion of wet waste to energy: Suitable feedstock and raw materials, Process flow and equipment. Overview of emerging technologies in chemical conversion: Catalytic depolymerization, Plasma gasification and Wet oxidation	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the importance and benefits of various waste-to-energy technologies in global and local contexts.	K2
CO2	Utilize various thermal conversion technologies for waste-to-energy.	K3
CO3	Utilize various biological conversion technologies for waste-to-energy	K3
CO4	Apply chemical conversion technologies for waste-to-energy	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							
CO2	3	3	3		3							
CO3	3	3	3		3							
CO4	3	3	3		3							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Waste Recovery and Management	Ajay, Parveen, A. Kumar, R. K. Mittal, R. Goel	CRC Press	2023
2	Waste to Energy Conversion Technology	N. Klinghoffer, M. Castaldi	Woodhead Publishing Series in Energy,	2013.
3	Waste-to-Energy	M. J. Rogoff, F. Screve	William Andrew	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Waste-to-Energy Approaches Towards Zero Waste	C. M. Hussain, S. Singh, L. Goswami	Elsevier	2021
2	Up-to-Date Waste-to-Energy Approach	P. Stehlik	Springer	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ch29/preview
2	https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-ch09/
3	https://archive.nptel.ac.in/noc/courses/noc21/SEM1/noc21-ch09/

SEMESTER 8

NON-CONVENTIONAL ENERGY SYSTEMS

Course Code	OECHT837	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the importance and role of non-conventional energy systems in sustainable development.
2. To study various non-conventional energy sources and their technologies.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Non-Conventional Energy Systems: Classification of Energy sources, Definition and importance of Non-Conventional Energy ,Global and national energy scenario, Environmental impacts of conventional energy sources, Need for non-conventional energy sources. India's non-conventional sources. Solar Energy: Solar Radiation, Measurement and estimation techniques, Solar Thermal Systems: Flat plate collectors, Concentrating collectors: Parabolic trough, dish, and solar tower. Applications: Solar water heaters, solar dryers, solar ponds. Solar Photovoltaic Systems: PV cell principles and materials (silicon, thin-film, organic), PV modules and arrays, solar farms, off-grid systems. Solar Power Plants.	10

2	Geothermal Energy: Energy Conversion Systems, energy extraction principles: Hot water and dry steam systems. Wind Energy: Wind Characteristics and Measurement, Wind Energy Conversion Systems, horizontal and vertical axis windmills, performance characteristics, Wind Power Generation.	8
3	Ocean Energy: Ocean Energy Resources: Tidal energy, Wave energy, Ocean thermal energy. Tidal Energy Conversion Systems: Tidal barrages and tidal stream systems, Wave Energy Conversion Systems: Oscillating water columns, point absorbers, and overtopping devices, operation. Ocean Thermal Energy Conversion (OTEC): Working principles and types (closed-cycle, open-cycle, hybrid). Energy from biomass: : Principles of Bio-Conversion, Anaerobic /aerobic digestion, pyrolysis, gasification, anaerobic digestion (biogas production), Biodiesels : Manufacture and characteristics.	9
4	Hydrogen Energy: Hydrogen Production Methods: Electrolysis, Thermochemical processes. Storage and Transportation: Compressed gas, liquid hydrogen, metal hydrides. Safety and infrastructure. Fuel Cells: Working principle, types, Characterization and durability of fuel cells, Applications.	9

Course Assessment Method

(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Classify various forms of Non-renewable and renewable energy and their importance.	K2
CO2	Explain solar energy and their technologies.	K2
CO3	Summarize Wind and Geothermal energy technologies	K2
CO4	Explain Ocean and Biomass energy technologies.	K2
CO5	Explain the principle of Hydrogen and Fuel cell technology.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					3	3					
CO2	3					3	3					
CO3	3					3	3					
CO4	3					3	3					
CO5	3					3	3					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy for a Sustainable World	Goldmberg J., Johansson, Reddy A.K.N. & Williams R.H.	John Wiley	
2	Renewable Energy Sources & Conversion Tech	Bansal N.K., Kleeman M. & Meliss M	Tata McGraw Hill	
3	Non-Conventional Energy Systems	Mittal K.M	Wheeler Pub	
4	Non-Conventional Energy Sources	Rai G.D	Khanna Pub.	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Technology	Venkataswarlu D	S. Chand	
2	Energy Technology	Rao S. & Parulekar B.B.	Khanna Pub.	
3	Solar Energy	Sukhatme S.P	Tata McGraw Hill	4 th edition 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1,2,3,4	https://onlinecourses.nptel.ac.in/noc22_ge14